

30-DAY CRASH COURSE

Computer Science XI

Complete preparatory material in 30 days — daily 3-hour plans, ★
Golden first, weekly checkpoints, mock paper and repair day.

Bilingual support: English + اردو

تیس دن کا کریش کورس — روزانہ منصوبہ، گولڈن، ماک پیپر

01 Computer Systems

02 Computational Thinking & Algorithm

03 Programming Fundamentals

04 Data and Analysis

05 Application and Impacts of Computing

06 Digital Literacy

XI

30-day calendar

Tick each day after the 3-hour block. Do not skip checkpoint or mock days.

Day	Focus	Kind	★ Golden today
Week 1			
Day 01	Computer Systems (1/5)	Teach	Analog and Digital Signals
Day 02	Computer Systems (2/5)	Teach	Logic Gates (Basic Gates: AND, OR, NOT)
Day 03	Computer Systems (3/5)	Teach	Logic Diagrams, Karnaugh Maps (K-Maps)
Day 04	Computer Systems (4/5)	Teach	Software Development Life Cycle (SDLC), The Waterfall Model
Day 05	Computer Systems (5/5)	Checkpoint	The Agile Model, Communication Models and the OSI 7-Layer Model
Week 2			
Day 06	Computational Thinking & Algorithm (1/3)	Teach	Algorithm, Algorithm vs Pseudocode, Decomposition
Day 07	Computational Thinking & Algorithm (2/3)	Teach	Sorting Algorithms: Bubble Sort, Sorting Algorithms: Selection Sort
Day 08	Computational Thinking & Algorithm (3/3)	Checkpoint	Searching Algorithms: Binary Search
Day 09	Programming Fundamentals (1/5)	Teach	—
Day 10	Programming Fundamentals (2/5)	Teach	Fundamentals of Python Programming, Variables, Data Types in Python, Operators and Operands: Arithmetic and Assignment
Week 3			
Day 11	Programming Fundamentals (3/5)	Teach	Relational, Logical and Membership Operators, Bitwise Operators, Control Structures, Selection Statements: if and if-else
Day 12	Programming Fundamentals (4/5)	Teach	Selection Statements: if-elif-else and Nested if, Repetition Statements: the for Loop and range(), The while Loop, break and continue, Nested Loops and for vs while

Day	Focus	Kind	★ Golden today
Day 13	Programming Fundamentals (5/5)	Checkpoint	—
Day 14	Data and Analysis (1/3)	Teach	Keys and Integrity Constraints
Day 15	Data and Analysis (2/3)	Teach	Entity Relationship Model (ER-Model), Referential Integrity
Week 4			
Day 16	Data and Analysis (3/3)	Checkpoint	Creating Queries in MS Access
Day 17	Application and Impacts of Computing (1/3)	Teach	Artificial Intelligence (AI), Internet of Things (IoT) & Components
Day 18	Application and Impacts of Computing (2/3)	Teach	Information Sources (Primary, Secondary, Tertiary)
Day 19	Application and Impacts of Computing (3/3)	Checkpoint	Assistive Technologies
Day 20	Digital Literacy (1/4)	Teach	Types of Data, Data-Collection Strategies, Interviews and Surveys
Week 5			
Day 21	Digital Literacy (2/4)	Teach	Prototypes, Observation and Simulations, Primary and Secondary Data
Day 22	Digital Literacy (3/4)	Teach	Case Study: Digital Inquiry Project, Steps 1 – 3: Advanced Search and Methodology, Steps 4 – 6: Survey Preparation and Primary Data Collection
Day 23	Digital Literacy (4/4)	Checkpoint	Steps 7 – 9: Secondary Data and Spreadsheet Organisation, Steps 10 – 12: Data Analysis, Conclusion and Recommendations, The Final Digital Artefact
Day 24	★ only · Chapters 1–2	Golden blitz	14 Golden cards
Week 6			
Day 25	★ only · Chapters 3–4	Golden blitz	16 Golden cards
Day 26	★ only · Chapters 5–6	Golden blitz	15 Golden cards
Day 27	Say every chapter recap aloud, book closed	Recap	All chapters
Day	Focus	Kind	★ Golden today
Day 28	2 hours 30 minutes · 75 marks · no notes	Mock	Whole syllabus
Day 29	Mark to the key. List every ★ Golden you missed.	Mark	Missed Goldens
Day 30	Redo the Day 29 list only. Sleep by 10 pm.	Repair	Missed Goldens only

How to run this 30-Day Crash Course

This booklet is a **complete 30-day exam plan**. Days 1–23 teach every topic as revision cards. The last day of each chapter is a timed checkpoint. Days 24–26 are ★ Golden blitz. Day 27 is recaps. Day 28 is the mock. Day 29 you mark. Day 30 you repair and stop.

یہ تیس دن کا کریش کورس ہے۔ پہلے ۲۳ دن سب موضوعات، ہر باب کے آخر میں ٹیسٹ، آخری ہفتہ گولڈن، ماک پیپر اور اصلاح۔

Every teaching day — 3 hours, no phone

1. **0–15 min.** Recite yesterday's ★ Golden definition and redraw its table from memory.
2. **15–105 min.** Study today's cards. Copy every table. Say the Urdu line aloud.
3. **105–135 min.** Closed-book drill at the end of the day. Mark at once.
4. **135–180 min.** Homework: one 5-mark answer using Define · Explain · Example · Diagram · Working.

If you fall behind: do only the ★ Golden cards that day. Never skip a checkpoint day. Never start a new chapter until you can say that chapter's Golden definitions without looking.

The last seven days

- **Days 24–26.** Golden blitz. No new non-Golden topics.
- **Day 27.** Recaps only. Close the book and say each chapter aloud.
- **Day 28.** Sit the mock in 2 hours 30 minutes. Hall conditions.
- **Day 29.** Mark with the key. Write a repair list of every Golden you missed.
- **Day 30.** Redo only that repair list. Sleep by 10 pm. Do not open a new topic.

Chapter 1: Computer Systems

Computer Systems (1/5) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Discrete and Continuous Quantities · Introduction to Digital Systems · Analog and Digital Signals · Boolean Algebra

★ **Must lock:** Analog and Digital Signals

If behind, skip non-Golden. Phone in another room.

1.1.1 Discrete and Continuous Quantities

Everything around you that can be measured or counted is a quantity.

- Quantities come in two kinds.
- Discrete quantities are separate, countable items with specific values.
- You can count them one by one: 1, 2, 3 ... There is nothing "in between".

Discrete (counted)	Continuous (measured)
Number of students in a classroom	Temperature of water
Number of cars in a parking lot	Speed of a moving car
Has fixed, separate values	Has any value in a range

Remember: Stairs vs Ramp. On stairs you can only stand on step 1, 2 or 3 — that is discrete data. On a ramp you can stand at any height — that is continuous data. Ask...

منفصل (Discrete) مقادیر وہ ہیں جنہیں ایک ایک کر کے گنا جا سکے، جیسے کلاس میں طلبہ کی تعداد۔

1.1.2 Introduction to Digital Systems

A digital system is an electronic system that uses only two values: 0 and 1.

- 0 represents OFF, LOW, or FALSE.
- 1 represents ON, HIGH, or TRUE.

Advantage	Detail
Reliable	Less affected by noise and interference.
Accurate	No loss of quality when copying data.
Easy to design	Only two states (0 and 1) to handle.
Programmable	Can be easily controlled by software.

Remember: Think of a bit as a light switch: it is either ON (1) or OFF (0). A digital system is a huge panel of such switches.

ڈیجیٹل نظام صرف دو قیمتیں استعمال کرتا ہے: 0 اور 1، جنہیں بٹ (Bit) کہتے ہیں۔

1.1.3 Analog and Digital Signals ★ Golden

A signal is a changing voltage (or other quantity) that carries information.

- Analog signals are continuous signals that change smoothly over time, like a sine wave.
- Examples: the human voice, a traditional mercury thermometer.
- Digital signals have only two levels: HIGH (1) and LOW (0).

Feature	Analog data	Digital data
Nature	Continuous flow.	Discrete, separate values (bits).
Wave shape	Sine wave (smooth curve).	Square wave (sudden jumps).
Reliability	Low (affected by noise).	High (resistant to noise).
Examples	Human voice, traditional thermometer.	Computer data.

Remember: Analog = A smooth curve. Digital = Doorsteps. Analog looks like ocean waves; digital looks like a castle wall (square teeth).

Exam: Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least...

اینالگ سگنل وقت کے ساتھ ہموار انداز میں بدلتا ہے، جیسے سائن ویو (Sine Wave)؛ انسانی آواز اس کی مثال ہے۔

1.1.4 Boolean Algebra

Boolean algebra was developed by the English mathematician George Boole.

- It is a kind of algebra that deals with only two values: TRUE (1) and FALSE (0).
- It is the foundation of digital electronics and computer programming — every if condition in a program is a small piece of Boolean algebra.

Concept	Description	Example
Variables	Letters that hold the value 0 or 1.	A, B, C, X, Y
Operations	Actions performed on variables.	AND, OR, NOT
Expressions	Combinations of variables and operations.	A AND B, $Y = A + B$

Trap: In Boolean algebra "+" means OR, not addition.

بولین الجبرا جارج بول (George Boole) نے بنایا۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. Is the number of pages in a book discrete or continuous?
2. What does a 1 represent in a digital system?
3. What shape of wave does a digital signal make?
4. Who developed Boolean algebra?

Answers: 1. Discrete (you count pages) 2. ON, HIGH or TRUE 3. A square wave 4. George Boole

TONIGHT'S 5-MARK

Write a 5-mark answer on **Analog and Digital Signals** using Define · Explain · Example · Diagram · Working.
Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 1: Computer Systems

Computer Systems (2/5) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Boolean Operations · Truth Tables · Logic Gates (Basic Gates: AND, OR, NOT) · Universal Logic Gates (NAND and NOR)

★ **Must lock:** Logic Gates (Basic Gates: AND, OR, NOT)

If behind, skip non-Golden. Phone in another room.

1.1.5 Boolean Operations

There are three basic Boolean operations. Every other operation can be made from them.

- Every other operation can be made from them.
- AND operation ($\cdot$) — gives output 1 only when ALL inputs are 1.
- Expression: $Y = A \cdot B$ (often written just AB).

Operation	Real-life analogy
AND	A door with two locks. The door opens only when both locks are unlocked.
OR	A room with two doors. You can enter if either door is open.
NOT	A room light switch. When the switch is ON the room is lit; when it is OFF the room is dark — the state is the opposite of "dark".

Remember: AND is "strict" — everyone must agree (all 1s). OR is "easy-going" — one 1 is enough. NOT is "the rebel" — it always says the opposite.

AND آپریشن تب ہی 1 دیتا ہے جب تمام ان پٹ 1 ہوں (دو تالوں والا دروازہ)۔

1.1.6 Truth Tables

A truth table shows all possible input combinations and the output for each one, for a logic gate or a circuit.

- Count the number of inputs (n).
- Calculate total rows = 2^n .
- List all input combinations in binary counting order (0, 1, 2, 3 ... in binary).
- Calculate the output for each combination.

A	B	Combination
0	0	Row 1
0	1	Row 2
1	0	Row 3
1	1	Row 4

Remember: Quick fill trick for 3 inputs: In column C write 0,1,0,1... (change every row). In column B write 0,0,1,1... (change every 2 rows). In column A write...

ٹروٹھ ٹیبل (Truth Table) کسی گیٹ یا سرکٹ کے تمام ممکنہ ان پٹ اور ان کے آؤٹ پٹ دکھاتا ہے۔

1.1.7 Logic Gates (Basic Gates: AND, OR, NOT) ★ Golden

Logic gates are electronic circuits that perform Boolean operations.

- They take one or more input bits and produce one output bit.
- They are the building blocks of all digital systems — a processor contains billions of them.
- The three basic gates match the three basic operations: the AND gate, the OR gate and the NOT gate (also called an inverter).

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Remember: Shortcut: AND output is 0 if any input is 0. OR output is 1 if any input is 1. Learn just the "special row": AND is 1 only at 11; OR is 0 only at 00.

Exam: You are often asked to "draw the symbol, write the expression and truth table" of a gate (2–3 marks each). Draw neat symbols with labelled inputs A, B and output Y....

لاجک گیٹس الیکٹرانک سرکٹ ہیں جو بولین آپریشن انجام دیتے ہیں اور تمام ڈیجیٹل نظاموں کی بنیادی اینٹیں ہیں۔

1.1.7 Universal Logic Gates (NAND and NOR)

NAND means "NOT AND": it is an AND gate followed by a NOT.

- NOR means "NOT OR": an OR gate followed by a NOT.
- On the symbol, the NOT is shown as a small bubble at the output.
- NAND and NOR are called universal gates because any logic circuit in the world can be built using only NAND gates (or only NOR gates).

A	B	A·B	Y
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

Remember: NAND is the exact opposite of AND (0 only at 11). NOR is the exact opposite of OR (1 only at 00). Just flip the AND/OR output column.

NAND کا مطلب ہے NOT AND اور NOR کا مطلب ہے NOT OR.

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. What symbol is used for the OR operation?
2. How many rows does a truth table with 4 inputs have?
3. What is the output of an OR gate when A = 0 and B = 0?
4. Why are NAND and NOR called universal gates?

Answers: 1. "+" 2. $2^4 = 16$ 3. 0 4. Any logic circuit can be built using only NAND gates or only NOR gates

TONIGHT'S 5-MARK

Write a 5-mark answer on **Logic Gates (Basic Gates: AND, OR, NOT)** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 1: Computer Systems

Computer Systems (3/5) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Advanced Logic Gates (XOR and XNOR) · Logic Expressions, Minterms and Maxterms · Logic Diagrams · Karnaugh Maps (K-Maps)

★ **Must lock:** Logic Diagrams, Karnaugh Maps (K-Maps)

If behind, skip non-Golden. Phone in another room.

1.1.7 Advanced Logic Gates (XOR and XNOR)

XOR (Exclusive OR) gives output 1 when the inputs are different.

- In terms of basic operations, $A \oplus B = A'B + AB'$.
- XNOR (Exclusive NOR) is XOR followed by NOT.
- It gives output 1 when the inputs are the same.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Remember: Memory trick: XOR outputs 1 when inputs are "eXclusively" different. XNOR outputs 1 when inputs are the same. XNOR is sometimes called the "equality gate".

XOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ مختلف ہوں، اور XNOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ ایک جیسے ہوں۔

1.1.8 – 1.1.10 Logic Expressions, Minterms and Maxterms

A logic expression (Boolean expression) combines variables with operations, for example $Y = A + B'C$.

1. Parentheses () → 2. NOT (') → 3. AND (·) → 4. OR (+)

- To evaluate it correctly you must follow the order of precedence, just like BODMAS in maths: 1.

A	B	C	B'	B'·C	Y = A + B'·C
0	0	0	1	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	1	0	0	1

Remember: SOP = 1s, POS = 0s. Minterm: 0 → bar, 1 → plain. Maxterm: the reverse (1 → bar, 0 → plain). Think "Min likes 1s, Max likes 0s".

ایکسپریشن حل کرتے وقت ترتیب یہ ہے: پہلے قوسین، پھر NOT، پھر AND اور آخر میں OR.

1.1.11 Logic Diagrams ★ Golden

A logic diagram shows how logic gates are connected to create a circuit.

- It is a "picture" of a Boolean expression.
- To draw one, use the order of precedence: the operation with the highest priority is drawn first (nearest the inputs), and the last...

Remember: Read the expression and ask "What is done last?" That gate goes on the far right, next to Y. Work backwards to the inputs. Count gates: $Y = A \cdot B + C$ needs 2...

Exam: Typical question: "Draw the logic diagram of $Y = \dots$ " and sometimes "also make its truth table" (4–5 marks). Write the precedence steps in one line, draw neat gate symbols...

لاچک ڈایاگرام دکھاتا ہے کہ گیٹس آپس میں کیسے جڑ کر سرکٹ بناتے ہیں۔

1.1.13 Karnaugh Maps (K-Maps) ★ Golden

A Karnaugh map (K-map) is a graphical tool used to simplify Boolean expressions.

- It is a grid of cells — one cell for each row of the truth table.
- The cells are arranged so that neighbouring (adjacent) cells differ by only ONE bit.
- This ordering is called Gray code.

	B = 0	B = 1
A = 0	Cell 0 (A'B')	Cell 1 (A'B)
A = 1	Cell 2 (AB')	Cell 3 (AB)

Remember: Gray code song: "zero-zero, zero-one, one-one, one-zero" — only one bit flips at each step. Imagine the K-map is drawn on a tube, so the left and right edges...

Exam: K-maps are a favourite long question (5–6 marks): "Simplify $F(A,B,C) = \sum m(\dots)$ using a K-map." Draw the map with labelled Gray-code headings, fill every cell (write the 0s...

کارنو میپ (K-Map) بولین ایکسپریشن کو آسان بنانے کا تصویری طریقہ ہے۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. What is $1 \oplus 1$?
2. Which comes first: AND or OR?
3. In $Y = A + B \cdot C$, which gate is drawn first?
4. In a 3-variable K-map, which column is next to $BC = 01$ on the right?

Answers: 1. 0 2. AND 3. The AND gate (for $B \cdot C$) 4. $BC = 11$

TONIGHT'S 5-MARK

Write a 5-mark answer on **Logic Diagrams** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 1: Computer Systems

Computer Systems (4/5) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: LogiSim Evolution: Simulation Tool · Building a Circuit in LogiSim · Software Development Life Cycle (SDLC) · The Waterfall Model

★ **Must lock:** Software Development Life Cycle (SDLC), The Waterfall Model

If behind, skip non-Golden. Phone in another room.

LogiSim Evolution: Simulation Tool

LogiSim Evolution is a free graphical tool for designing and simulating logic circuits.

- "Simulating" means the computer pretends to be the real circuit: you can switch inputs on and off and watch the output change.
- It lets you see the truth tables and Boolean expressions you learn in theory.

Area	What it does
Menu Bar (top)	File, Edit, Project, Simulate menus — open/save files and control the simulation.
Toolbar	Quick access to standard tools: Wiring, Poke, and AND, OR, NOT gates.
Explorer Panel (left)	Contains the component libraries: Gates, I/O, Wiring.
Attribute Table (bottom left)	Adjust the properties of the selected gate, e.g. Number of Inputs or Facing direction.
Drawing Canvas (right)	The dotted grid where you place and wire the gates.

Remember: Colours like a traffic light: Green means "working" (bright = 1, dark = 0). Blue means "the wire is lonely — not connected". Red means "stop, something is..."

لاجی سیم ایوولوشن (LogiSim Evolution) لاجک سرکٹ بنانے اور چلا کر دیکھنے کا سافٹ ویئر ہے۔

Building a Circuit in LogiSim

Poke A	Poke B	Y shows	Output wire colour
0	0	0	Dark green
0	1	0	Dark green
1	0	0	Dark green
1	1	1	Bright green

Remember: Input pin = square (you push it). Output pin = circle (it shows a light). Poke = finger — you "poke" inputs to change them.

لاجی سیم میں $Y = A \cdot B$ بنانے کے لیے AND گیٹ رکھیں، بائیں جانب دو ان پٹ پن (چوکور) اور دائیں جانب ایک آؤٹ پٹ پن (گول) رکھیں، پھر وائر...

1.2.1 – 1.2.3 Software Development Life Cycle (SDLC) ★ Golden

The Software Development Life Cycle (SDLC) is a structured process that guides software development from beginning to end.

1. Requirement Analysis → 2. System Design → 3. Implementation → 4. Testing → 5. Deployment → 6. Maintenance

- It ensures clear goals, better planning, risk reduction and high quality.
- Without it, teams forget features, go over budget and deliver buggy software.
- SDLC has 6 phases: 1.

Phase	Activities and details	Deliverable (what it produces)
1. Requirement Analysis	Meet clients, understand their needs, list the features. This is the most important phase — mistakes here spread to every later phase.	SRS Document (Software Requirements Specification)
2. System Design	Create the High-Level Design (overall architecture), the Low-Level Design (database, UI screens) and flowcharts.	Design Document
3. Implementation	Write the actual code in programming languages (Python, Java, etc.) based on the design documents.	Working Source Code

Remember: Remember the order with "Real Developers Implement, Test, Deploy, Maintain" (R-D-I-T-D-M). Black box = a closed box (you can't see inside). White box = a glass...

Exam: "Explain the phases of SDLC" is a very common long question (5–8 marks). Name all 6 phases in order, write 1–2 lines of activities for each, and state the deliverable of...

ایس ڈی ایل سی (SDLC) سافٹ ویئر بنانے کا منظم طریقہ ہے جس کے چھ مراحل ہیں: ضروریات کا تجزیہ، سسٹم ڈیزائن، کوڈنگ (Implementation)۔۔۔

1.2.4 The Waterfall Model ★ Golden

The Waterfall model is a linear, sequential approach to the SDLC.

- Each phase must be completed before the next one begins, like water falling down steps.
- There is no easy way to go back to a previous phase.

Advantages (Pros)	Disadvantages (Cons)
Simple to understand and manage.	Not flexible to changes.
Well documented at every step.	Testing happens very late in the process.

Remember: Waterfall = water only flows downwards. Once water has fallen it cannot climb back up — once a phase is finished you do not go back.

Exam: Expect "Explain the Waterfall model with its advantages and disadvantages" or "When should the Waterfall model be used?" Draw the step diagram, list the three "when to..."

واترفال (آبشار) ماڈل ایک سیدھا اور ترتیب وار طریقہ ہے؛ ہر مرحلہ مکمل ہونے کے بعد ہی اگلا شروع ہوتا ہے اور پیچھے جانا مشکل ہوتا ہے۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. Where do you change the number of inputs of a gate?
2. Which icon is the input pin: square or circle?
3. Which phase produces the SRS document?
4. Why is the Waterfall model called "sequential"?

Answers: 1. In the Attribute Table (bottom left) 2. Square (output pin is the circle) 3. Requirement Analysis 4. Because each phase must be completed before the next begins, in a fixed order

TONIGHT'S 5-MARK

Write a 5-mark answer on **Software Development Life Cycle (SDLC)** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 1: Computer Systems

Computer Systems (5/5) · Checkpoint · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–90 min remaining cards · 90–130 min closed-book chapter checkpoint · 130–180 min mark and rewrite missed Golden.

Cover: The Agile Model · Waterfall vs Agile: Comparison and Case Studies · Communication Models and the OSI 7-Layer Model · TCP/IP Model vs OSI Model

★ **Must lock:** The Agile Model, Communication Models and the OSI 7-Layer Model

If behind, skip non-Golden. Phone in another room.

1.2.4 The Agile Model ★ Golden

The Agile model is an iterative approach: the software is built in small, repeated cycles called sprints (usually 2–4 weeks long).

Plan → Design → Build → Test → Review & feedback → Next sprint ↻

- Each sprint plans, builds and tests a small part of the software, and then shows it to the customer.
- Customer feedback is gathered continuously and used in the next sprint.
- Plan→Design→Build→Test→Review & feedback→Next sprint ↻

Advantages (Pros)	Disadvantages (Cons)
Highly flexible to changes.	Less documentation.
Early delivery of working software.	Unpredictable end-time and cost.

Remember: Agile = agile athlete: quick and able to change direction. Think of making a pizza where the customer tastes each slice and you adjust the next one.

Exam: Questions ask you to define Agile, explain sprints, list pros/cons, or say when to use it. Always mention the words "iterative", "sprints (2–4 weeks)" and "continuous..."

ایجائل (Agile) ماڈل میں سافٹ ویئر چھوٹے چھوٹے ادوار میں بنتا ہے جنہیں اسپرینٹ (Sprint) کہتے ہیں، جو عموماً دو سے چار ہفتے کے ہوتے ہیں۔

1.2.5 – 1.2.7 Waterfall vs Agile: Comparison and Case Studies

Both models follow the same SDLC phases. The difference is how they move through them: Waterfall goes through each phase once, in a straight line; Agile goes through all phases many times, in small loops.

- The difference is how they move through them: Waterfall goes through each phase once, in a straight line; Agile goes through all phases...

Dimension	Waterfall model	Agile model
Flexibility	Rigid. Hard to make changes later.	Very flexible. Adapts to changes easily in the next sprint.
Customer involvement	Only at the start (requirements) and the end (delivery).	Continuous throughout the project (after every sprint).
Delivery of software	Single delivery at the very end.	Frequent, partial deliveries after every sprint.
Testing	Done late, in a dedicated testing phase.	Continuous testing within every sprint.
Documentation	Heavy documentation required.	Light documentation. Focus is on working software.

Remember: One question decides it: "Will the requirements change?" No → Waterfall. Yes / not sure → Agile.

وائرفال ماڈل سخت (Rigid) ہے، گاہک صرف شروع اور آخر میں شامل ہوتا ہے، سافٹ ویئر ایک ہی بار آخر میں ملتا ہے اور دستاویزات زیادہ ہوتی

ہیں۔

1.3.1 – 1.3.4 Communication Models and the OSI 7-Layer Model ★ Golden

A communication model explains how data travels from one device to another through a network.

- Because sending data is complicated, the job is split into layers; each layer does one part of the job and passes the data to the next...
- The OSI (Open Systems Interconnection) model was developed by ISO (International Organization for Standardization) in 1984.
- It divides communication into 7 layers.

Layer	Function	Protocols / examples
7. Application	Provides network services directly to user applications (browser, email program).	HTTP, HTTPS, FTP, SMTP, DNS
6. Presentation	Data translation, encryption (security) and compression.	SSL/TLS, JPEG, MP3, ASCII
5. Session	Opens, maintains and closes communication sessions.	NetBIOS, RPC
4. Transport	Reliable end-to-end delivery, error checking, port numbering.	TCP (reliable), UDP (fast)

Remember: Mnemonic (bottom to top, 1 → 7): "Please Do Not Throw Sausage Pizza Away" = Physical, Data Link, Network, Transport, Session, Presentation, Application. (Top...)

Exam: A guaranteed long question: "Explain the seven layers of the OSI model." Draw the 7-layer stack (numbered, layer 7 on top), then write one function and one or two...

کمیونیکیشن ماڈل بتاتا ہے کہ ڈیٹا نیٹ ورک کے ذریعے ایک ڈیوائس سے دوسری تک کیسے پہنچتا ہے۔

1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

The OSI model is mainly a reference model used for teaching and design.

- The TCP/IP model is the practical model used on the real Internet.
- It condenses the 7 OSI layers into just 4 layers.

TCP/IP model (4 layers)	Corresponds to OSI layers
1. Application	Application, Presentation, Session (7, 6, 5)
2. Transport	Transport (4)
3. Internet	Network (3)
4. Network Access	Data Link, Physical (2, 1)

Remember: Count it as 3-1-1-2: the TCP/IP Application layer takes 3 OSI layers, Transport takes 1, Internet takes 1, Network Access takes 2. Total 3+1+1+2 = 7.

ٹی سی پی / آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. What is a sprint and how long does it usually last?
2. In which model is the customer involved only at the start and the end?
3. Which OSI layer is responsible for encryption and compression?
4. How many layers does the TCP/IP model have?

Answers: 1. A short development cycle that delivers a working part of the software; usually 2–4 weeks 2. Waterfall 3. Presentation layer (6) 4. Four

TONIGHT'S 5-MARK

Write a 5-mark answer on **The Agile Model** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

CHECKPOINT TEST

40 minutes, closed book. Chapter 1: Computer Systems. 8 MCQ + 3 short. Mark below 60% means tomorrow morning is Golden repair, not a new chapter.

Multiple choice

1. Which of the following is a continuous quantity?
 - a. Number of students in a class
 - b. Number of cars in a parking lot
 - c. Temperature of water
 - d. Number of books on a shelf
2. In a digital system, the value 1 represents:
 - a. OFF
 - b. LOW
 - c. FALSE
 - d. ON / HIGH / TRUE
3. A square wave represents a:
 - a. Digital signal
 - b. Analog signal
 - c. Sine wave
 - d. Noise signal
4. Boolean algebra was developed by:
 - a. Charles Babbage
 - b. George Boole
 - c. Alan Turing
 - d. Maurice Karnaugh
5. How many rows does a truth table with 4 inputs have?
 - a. 4
 - b. 8
 - c. 16
 - d. 32
6. Which gate gives output 1 only when all inputs are 1?
 - a. OR
 - b. AND
 - c. NOR
 - d. XOR
7. Which pair of gates are called universal gates?
 - a. AND and OR
 - b. XOR and XNOR
 - c. NOT and AND
 - d. NAND and NOR
8. What is the output of a NOR gate when $A = 0$ and $B = 0$?
 - a. 1
 - b. 0
 - c. Unknown
 - d. Depends on the gate size

Short questions

1. Define discrete and continuous quantities. Give two examples of each.
2. State four advantages of digital systems.
3. Differentiate between analog and digital signals (any three points).

Checkpoint answers — mark after the 40 minutes.

MCQ: 1. c 2. d 3. a 4. b 5. c 6. b 7. d 8. a

1. Discrete quantities are separate, countable items with specific values (number of students in a classroom, number of cars in a parking lot). Continuous quantities can take any value within a range and flow without breaks (temperature of water, speed of a moving car).
2. Reliable (less affected by noise and interference); accurate (no loss of quality when copying); easy to design (only two states, 0 and 1); programmable (easily controlled by software).
3. Nature: analog is a continuous flow; digital has discrete values (bits, HIGH/LOW). Shape: analog is a sine wave; digital is a square wave. Reliability: analog is low (affected by noise); digital is high (resistant to noise). Examples: human voice vs computer data.

Chapter 2: Computational Thinking & Algorithm

Computational Thinking & Algorithm (1/3) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Computational Thinking · Algorithm · Algorithm vs Pseudocode · Decomposition · Pattern Recognition

★ **Must lock:** Algorithm, Algorithm vs Pseudocode, Decomposition

If behind, skip non-Golden. Phone in another room.

2.1 Computational Thinking

Computational Thinking (CT) is a systematic approach to solving problems.

Problem → Decomposition → Pattern recognition → Abstraction → Algorithm → Program

- Decomposition — breaking a big problem down into smaller parts.
- Pattern recognition — identifying things that repeat or follow a rule.
- Abstraction — hiding (ignoring) details that do not matter.
- Algorithms — developing clear step-by-step instructions to solve the problem.

Remember: Remember the four parts of CT with the phrase "Don't Pick Any Apple": Decomposition, Pattern recognition, Abstraction, Algorithm.

کمپیوٹیشنل تھنکنگ (Computational Thinking) مسائل حل کرنے کا ایک منظم طریقہ ہے۔

2.2 Algorithm ★ Golden

An algorithm is a step-by-step logical procedure used to solve a problem.

- Input — the data it starts with.
- Clear steps — each step has only one meaning.
- A definite end — it must finish after a limited number of steps.
- Output — the result it produces.

Reason	What it means for you
Structured approach	Algorithms provide a structured approach to problem-solving, making complex tasks manageable.
Optimisation	They optimise time and memory usage, which is crucial for large-scale applications (for example, searching millions of records).
Reusability	They can be reused for similar problems, saving time and effort. A sorting algorithm for marks can also sort prices.
Accuracy and consistency	They ensure accuracy and consistency, reducing human error. The same steps always give the same result.
Blueprint	They act as a blueprint (a plan) before actual code implementation, just like an architect's map before building a house.

Remember: Remember the five reasons as S-O-R-A-B: Structured, Optimised, Reusable, Accurate, Blueprint.

Exam: Very common short question: "Define algorithm and write its importance." Write the definition word for word — a step-by-step logical procedure used to solve a problem;...

الگورتھم (Algorithm) کسی مسئلے کو حل کرنے کا مرحلہ وار منطقی طریقہ ہے۔

2.2.2 Algorithm vs Pseudocode ★ Golden

Both algorithms and pseudocode are used in problem-solving, but they serve different purposes in the design phase.

Problem → Algorithm (what) → Pseudocode (how) → Program (code) → Output

- The algorithm says what must be done.
- The pseudocode shows how it could be implemented, using programming-like keywords such as IF, FOR and WHILE, but it still cannot be run on...

Point	Algorithm	Pseudocode
Definition	A step-by-step logical procedure to solve a problem.	A human-readable representation of an algorithm using plain English & keywords.
Nature	Conceptual and language-independent.	Semi-formal and structured, but not executable code.
Purpose	To define what needs to be done logically.	To present how it can be implemented in a readable format.
Form	Usually described in plain steps, bullets or numbered points.	Written using structured constructs (IF, FOR, WHILE) like programming.
Focus	Focuses on the logic and steps of the problem.	Focuses on representing steps clearly before coding.
Audience	For anyone trying to understand the logic (including non-programmers).	Mainly for programmers or students preparing for coding.
Format rules	No fixed format; can be informal.	Follows standard pseudocode conventions for clarity.

Remember: Teacher's analogy: an algorithm is like a recipe ("Bake a cake at 350 degrees"). Pseudocode is like a structured kitchen manual using standard verbs (SET...

Exam: "Differentiate between algorithm and pseudocode" is a favourite 4–5 mark question. Draw a two-column table and give at least five points (definition, nature, purpose,...

الگورتھم بتاتا ہے کہ کیا کرنا ہے، جبکہ سوڈو کوڈ (Pseudocode) بتاتا ہے کہ اسے کیسے کیا جا سکتا ہے۔

2.3.1 Decomposition ★ Golden

Decomposition is a problem-solving technique in which a complex task or system is broken down into smaller, manageable parts.

Vocabulary → Grammar rules → Tenses → Sentence building

- By dividing a big problem into simpler steps, it becomes easier to understand, develop and solve.
- Each small part is called a sub-task (or sub-problem).
- You solve each sub-task separately, then join the solutions to solve the whole problem.

Sub-task	Description
Learn vocabulary	Learn the words for people, actions and things (e.g., "I", "eat", "apple").
Grammar rules	Learn the sentence structure: Subject → Verb → Object.
Learn tenses	Present, past, future (e.g., "I eat" vs. "I ate").
Sentence building	Combine words using the rules to create valid sentences: "I ate an apple."

Remember: Think of eating a big roti: you cannot eat it in one bite, so you break it into small pieces. Decomposition = "break the roti".

Exam: Expect "Define decomposition with a real-life example" or "Explain the benefits of decomposition". Give the definition, draw the small tree (Complex Problem → Sub-task...

ڈیکمپوزیشن (Decomposition) کا مطلب ہے کسی بڑے اور پیچیدہ مسئلے کو چھوٹے اور آسان حصوں میں تقسیم کرنا۔

2.3.2 Pattern Recognition

Pattern recognition is the process of identifying regularities, repetitions or rules in data.

- What is repeating?
- What changes in a predictable way?
- Can I generalise this (write one rule that works for every case)?

Row	Stars
1	1
2	2
3	3
...	...
N	N

```
print("*")
print("**")
print("***")
print("****")
print("*****")
```

Remember: Whenever you see "the same thing again and again, with a small change each time", think loop. Pattern recognition is the habit of spotting that.

پیٹرن ریکگنیشن (Pattern Recognition) کا مطلب ہے ڈیٹا میں دہرائی جانے والی چیزوں یا اصولوں کو پہچاننا۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. Name the four parts of computational thinking.
2. Why is an algorithm called "language-independent"?
3. Which one — algorithm or pseudocode — uses keywords like FOR and WHILE?
4. What is each smaller part of a decomposed problem called?
5. Which programming concept is based on pattern recognition?

Answers: 1. Decomposition, pattern recognition, abstraction, algorithms 2. Because it describes only the logic; it is not tied to Python, C++ or any other language 3. Pseudocode 4. A sub-task (sub-problem) 5. Loops

TONIGHT'S 5-MARK

Write a 5-mark answer on **Algorithm** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 2: Computational Thinking & Algorithm

Computational Thinking & Algorithm (2/3) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Abstraction · Sorting and Searching Algorithms · Sorting Algorithms: Bubble Sort · Sorting Algorithms: Selection Sort

★ **Must lock:** Sorting Algorithms: Bubble Sort, Sorting Algorithms: Selection Sort

If behind, skip non-Golden. Phone in another room.

2.3.3 Abstraction

Abstraction means hiding unnecessary features and focusing only on the essential ones needed to solve a problem.

Boil water → Add tea → Steep → Pour → Serve

- Necessary steps — the core logic that never changes.
- Changes based on the situation — these become variables (inputs) in the algorithm.
- Abstraction items (noise) — details you can ignore completely.

1. Necessary steps (core logic)	2. Changes based on situation (variables)	3. Abstraction items (ignore / noise)
1. Boil water 2. Add tea leaves/bag 3. Steep for a few minutes 4. Pour into cup 5. Serve	Ingredients (black, green, milk) Additions (sugar, lemon) Tools (stove vs kettle)	Brand of kettle Cup colour or design Where tea is made (kitchen, office)

Remember: Abstraction = "Keep the skeleton, drop the decoration." Ask: "If I remove this detail, does the solution still work?" If yes, it is noise.

ایبستریکشن (Abstraction) کا مطلب ہے غیر ضروری تفصیلات کو چھپا کر صرف ضروری باتوں پر توجہ دینا۔

2.4 Sorting and Searching Algorithms

Sorting and searching are both fundamental techniques of computer science.

temp = A (temp = 8) → A = B (A = 4) → B = temp (B = 8)

- They work on lists of the same data type (for example, a list of all integers or a list of all names).
- Sorting Searching Arranging a collection of items in an order.
- The order may be ascending (0–9, A–Z) or descending (9–0, Z–A).

Sorting	Searching
Arranging a collection of items in an order. The order may be ascending (0–9, A–Z) or descending (9–0, Z–A). Used to organise data.	Locating a specific element within a list to find its position or check its existence (is it there or not?).
Examples: Bubble Sort, Selection Sort	Examples: Linear Search, Binary Search
Real life: merit list arranged from highest to lowest marks.	Real life: finding your roll number in the merit list.

Trap: Position 1 in the textbook algorithm is index 0 in Python.

سارٹنگ (Sorting) کا مطلب ہے ڈیٹا کو صعودی (چھوٹے سے بڑے) یا نزولی (بڑے سے چھوٹے) ترتیب میں رکھنا۔

2.4.1 Sorting Algorithms: Bubble Sort ★ Golden

Bubble Sort is an elementary sorting algorithm that works by repeatedly comparing adjacent (side-by-side) elements and swapping them if they are in the wrong order.

Compare DATA[j] and DATA[j+1] → Left > Right? → Yes: swap → Move one step right → End of pass: largest is at the end → Repeat N-1 passes

- In each pass, the largest remaining value "bubbles up" to the end of the list, like an air bubble rising to the top of water.
- So after pass 1 the largest value is in the last place, after pass 2 the second largest is in second-last place, and so on.
- A list of N items needs N - 1 passes, and pass K makes N - K comparisons.

Comparison	Pair	Left > Right?	Action	List after
1st vs 2nd	[8, 4], 1, 9, 3	8 > 4 Yes	Swap	4, 8, 1, 9, 3
2nd vs 3rd	4, [8, 1], 9, 3	8 > 1 Yes	Swap	4, 1, 8, 9, 3
3rd vs 4th	4, 1, [8, 9], 3	8 > 9 No	No change	4, 1, 8, 9, 3
4th vs 5th	4, 1, 8, [9, 3]	9 > 3 Yes	Swap	4, 1, 8, 3, 9

Remember: Bubble = neighbours. Bubble Sort only ever compares neighbours (j and j+1). Each pass pushes the biggest bubble to the end. For descending order, just change >...

Exam: The most common question: "Sort the list ... using Bubble Sort. Show each pass." Draw one table per pass with columns Pair, Compare, Action, List after. Mark the element...

ببل سارٹ (Bubble Sort) میں ساتھ ساتھ موجود دو عناصر کا موازنہ کیا جاتا ہے اور اگر وہ غلط ترتیب میں ہوں تو ان کی جگہ بدل دی جاتی ہے۔

2.4.1 Sorting Algorithms: Selection Sort ★ Golden

Selection Sort is a simple sorting algorithm that works by repeatedly selecting the smallest element from the unsorted portion of the list and moving it to its correct position at the front.

Fix position K → Compare DATA[K] with DATA[PTR], PTR = K+1 ... N → DATA[K] > DATA[PTR]? swap → End of pass: smallest is at K → K = K + 1

- Think of it as filling seats from the front: first find who belongs in position 1, then position 2, and so on.
- In the textbook version, the element at position K is compared with every element after it; whenever a smaller one is found, the two are...
- So at the end of pass K, the K-th smallest value sits in position K.

Compare	List	Pos1 > PosX?	Action	List after
Pos1 vs Pos2	[8], [4], 1, 9, 3	8 > 4 Yes	Swap	4, 8, 1, 9, 3
Pos1 vs Pos3	[4], 8, [1], 9, 3	4 > 1 Yes	Swap	1, 8, 4, 9, 3
Pos1 vs Pos4	[1], 8, 4, [9], 3	1 > 9 No	No change	1, 8, 4, 9, 3
Pos1 vs Pos5	[1], 8, 4, 9, [3]	1 > 3 No	No change	1, 8, 4, 9, 3

Remember: Teacher's tip — spot the difference: in Selection Sort the inner loop starts at outer + 1 and compares with the fixed position DATA[outer]. In Bubble Sort the...

Exam: Typical questions: "Sort ... using Selection Sort, showing all steps", "Differentiate between Bubble Sort and Selection Sort" and "Why is Selection Sort not stable?" In...

سلیکشن سارٹ (Selection Sort) میں فہرست کے غیر ترتیب شدہ حصے سے سب سے چھوٹا عدد چن کر اسے سامنے اس کی درست جگہ پر رکھا جاتا ہے۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. What is "noise" in abstraction?
2. Arrange C, A, D, B in descending order.
3. How many passes and comparisons does Bubble Sort need for 6 elements?
4. In Selection Sort, where does the inner loop start?

Answers: 1. Unnecessary details that do not affect the solution 2. D, C, B, A 3. 5 passes; 5 + 4 + 3 + 2 + 1 = 15 comparisons 4. At outer + 1 (PTR = K + 1)

TONIGHT'S 5-MARK

Write a 5-mark answer on **Sorting Algorithms: Bubble Sort** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 2: Computational Thinking & Algorithm

Computational Thinking & Algorithm (3/3) · Checkpoint · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–90 min remaining cards · 90–130 min closed-book chapter checkpoint · 130–180 min mark and rewrite missed Golden.

Cover: Searching Algorithms: Linear Search · Searching Algorithms: Binary Search · Evaluation Criteria of Algorithm · How to Decide Which Algorithm to Use

★ **Must lock:** Searching Algorithms: Binary Search

If behind, skip non-Golden. Phone in another room.

2.4.2 Searching Algorithms: Linear Search

Linear Search, also known as sequential search, works by examining each element in the list one by one, from the start, until the target is found (or the list ends).

PTR = first → **DATA[PTR] = FindValue?** → **Yes: ValueFound = 1, exit** → **No: PTR = PTR + 1** → **List ended? print "Not Found"**

- It works on both sorted and unsorted data.
- A flag variable (here ValueFound) starts at 0 and becomes 1 when the item is found.
- After the loop, the flag tells us whether to print the item or "Not Found".

Step	Element checked	Condition (== 7)	Result
1	8	8 == 7 ?	Not found
2	3	3 == 7 ?	Not found
3	6	6 == 7 ?	Not found
4	1	1 == 7 ?	Not found
5	7	7 == 7 ?	Found! (Exit loop)

Remember: Linear Search is like looking for your friend in a queue by checking every face from the front. Best case: 1 check (first person). Worst case: N checks (last...)

لینیئر سرچ (Linear Search) کو ترتیب وار تلاش (Sequential Search) بھی کہتے ہیں۔

2.4.2 Searching Algorithms: Binary Search ★ Golden

Binary Search is a highly efficient search algorithm that works by repeatedly dividing the search interval in half.

BEG = first, END = last → **MID = INT((BEG+END)/2)** → **Item = DATA[MID]? Found** → **Item < DATA[MID]: END = MID-1** → **Else: BEG = MID+1** → **Repeat while BEG ≤ END**

- BEG (low) — first position of the current part.
- END (high) — last position of the current part.
- MID = INT((BEG + END) / 2) — the middle position. INT means "drop the decimal part".
- Equal → found, stop.

Step	BEG (low)	END (high)	MID	DATA[MID]	Compare	Action
1	1	9	INT(10/2) = 5	5	7 > 5	BEG = 6 (go right)
2	6	9	INT(15/2) = 7	7	7 = 7	Found, LOC = 7

Remember: Binary Search is how you find a word in a dictionary: open the middle, decide "before or after?", and throw away half the book each time. "Sorted first, then..."

Exam: You will usually be given a sorted list and an item and asked to "search using Binary Search, showing all steps". Always draw a table with BEG, END, MID, DATA[MID],...

بائنری سرچ (Binary Search) ایک تیز رفتار طریقہ ہے جو ہر قدم پر تلاش کے حصے کو آدھا کر دیتا ہے۔

2.5 Evaluation Criteria of Algorithm

Many algorithms can solve the same problem.

- To judge which is better, we evaluate them on three criteria: correctness, efficiency and clarity.

Input	Operation (divide by 2, find remainder)	Remainder == 0?	Result
3	3 / 2 (Quotient = 1, Remainder = 1)	No	Odd
6	6 / 2 (Quotient = 3, Remainder = 0)	Yes	Even
15	15 / 2 (Quotient = 7, Remainder = 1)	No	Odd

```
for number in [3, 6, 15]:
    remainder = number % 2
    if remainder == 0:
        print(number, "remainder", remainder, "→ Even")
    else:
        print(number, "remainder", remainder, "→ Odd")
```

Remember: Remember C-E-C: Correct (right answer, always finishes), Efficient (fast and light), Clear (easy to read).

کسی الگورتھم کو تین معیارات پر پرکھا جاتا ہے: درستگی (Correctness) یعنی درست نتیجہ دے، غیر معمولی صورتوں کو سنبھالے اور ہمیشہ ختم...

2.6 How to Decide Which Algorithm to Use

Before choosing an algorithm, ask three questions: What is the input size?

What is the task? → Need order? → Sorting algorithm → Need to find? Is data sorted? → No → Linear Search → Yes → Binary Search

- What is the input size? (10 items or 10 million?)
- Is the data sorted?
- Do we need speed or memory efficiency?
- If searching on unsorted data → use Linear Search.

Algorithm	Type	Best case	Worst case	Extra memory	Needs sorted data?	Stable?
Bubble Sort	Sorting	$O(N^2)$ as written; $O(N)$ with an early-stop check	$O(N^2)$	None — $O(1)$	—	Yes
Selection Sort	Sorting	$O(N^2)$	$O(N^2)$	None — $O(1)$	—	No
Linear Search	Searching	$O(1)$ (first item)	$O(N)$ (last item / absent)	None — $O(1)$	No	—
Binary Search	Searching	$O(1)$ (item at first MID)	$O(\log N)$	None — $O(1)$	Yes	—

Remember: Two keywords decide most exam answers: "sorted?" and "how big?". Unsorted or small → Linear. Sorted and big → Binary.

الگورتھم چننے سے پہلے دیکھیں کہ ڈیٹا کتنا بڑا ہے، ترتیب شدہ ہے یا نہیں، اور ہمیں رفتار چاہیے یا میموری کی بچت۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

- What is another name for Linear Search?
- What is the crucial condition for Binary Search to work?
- Name the three evaluation criteria of an algorithm.
- Which search would you use on an unsorted list of phone contacts?

Answers: 1. Sequential search 2. The list must be sorted 3. Correctness, efficiency, clarity 4. Linear Search

TONIGHT'S 5-MARK

Write a 5-mark answer on **Searching Algorithms: Binary Search** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

CHECKPOINT TEST

40 minutes, closed book. Chapter 2: Computational Thinking & Algorithm. 8 MCQ + 3 short. Mark below 60% means tomorrow morning is Golden repair, not a new chapter.

Multiple choice

- An algorithm is best defined as:
 - a programming language
 - a step-by-step logical procedure to solve a problem
 - a type of computer hardware
 - compiled machine code
- Pseudocode is:
 - executable machine code
 - a symbol used in flowcharts
 - a Python library
 - a human-readable representation of an algorithm using plain English and keywords
- Breaking a complex problem into smaller, manageable parts is called:
 - decomposition
 - abstraction
 - pattern recognition
 - compilation
- In the "making tea" example, ignoring the brand of the kettle is an example of:
 - decomposition
 - sorting
 - abstraction
 - searching
- Pattern recognition forms the foundation of:
 - variables
 - loops
 - comments
 - input devices
- Bubble Sort works by comparing:
 - the first and last elements
 - every element with the middle element
 - adjacent elements
 - random pairs of elements
- After the first pass of Bubble Sort (ascending) on 8, 4, 1, 9, 3, the list is:
 - 1, 3, 4, 8, 9
 - 4, 8, 1, 9, 3
 - 1, 8, 4, 9, 3
 - 4, 1, 8, 3, 9
- Binary Search works only on a list that is:
 - sorted
 - unsorted
 - stored in a linked list
 - of strings only

Short questions

- Define computational thinking and name its four components.
- Write any four points on the importance of algorithms.
- Give three differences between an algorithm and pseudocode.

Checkpoint answers — mark after the 40 minutes.

MCQ: 1. b 2. d 3. a 4. c 5. b 6. c 7. d 8. a

1. Computational thinking is a systematic approach to solving problems. Its four components are decomposition (breaking down), pattern recognition (finding repetitions), abstraction (hiding unnecessary details) and algorithms (writing step-by-step solutions).
2. Algorithms (1) provide a structured approach that makes complex tasks manageable; (2) optimise time and memory usage, crucial for large applications; (3) can be reused for similar problems; (4) ensure accuracy and consistency, reducing human error; (5) act as a blueprint before coding. (Any four.)
3. (1) An algorithm is a step-by-step logical procedure; pseudocode is a human-readable representation of it using English and keywords. (2) An algorithm is conceptual and language-independent; pseudocode is semi-formal and structured but not executable. (3) An algorithm has no fixed format and is for anyone; pseudocode uses IF/FOR/WHILE, follows conventions and is mainly for programmers.

Chapter 3: Programming Fundamentals

Programming Fundamentals (1/5) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Programming · Programming Language Types · Career Growth with Python · Integrated Development Environment (IDE) · Visual Studio Code (VS Code)

★ **Must lock:** none — still copy every table

If behind, skip non-Golden. Phone in another room.

3.1 Programming

Computers are "dumb" machines. They are very fast, but they cannot think or do anything on their own.

- A set of instructions given to a computer is called a program.
- The language used to write these instructions is called a programming language.
- Writing programs is called programming, and the person who writes them is a programmer.

```
print("Hello")
```

Remember: Program = recipe, programmer = recipe writer, computer = cook who follows every line exactly. If the recipe is wrong, the food is wrong. The cook does not fix...

کمپیوٹر خود سے کچھ نہیں سوچ سکتا۔

3.1.1 Programming Language Types

Programming languages are classified into three main types, depending on how close they are to the computer's own language (binary 0s and 1s) or to human language.

Type	Description	Examples
Low level	Closest to the native language of the computer (binary 0s and 1s). Very difficult for humans to read and write.	Machine language (binary)
Mid level	Bridges the gap. It can access hardware but is easier to read than binary because it uses short words called mnemonics (e.g. MOV , ADD).	Assembly language
High level	Designed to be easy for humans to read and write. Uses English-like words.	C++, Java, Python

Remember: "Low = Like the machine, High = Human." Mid level sits in the Middle and uses Mnemonics (both start with M).

پروگرامنگ زبانوں کی تین اقسام ہیں: لو لیول (Low Level) جو بائری 0 اور 1 میں ہوتی ہے، مڈ لیول (Mid Level) یعنی اسمبلی جو مختصر الف...

3.2 Career Growth with Python

Python is one of the most in-demand languages in the world.

- Software Development: build desktop and web applications.
- Data Science and Data Analysis: analyse data, create reports and make predictions.
- Artificial Intelligence and Machine Learning: develop intelligent systems, chatbots and recommendation algorithms (like "videos you may...")
- Web Development: use frameworks such as Django and Flask to build websites.

```
# Python
print("Hello")
```

Remember: Six careers — remember "S-D-A-W-A-C": Software, Data, AI, Web, Automation, Cybersecurity.

پائتھن سیکھ کر آپ سافٹ ویئر ڈیولپمنٹ، ڈیٹا سائنس، مصنوعی ذہانت (AI)، ویب ڈیولپمنٹ، آٹومیشن اور سائبر سیکیورٹی جیسے شعبوں میں کیریئر بنا سکتے

ہیں۔

3.3 Integrated Development Environment (IDE)

An IDE (Integrated Development Environment) is a software tool that helps programmers write, run and manage code in one place.

- Code Editor: where you write and edit code.
- Syntax Highlighting: colours different parts of the code (keywords, strings, numbers) to make it easier to read.
- Auto-completion: suggests code while you type (type `print` and it offers `print()`).
- Run/Execute Tool: runs the program with one click.

Remember: IDE features — "E-S-A-R-D": Editor, Syntax highlighting, Auto-completion, Run, Debugger.

آئی ڈی ای (IDE) ایک ایسا سافٹ ویئر ہے جس میں کوڈ لکھنا، چلانا اور غلطیاں درست کرنا ایک ہی جگہ ممکن ہوتا ہے۔

3.3.1 Visual Studio Code (VS Code)

VS Code is a free, lightweight and powerful code editor made by Microsoft and widely used for Python.

Install Python → Install VS Code → Extensions: install "Python" (Microsoft) → Explorer: new file `hello.py` → Type code → Click ► Run

- By default it is not a full IDE, but it becomes one when you install extensions (add-ons), such as the Python extension by Microsoft.

Element	Description
1. Top Menu Bar	Control over files, editing, navigation and workspace (File, Edit, Selection, View, Go, Run).
2. Side Bar	Contains the core tools: Explorer, Search, Source Control, Run & Debug and Extensions.
3. Explorer	Shows project files and folders. Helps you create, open and manage your Python files.
4. Search	Finds and replaces text in the open file or across the whole project.
5. Source Control	Connects with Git (a version control system) to track changes, undo mistakes and collaborate.
6. Run & Debug	Runs code, finds errors, pauses at specific lines and lets you inspect variables.
7. Extensions	Like an App Store for VS Code. Add features such as the Python extension, Prettier or Python Debugger.

```
print("Hello, World!")
```

Remember: Extensions are the "App Store" of VS Code: just as apps give your phone new abilities, extensions give VS Code new abilities.

وی ایس کوڈ (VS Code) ایک مفت اور ہلکا پھلکا کوڈ ایڈیٹر ہے۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. What is a program?
2. Which level is MOV AX, BX?
3. Name two Python web frameworks.
4. What does "integrated" mean in IDE?
5. Which VS Code element connects to Git?

Answers: 1. A set of instructions given to a computer to perform a task 2. Mid level (assembly language) 3. Django and Flask 4. Many programming tools (editor, runner, debugger) are combined in one program 5. Source Control

TONIGHT'S 5-MARK

Write a 5-mark answer on **Programming** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 3: Programming Fundamentals

Programming Fundamentals (2/5) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Fundamentals of Python Programming · Variables · Data Types in Python · Input and Output Handling · Operators and Operands: Arithmetic and Assignment

★ **Must lock:** Fundamentals of Python Programming, Variables, Data Types in Python, Operators and Operands: Arithmetic and Assignment

If behind, skip non-Golden. Phone in another room.

3.4 Fundamentals of Python Programming ★ Golden

The basic structure of a Python program is simple and clear.

- A program takes data, processes it using statements, and finally shows output.
- Python runs the statements top to bottom, one after another (sequentially).
- A statement is one complete instruction, usually written on one line, such as `print("Hi")` or `x = 5`.

```
# This is a comment. It explains the code below.
print("Hello, World!")
if 5 > 2:
    # The space before 'print' is the indentation
    print("Five is greater than two!")
```

Remember: Think of indentation like the points and sub-points in your notebook. Sub-points are written a little to the right to show they "belong" to the main point. A...

Exam: Common questions: "What is the purpose of comments?", "Why is indentation important in Python?", "How is Python different from C++/Java in defining blocks?" Write:...

پائتھن پروگرام اوپر سے نیچے ترتیب سے چلتا ہے۔

3.4.1 Variables ★ Golden

A variable is a named space in computer memory used to store data.

- Can contain letters, numbers and underscores (_) only — no spaces or symbols like -, @, \$.
- Must start with a letter or underscore, NOT a number (1name is invalid).
- Are case-sensitive: age and Age are two completely different variables.
- Cannot be a Python keyword (a reserved word such as if, for, while, else, True).

```
name = "Ali"
age = 16
my_height = 5.7
score1 = 100
```

Remember: Choose meaningful names: total_marks is far clearer than tm or x. Use lowercase words joined by underscores (this style is called snake_case).

Exam: A favourite question: "Which of these variable names are valid? Give reasons." Check each name against all four rules and state the rule broken, e.g. "2nd_place —...

ویری ایبل (Variable) میموری میں ایک نام والی جگہ ہے جس میں ڈیٹا رکھا جاتا ہے، بالکل ایک لیبل لگے ڈیے کی طرح۔

3.4.1 Data Types in Python ★ Golden

A data type specifies the kind of data a variable holds — a whole number, a decimal, text, True/False, or a collection of items.

- The type decides what you can do with the data (you can divide numbers, but not names).
- Python has several built-in data types:

Name	Type	Description	Example
Integer	<code>int</code>	Whole numbers without decimals (positive, negative or zero).	<code>age = 16</code>
Floating point	<code>float</code>	Numbers with a decimal point.	<code>height = 5.7</code>
String	<code>str</code>	Text data, enclosed in quotes (" " or ' ').	<code>name = "Sammy" or year = "2026"</code>
Boolean	<code>bool</code>	Logical value: True or False (capital T / F).	<code>is_student = True</code>
List	<code>list</code>	Ordered, changeable collection in square brackets [].	<code>marks = [85, 90, 75]</code>
Tuple	<code>tuple</code>	Ordered, unchangeable sequence in parentheses ().	<code>coords = (10, 20)</code>
Dictionary	<code>dict</code>	Key : value pairs in curly braces { }.	<code>student = {"name": "Ahmed", "age": 16}</code>
Set	<code>set</code>	Unordered collection of unique items (no duplicates).	<code>vowels = {'a', 'e', 'i'}</code>

Remember: Brackets decide the collection: [] List (changeable), () Tuple (locked), { } with key:value = Dictionary, { } with single items = Set (unique). Think: a list...

Exam: Expect: "Name any four data types with examples" or "Differentiate list and tuple". For a difference question, use a two-column table: list — [], changeable, e.g. marks...

ڈیٹا ٹائپ (Data Type) بتاتی ہے کہ ویری ایبل میں کس قسم کا ڈیٹا ہے۔

3.4.2 Input and Output Handling

Output — the `print()` function. A function is a named, ready-made piece of code that does a job when you call it with brackets.

- Items separated by commas are printed with one space between them.
- An f-string (a string with f before the quote) lets you put variables directly inside { }. It is the fastest and most readable way to...

Remember: "input() always gives a string." If you plan to do maths, wrap it: `int(input(...))` for whole numbers, `float(input(...))` for decimals. Read it from the inside...

`print()` اسکرین پر معلومات دکھاتا ہے اور `input()` صارف سے معلومات لیتا ہے۔

3.4.3 Operators and Operands: Arithmetic and Assignment ★ Golden

An operand is the value or variable being worked on.

- An operator is the symbol that performs the action.
- Together they form an expression, which Python evaluates to a single value.
- In `10 + 20`: 10 is an operand, + is the operator, 20 is an operand, and the result is 30.

Operator	Purpose	Example	Result
+	Addition	5 + 3	8
-	Subtraction	10 - 4	6
*	Multiplication	4 * 2	8
/	Division (result is always a float)	9 / 2	4.5
//	Floor division (drops the decimal part)	9 // 2	4
%	Modulus (remainder)	9 % 2	1
**	Exponentiation (power)	2 ** 3	8

```
print(5 + 3)
print(10 - 4)
print(4 * 2)
print(9 / 2)
print(9 // 2)
print(9 % 2)
print(2 ** 3)
print(8 / 2)
```

Remember: Three kinds of division: / = exact answer (float), // = quotient only ("floor"), % = remainder only. A number is even when `n % 2 == 0`.

Exam: You will be asked to evaluate expressions such as `17 // 5`, `17 % 5`, `2 ** 4` or to trace compound assignment. Show each step and write the final value with its type (e.g....

آپریٹنڈ (Operand) وہ ویلیو ہے جس پر عمل ہوتا ہے اور آپریٹر (Operator) وہ علامت ہے جو عمل کرتی ہے۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. Predict the output:
2. Valid or invalid? `_total`, `roll-no`, `class`, `marks2026`, `2marks`
3. What is the data type of each? `3.0`, `"3"`, `3`, `False`, `[3]`
4. Predict the output:
5. Predict the output:

Answers: 1. B then D on separate lines (line 1 is a comment; `3 > 7` is false so C is skipped) 2. `_total` valid; `roll-no` invalid (hyphen); `class` invalid (keyword); `marks2026` valid; `2marks` invalid (starts with a digit) 3. float, str, int, bool, list 4. 3333 then 12 (a string times a number repeats the string) 5. 3 1 3.5 then 49

TONIGHT'S 5-MARK

Write a 5-mark answer on **Fundamentals of Python Programming** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 3: Programming Fundamentals

Programming Fundamentals (3/5) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Relational, Logical and Membership Operators · Bitwise Operators · Control Structures · Sequence · Selection Statements: if and if-else

★ **Must lock:** Relational, Logical and Membership Operators, Bitwise Operators, Control Structures, Selection Statements: if and if-else

If behind, skip non-Golden. Phone in another room.

3.4.3 Relational, Logical and Membership Operators ★ Golden

3. Relational (comparison) operators compare two values and return a Boolean (True or False).

- Relational (comparison) operators compare two values and return a Boolean (True or False).
- They are used in conditions.

Op	Description	Example (a = 10, b = 5)	Result
<code>=</code>	Equal to	<code>a == b</code>	False
<code>≠</code>	Not equal to	<code>a != b</code>	True
<code>></code>	Greater than	<code>a > b</code>	True
<code><</code>	Less than	<code>a < b</code>	False
<code>≥</code>	Greater than or equal to	<code>a ≥ 10</code>	True
<code>≤</code>	Less than or equal to	<code>a ≤ b</code>	False

```
a = 10
b = 5
print(a == b)
print(a != b)
print(a > b)
print(a < b)
print(a ≥ 10)
print(a ≤ b)
```

Remember: AND is strict, OR is generous. AND is like a teacher who needs both homework and attendance; OR is like a shop that accepts cash or card. Order of evaluation:...

Exam: Typical: "Evaluate `a > b` and `b > 3` for `a = 10`, `b = 5`" or "Write the truth table of and/or". Write each sub-condition's value first, then combine. Remember: `==` compares,...

ریلیشنل آپریٹرز (Relational Operators) دو ویلیوز کا موازنہ کر کے True یا False دیتے ہیں۔

3.4.3 Bitwise Operators ★ Golden

6. Bitwise operators treat numbers as binary digits (bits: 0s and 1s) and work on them bit by bit, column by column.

- AND (&) — gives 1 only if both bits are 1, otherwise 0.
- OR (|) — gives 1 if at least one bit is 1.
- XOR (^) — gives 1 if the bits are different, 0 if they are the same.
- NOT (~) — inverts all bits (0 → 1, 1 → 0). In Python the result is the number made negative, minus 1: `~x = -(x + 1)`. So `~10` gives `-11`.

Bit A	Bit B	A & B	A B	A ^ B	~A
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

```
a = 10
b = 6
print(bin(a), bin(b))
print(a & b)
print(a | b)
print(a ^ b)
print(~a)
```

Remember: Shortcuts: $x > n = x // 2^n$, $\sim x = -x - 1$. For AND/OR/XOR always write both numbers in binary, one under the other, with the same number of bits.

Exam: Exams ask you to "evaluate $12 \& 10$, $12 | 10$, $12 \wedge 10$, ~ 5 , $7 >> 1$ showing binary working". Marks are given for the binary conversion and column working, not just the...

بٹ وائز آپریٹرز (Bitwise Operators) اعداد کو بائیری میں بدل کر ہر بٹ پر الگ الگ کام کرتے ہیں۔

3.5 Control Structures ★ Golden

Control structures decide the order in which statements are executed in a program (the "flow of control").

- Sequence — do the steps one after another, top to bottom.
- Selection — choose a path depending on a condition (if, else).
- Repetition — repeat steps (loops: for, while).

Remember: Remember "S-S-R": Sequence, Selection, Repetition — "Step, Select, Repeat".

Exam: "Name and explain the three basic control structures" is a very common short question. Give a one-line definition and a two-line Python example for each.

کنٹرول اسٹرکچرز (Control Structures) طے کرتے ہیں کہ پروگرام کی ہدایات کس ترتیب سے چلیں گی۔

3.5.1 Sequence

Sequence means statements are executed one after another, strictly from top to bottom, in the exact order they are written.

- This is the default behaviour of Python.
- No statement is skipped or repeated.

```
print("Enter two numbers")
a = int(input("Enter first number: ")) # Step 1: Input
b = int(input("Enter second number: ")) # Step 2: Input
total = a + b # Step 3: Process
print("Sum =", total) # Step 4: Output
```

Remember: Sequence is like reading a page: line 1, then line 2, then line 3. Most programs follow the pattern Input → Process → Output.

سیکوئنس (Sequence) کا مطلب ہے کہ ہدایات بالکل اسی ترتیب سے ایک کے بعد ایک اوپر سے نیچے چلتی ہیں جس ترتیب سے لکھی گئی ہیں۔

3.5.2 Selection Statements: if and if-else ★ Golden

Selection changes the flow of a program based on a condition — an expression that is either True or False.

- It helps a program respond intelligently to different situations.
- The if statement executes a block of code only if the condition is True.
- If the condition is False, the block is completely skipped.

```
if condition:
    statement(s) # runs only when condition is True
```

Remember: Formatting rule: put a colon (:) after if and else, and indent the code inside the blocks. else never has a condition of its own.

Exam: Asked as: "Write a program that inputs a number and prints whether it is even or odd / positive or negative / pass or fail." Write the full program with `int(input())`,...

سلیکشن (Selection) میں پروگرام شرط کی بنیاد پر فیصلہ کرتا ہے۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. Predict the output:
2. Predict the output (show the binary working):
3. Name the three basic control structures.
4. Predict the output:
5. Predict the output when marks = 32:

Answers: 1. True and True. (True and False = False; False or True = True. not True = False; $2 == 2$. 8 14 6 and -6 3 24. $12 = 1100$, $10 = 1010$: AND = $1000 = 8$, OR = $1110 = 14$, XOR = $0110 = 6$. $\sim 5 = -(5 + 1) = -6$. $7 = 111 \rightarrow \dots$
3. Sequence, selection, repetition 4. 10 8 — y was calculated when x was still 4; changing x later does not change y 5. Fail then Done

TONIGHT'S 5-MARK

Write a 5-mark answer on **Relational, Logical and Membership Operators** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 3: Programming Fundamentals

Programming Fundamentals (4/5) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Selection Statements: if-elif-else and Nested if · Repetition Statements: the for Loop and range() · The while Loop, break and continue · Nested Loops and for vs while · Practical Activity: Shopping Bill Calculator

★ **Must lock:** Selection Statements: if-elif-else and Nested if, Repetition Statements: the for Loop and range(), The while Loop, break and continue, Nested Loops and for vs while

If behind, skip non-Golden. Phone in another room.

3.5.2 Selection Statements: if-elif-else and Nested if ★ Golden

3. The if-elif-else statement handles many conditions.

- The if-elif-else statement handles many conditions.
- (elif is short for "else if".) Python checks the conditions from top to bottom.
- As soon as it finds a True condition, it runs that block and ignores all the rest.
- If none is True, the else block runs.

marks	>= 80?	>= 70?	>= 60?	Output
92	True	(not checked)	(not checked)	A-1 Grade
75	False	True	(not checked)	A Grade
60	False	False	True	B Grade
41	False	False	False	Fail or below B

Remember: In an if-elif-else ladder, put the strictest condition first (highest marks first). Think of a ladder: you step off at the first rung that fits.

Exam: Grade-calculator programs (if-elif-else) and "largest of three numbers" are very common long questions. Check conditions in the right order, indent nested blocks...

if-elif-else میں کئی شرطیں اوپر سے نیچے ترتیب سے دیکھی جاتی ہیں، اور جو شرط سب سے پہلے درست ہو اسی کا بلاک چلتا ہے، باقی سب چھوڑ دی جاتی ہیں۔

3.5.3 Repetition Statements: the for Loop and range() ★ Golden

A repetition statement (loop) lets a block of code run again and again.

- This greatly reduces the need to write the same code many times.
- Each single run of the loop body is called an iteration.
- The for loop is used when the exact number of iterations is known.

Syntax	How it works	Example	Values produced
range(stop)	From 0 up to, but not including , stop.	range(3)	0 1 2
range(start, stop)	From start up to, but not including, stop.	range(2, 5)	2 3 4
range(start, stop, step)	From start towards stop, jumping by step.	range(1, 6, 2)	1 3 5

```
for i in range(3):
    print(i)
```

Remember: range stops one short: range(1, 11) gives 1 to 10. To include a number n, write range(..., n + 1). Number of values in range(a, b) = b - a.

Exam: Expect "predict the output" questions on range() with a step, and programs such as "print a table", "sum of first n numbers", "even numbers from 1 to 20". Always state...

لوپ (Loop) کوڈ کے ایک بلاک کو بار بار چلاتا ہے۔

3.5.3 The while Loop, break and continue ★ Golden

2. The while loop repeats a block of code as long as a condition remains True.

- break — exits (stops) the loop entirely and immediately.
- continue — skips the rest of the current iteration and jumps to the next one.

count	count <= 3?	Printed	count after update
1	True	1	2
2	True	2	3
3	True	3	4
4	False	—	loop ends

```
count = 1
while count <= 3:
    print(count)
    count += 1 # VERY IMPORTANT: update the condition!
```

Remember: break = "exit the room"; continue = "skip this turn". For every while loop, check the I-T-U rule: Initialise, Test, Update.

Exam: "Differentiate break and continue with examples" and "What is an infinite loop?" are frequent. Use the two side-by-side programs above with their outputs — examiners...

while لوپ اس وقت تک چلتا رہتا ہے جب تک شرط درست (True) رہے، اور یہ تب استعمال ہوتا ہے جب پہلے سے معلوم نہ ہو کہ کتنی بار دہرانا ہے۔

3.5.3 Nested Loops and for vs while ★ Golden

A nested loop is a loop inside another loop.

- The inner loop runs completely, from start to finish, for every single iteration of the outer loop.
- If the outer loop runs 2 times and the inner loop runs 3 times, the inner body runs $2 \times 3 = 6$ times.

for loop	while loop
Used when the number of iterations is known .	Used when the number of iterations is unknown (condition-based).
Automatically updates the loop variable.	Requires a manual update of the counter (risk of an infinite loop).
Cleaner and shorter for counting tasks.	Better for dynamic situations (like "keep playing until the user says Stop").
for i in range(1, 4): print(i)	i = 1 / while i <= 3: / print(i) / i += 1

```
for outer in range(1, 3): # runs 2 times (1, 2)
    for inner in range(1, 4): # runs 3 times per outer (1, 2, 3)
        print(f"Outer: {outer}, Inner: {inner}")
```

Remember: Ask yourself: "Do I know how many times?" Yes → for. "Do I only know when to stop?" → while. Nested loops are like a clock: the minute hand (inner loop) goes...

Exam: "Differentiate between for and while loops" (give three points and an example of each) and "How many times will the inner statement run?" (multiply the counts) are...

نیسٹڈ لوپ (Nested Loop) میں ایک لوپ دوسرے لوپ کے اندر ہوتا ہے، اور باہر والے لوپ کے ہر چکر میں اندر والا لوپ شروع سے آخر تک مکمل چلتا ہے۔

Practical Activity: Shopping Bill Calculator

This program combines all three control structures: sequence (steps in order), selection (checking "done", checking the price, checking for a discount) and repetition (a while loop that keeps adding items).

- It uses a flag — a Boolean variable (shopping) that controls when the loop stops.
- item.lower() converts the text to lowercase so that "done", "Done" and "DONE" all work.

Step	item	price	total	count
1	Rice	3000.0	3000.0	1
2	Soap	-50.0 (invalid, not added)	3000.0	1
3	Oil	2500.0	5500.0	2
4	DONE → shopping = False	—	5500.0	2

Remember: Label the parts of any program with comments, as here: # REPETITION, # SELECTION. It helps you, your teacher and the examiner see the structure.

یہ شاپنگ بل پروگرام تینوں کنٹرول اسٹرکچرز کو ملا کر بنایا گیا ہے: while لوپ سے اشیاء بار بار شامل ہوتی ہیں، if-else سے غلط قیمت رو...

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. Predict the output:
2. Predict the output:
3. Predict the output:
4. Predict the output:
5. What will "Final to Pay" be if the user buys one item for 4000 and then types done?

Answers: 1. Warm ($30 > 35$ is False; $30 > 25$ is True, so the rest are skipped even though $30 > 15$ is also True) 2. 10 7 4 1 3. 27 (n: $1 \rightarrow 3 \rightarrow 9 \rightarrow 27$; $27 < 20$ is False, so the loop stops) 4. 6 (3×2) 5. Final to Pay: 4000.0 (4000 is less than 5000, so no discount)

TONIGHT'S 5-MARK

Write a 5-mark answer on **Selection Statements: if-elif-else and Nested if** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 3: Programming Fundamentals

Programming Fundamentals (5/5) · Checkpoint · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–90 min remaining cards · 90–130 min closed-book chapter checkpoint · 130–180 min mark and rewrite missed Golden.

Cover: Libraries in Python · Built-in Libraries · Third-Party Libraries · Debugging Code in Python

★ **Must lock:** none — still copy every table

If behind, skip non-Golden. Phone in another room.

3.6 Libraries in Python

A library (also called a module) is a collection of pre-written code that offers ready-made functions.

- Instead of writing a square-root algorithm from scratch, you import math and call `math.sqrt()`.
- Libraries save time, reduce bugs and let you stand on other programmers' work.
- There are five common ways to import a library.

Method	Syntax	How you call the function
1. Entire module	<code>import math</code>	<code>math.sqrt(25)</code>
2. Alias (nickname)	<code>import math as m</code>	<code>m.sqrt(16)</code>
3. Specific function	<code>from math import sqrt</code>	<code>sqrt(36)</code>
4. Several functions	<code>from math import sqrt, pi</code>	<code>sqrt(9)</code> , <code>pi</code>
5. Everything (*)	<code>from math import *</code>	<code>sqrt(64)</code> — avoid this

Trap: `from math import *` dumps every name into your program.

لائبریری پہلے سے لکھا ہوا کوڈ ہے جسے import سے پروگرام میں لایا جاتا ہے۔

3.6.1 Built-in Libraries

Built-in libraries come pre-installed with Python.

- You only import them; you do not run pip.
- Three that every Class XI student must know are `math`, `random` and `datetime`.

```
import math
print(math.pi)
print(math.ceil(4.3))
print(math.floor(4.9))
print(math.pow(2, 4))
print(math.sqrt(81))
```

Remember: `ceil` always rounds up (ceiling); `floor` always rounds down. `round()` (built-in, not in `math`) goes to the nearest integer and is a different function.

`math` حساب کے لیے، `random` بے ترتیب عدد کے لیے اور `datetime` تاریخ وقت کے لیے بنی ہوئی لائبریریاں ہیں۔

3.6.2 Third-Party Libraries

Third-party libraries are written by the community.

- They are not bundled with Python, so you install them from a terminal with `pip install library_name` and then import them as usual.

Library	Primary use
NumPy	Heavy numerical computing and large arrays
Pandas	Data analysis; table-like data (CSV / Excel)
Matplotlib	Data visualisation — graphs and charts
TensorFlow / PyTorch	Artificial intelligence and machine learning

```
pip install pandas
import pandas as pd
```

Trap: `pip` is a terminal command, not a Python statement.

تھرڈ پارٹی لائبریریاں پائتھن کے ساتھ نہیں آتیں، انہیں `pip install` سے انسٹال کیا جاتا ہے۔

3.7 Debugging Code in Python

Debugging is the process of identifying, tracing and fixing bugs so a program works correctly.

- Every program you write will have bugs.
- The skill is not "never make a mistake"; it is "find the mistake quickly".
- Python errors fall into three families:

Type	When it appears	What you notice	Example
Syntax error	Before the program even starts	Python refuses to run	<code>print("Hello" — missing)</code>
Runtime error (exception)	While the program is running	It crashes with a traceback	<code>x = 10 / 0 → ZeroDivisionError</code>
Logical error	Program finishes "successfully"	The output is wrong	<code>area = length + width</code> instead of <code>*</code>

```
x = int("abc")
```

Remember: Fix syntax errors first (the program cannot start). Then run it. Then check the output against a hand-worked example to catch logical errors.

Exam: Examiners love "identify the error type" questions. Ask: (1) Does it refuse to start? → syntax. (2) Does it crash mid-run? → runtime. (3) Does it run but give a wrong...

ڈیبنگ یعنی غلطی ڈھونڈ کر ٹھیک کرنا۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. Write the line that lets you call ceil without writing math. in front.
2. What does `math.ceil(4.0)` return?
3. Which library would you choose to draw a bar chart of class marks?
4. `print("Hi" — which error type?`

Answers: 1. `from math import ceil` 2. 4 (already an integer) 3. Matplotlib 4. Syntax error

TONIGHT'S 5-MARK

Write a 5-mark answer on **Libraries in Python** using Define · Explain · Example · Diagram · Working.
Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

CHECKPOINT TEST

40 minutes, closed book. Chapter 3: Programming Fundamentals. 8 MCQ + 3 short. Mark below 60% means tomorrow morning is Golden repair, not a new chapter.

Multiple choice

1. Python is best described as a:
 - a. low-level compiled language
 - b. high-level interpreted language
 - c. machine language
 - d. hardware description language
2. `input("Age: ")` always returns a:
 - a. int
 - b. float
 - c. str
 - d. bool
3. The value of `17 // 5` is:
 - a. 3.4
 - b. 2
 - c. 3
 - d. 3.0
4. The value of `17 % 5` is:
 - a. 2
 - b. 3
 - c. 3.4
 - d. 0
5. `2 ** 3` equals:
 - a. 6
 - b. 8
 - c. 9
 - d. 5
6. Which operator is bitwise AND?
 - a. `and`
 - b. `&&`
 - c. `&`
 - d. `^`
7. `list(range(2, 5))` is:
 - a. [2, 3, 4, 5]
 - b. [2, 3, 4]
 - c. [2, 5]
 - d. [0, 1, 2, 3, 4]
8. A `while` loop is preferred when:
 - a. you know the exact number of repeats
 - b. you must wait for a condition
 - c. you never need a counter
 - d. the body must run exactly once

Short questions

1. Differentiate between a compiler and an interpreter.
2. Why must the result of `input()` often be converted with `int()`?
3. State the difference between `/` and `//` with one example each.

Checkpoint answers — mark after the 40 minutes.

MCQ: 1. b 2. c 3. c 4. a 5. b 6. c 7. b 8. b

1. A compiler translates the whole program into machine code before it runs (C, C++). An interpreter translates and runs line by line (Python). Compiled programs often run faster; interpreted ones are easier to test.
2. `input()` always returns a string. Adding strings concatenates them ("`2`"+"`3`" is "`23`"), so numeric work needs `int()` or `float()`.
3. `/` is true division: `17/5` is `3.4`. `//` is integer (floor) division: `17//5` is `3`.

Chapter 4: Data and Analysis

Data and Analysis (1/3) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Data, Information and Database · Database Management System (DBMS) · Database Components (Tables, Records, Fields) · Keys and Integrity Constraints · Relational Database Management System (RDBMS)

★ **Must lock:** Keys and Integrity Constraints

If behind, skip non-Golden. Phone in another room.

4.1.1 Data, Information and Database

Data are unprocessed facts, figures or observations.

Data → Processing → Information → Decision

- They can be quantitative (numbers: 80, 75, 90) or qualitative (descriptions: the colours of cars in a parking lot).
- Data on its own is not very useful — a list of marks does not yet tell you whether the class is strong.
- When you organise, process and interpret data, it becomes information.

Aspect	Data	Information
Definition	Raw facts and figures	Processed, meaningful data
Example	85, 90, 75	Average marks = 83.3%
Usefulness	Little, on its own	Useful for decisions

Remember: Exam line: Data is raw; information is processed data used for decision-making. Always add one pair of examples.

ڈیٹا خام حقائق ہیں، معلومات وہی حقائق ہیں جب انہیں ترتیب دے کر معنی دے دیا جائے۔

4.1.2 Database Management System (DBMS)

A DBMS is software that lets you create, manage and interact with databases.

- Less redundancy. The department name is stored once, not in every student row.
- Consistency. Change a phone number once; every screen that uses it shows the new number.
- Security. Only authorised users see sensitive fields (an admin sees marks; office staff see names).
- Integrity. Rules stop duplicate roll numbers and empty primary keys.

ڈی بی ایم ایس وہ سافٹ ویئر ہے جو صارف اور ڈیٹا بیس کے درمیان پل کا کام کرتا ہے۔

4.2 Database Components (Tables, Records, Fields)

The basic structure is a table (also called a relation): rows and columns, like a spreadsheet, holding facts about one kind of thing (Students, Teachers, Books).

- A record (row, tuple) is one complete instance — everything about Ali.
- A field (column, attribute) is one kind of fact — the whole Name column, or the whole Grade column.

Roll No	Name	Course	Grade
101	Ali	CS	A
102	Sara	IT	B+
103	Ahmed	CS	A

Remember: Memory: row = record = one person. column = field = one question on the admission form.

ٹیبل قطاروں اور کالموں میں ڈیٹا رکھتا ہے۔

4.3.1 Keys and Integrity Constraints ★ Golden

A key is a field (or a set of fields) that uniquely identifies records and links tables.

- Integrity constraints are the rules that keep those keys honest.
- Any field (or combination) that could uniquely identify a row.
- A table may have several.

Table	Field	Role
Student	RollNo	Primary key
Student	BFormNo	Alternate key
Result	RollNo	Foreign key → Student.RollNo
Result	ExamID + RollNo	Could be a composite primary key

Trap: A phone number is a bad PK: people change numbers, two siblings may share one, and some students have none (NULL).

Exam: Board questions: "Differentiate primary and foreign key" (table of 4 points) and "Why can a PK not be NULL?" Because a NULL would mean "this row has no identity", so the...

کینڈیڈیٹ کی وہ فیلڈ ہے جو ریکارڈ کی شناخت کر سکے۔

4.3 Relational Database Management System (RDBMS)

An RDBMS stores data in several tables and uses primary and foreign keys to keep relationships.

- Reduces redundancy by splitting data into related tables.
- Supports many users at once, with permissions.
- Is queried with SQL (Structured Query Language) — a standard language of SELECT, INSERT, UPDATE, DELETE.

آر ڈی بی ایم ایس کئی ٹیبلز میں ڈیٹا رکھتا ہے اور کیز سے ان کے تعلقات برقرار رکھتا ہے۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. Is "Nadia, Class 11, 72 marks" data or information?
2. Name two DBMS programs.
3. How many records and how many fields are in the table above?
4. Can a table have two primary keys?
5. What language do you use to query an RDBMS?

Answers: 1. Data (raw facts about one student) 2. MS Access, MySQL (Oracle, SQLite also) 3. 3 records, 4 fields 4. No — only one PK (it may be composite) 5. SQL

TONIGHT'S 5-MARK

Write a 5-mark answer on **Keys and Integrity Constraints** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 4: Data and Analysis

Data and Analysis (2/3) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Entity Relationship Model (ER-Model) · Referential Integrity · Relational Schema Development · Case Study — Library Management System · Database Objects (Tables, Forms, Queries, Reports)

★ **Must lock:** Entity Relationship Model (ER-Model), Referential Integrity

If behind, skip non-Golden. Phone in another room.

4.4 Entity Relationship Model (ER-Model) ★ Golden

The ER model is the architect's blueprint: you draw what will be stored and how it connects before you create tables in Access.

Component	Meaning	Symbol	Example
Entity	A real-world object we store data about	Rectangle	STUDENT, TEACHER, COURSE
Attribute	A property of an entity. PK is underlined	Oval	StudentID, Name, Class
Relationship	How two entities are connected	Diamond	Enrolls, Teaches, Has

Trap: The M:N rule.

Exam: Always state cardinality on the diagram (1 and M at the ends of the line). For M:N, the long question is "how do you resolve it?" — draw the junction entity with two...

ای آر ماڈل ڈیٹا بیس کا خاکہ ہے۔

4.5 Referential Integrity ★ Golden

Referential integrity means every foreign-key value in a child table must match an existing primary-key value in the parent — or be NULL if the design allows it.

- If Result.StudentID is a FK to Student.StudentID, you cannot type Result for student 999 if 999 is not in Student.
- You also cannot delete student 101 while results for 101 still exist — unless you first handle those child rows.

ID	Name
101	Ali
102	Sara

Trap: Cascade Delete is powerful and dangerous.

Exam: Write: definition, why it matters (no orphans), then one numbered example of Cascade Update and one of Cascade Delete. Mention parent vs child.

ریفرنشل انٹیگریٹی کا مطلب ہے بچہ ٹیبل کی فارن کی ہمیشہ والدین کی موجود پرائمری کی طرف اشارہ کرے۔

4.6 Relational Schema Development

A relational schema is the formal written design of the database: table names, fields, primary keys, foreign keys and how tables relate.

- You write it before clicking around in Access.
- It defines (1) how data is split into tables, (2) which columns each table has, (3) which fields are PK/FK, (4) the relationships.
- Benefits: less redundancy, integrity planned in advance, easier to grow later.

```
Department (DeptID PK, DeptName)
Student    (RollNo PK, Name, Class, DeptID FK → Department)
Course     (CourseID PK, Title, Credit, DeptID FK → Department)
```

Remember: Skipping the schema is like building a house without a drawing. You get messy, broken structures that are painful to repair.

ریلیشنل اسکیمہ ڈیٹا بیس کا لکھا ہوا نقشہ ہے: کون سے ٹیبل، کون سے فیلڈ، کون سی کیز، کون سے تعلقات۔

4.7 Case Study — Library Management System

A college library needs a database. Walk the seven steps examiners expect.

- A college library needs a database.
- Walk the seven steps examiners expect.
- A book has ID, title, author, publisher.
- A member (student/teacher) has ID, name, contact.

BorrowID	MemberID	BookID	IssueDate	ReturnDate
5001	102	B-17	2026-10-05	

Member (MemberID PK, Name, ContactInfo)
 Book (BookID PK, Title, Author, Publisher)
 BorrowRecord (BorrowID PK, MemberID FK, BookID FK, IssueDate, ReturnDate)

لائبریری میں ممبر اور کتاب کا تعلق بہت-سے-بہت-سے ہے، اس لیے بورو ریکارڈ نامی پل ٹیبل بناتے ہیں جس میں دونوں کی فارن کیز اور اجراء.../...

4.8 Database Objects (Tables, Forms, Queries, Reports)

Four objects you will meet in Access. Learn the one-line jobs cold:

- Learn the one-line jobs cold:

Object	Job	You ...
Table	Stores the data (foundation; every other object depends on it)	keep the actual records here
Form	Friendly screen for enter / edit / view (text boxes, drop-downs)	type data in; fewer mistakes than a raw grid
Query	Ask a question: filter, sort, calculate	e.g. "students above 80%"
Report	Formatted output for printing or sharing	read only — never type into a report

Remember: Store → Enter → Search → Print. Tables, Forms, Queries, Reports.

ٹیبل ڈیٹا رکھتی ہے، فارم سے اندراج ہوتا ہے، کوئری سوال پوچھ کر چھاننتی ہے، رپورٹ چھپائی کے لیے سنواری ہوئی پیشکش ہے — اس میں ڈیٹا بھرا نہیں جاتا۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. What symbol is used for a relationship?
2. What is an orphan record?
3. Name two things a schema must show.
4. Why is BorrowRecord needed?
5. Which object would you use for a printed fee slip?

Answers: 1. Diamond 2. A child row whose FK value does not exist in the parent 3. Tables/fields and keys/relationships 4. To break M:N into two 1:M links 5. Report

TONIGHT'S 5-MARK

Write a 5-mark answer on **Entity Relationship Model (ER-Model)** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 4: Data and Analysis

Data and Analysis (3/3) · Checkpoint · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–90 min remaining cards · 90–130 min closed-book chapter checkpoint · 130–180 min mark and rewrite missed Golden's.

Cover: Creation of Tables in Microsoft Access · Designing Forms for Data Manipulation · Creating Queries in MS Access · Data Summarization and Statistical Analysis · Data Visualization in MS Access (Charts)

★ **Must lock:** Creating Queries in MS Access

If behind, skip non-Golden. Phone in another room.

4.9 Creation of Tables in Microsoft Access

Plan first (4.9.1): purpose, field list, data type of each field, which field is the PK.

- Create tab → Table.
- A blank grid appears.
- Double-click Field1, Field2, ... to rename.
- Type data; Access guesses the data type.

Data type	Use it for
Short Text	Names, addresses, phone numbers (you do not add phone numbers). Up to 255 characters.
Number	Values you calculate with: Age, Marks, Quantity.
Date/Time	Date of birth, admission date, appointment time.
Currency	Fees, salary.
AutoNumber	Access fills a unique sequential number — excellent PK.

Trap: Phone numbers look numeric but are Short Text.

پہلے ٹیبل کا منصوبہ بنائیں۔

4.10 Designing Forms for Data Manipulation

A form is the screen people actually type into.

- Create tab → Form Wizard.
- Select the table (e.g. Student).
- Move fields with > (one) or >> (all).
- Choose layout. Columnar = one record at a time, best for data entry.

فارم اندراج کی سادہ اسکرین ہے۔

4.11 Creating Queries in MS Access ★ Golden

A query retrieves, filters and analyses data without changing the original tables.

- Create tab → Query Design.
- Show Table → add Student → Close.
- Double-click the fields you want; they drop into the grid.
- Red Run (!) button executes it.

Goal	Where to type	Example
Single criterion	Criteria row, one column	Department → "Computer Science"
AND (both must match)	Same Criteria row, two columns	Name "Ali" and Department "Computer Science"
OR (either may match)	Criteria row and the or: row	Department "IT" or "Computer Science"

Trap: Text criteria need quotes: "IT".

Exam: Remember: AND = same row in the grid; OR = different rows. Examiners draw a blank grid and ask you to fill Criteria and or:.

کوٹری ٹیبل کو بدلے بغیر سوال پوچھتی ہے۔

4.11.4 Data Summarization and Statistical Analysis

Grouping combines rows that share a value (all CS students together) and then an aggregate calculates a number for the group.

- In Access click Totals (Σ) to reveal the Total row in the grid.

Function	What it calculates
COUNT()	How many records (students in IT)
SUM()	Total of a number field (fees collected)
AVG()	Mean
MAX() / MIN()	Highest / lowest in the group

```
SELECT Department, AVG(Marks) AS AvgOfMarks
FROM Student
GROUP BY Department;
```

گروپ بائی ایک جیسے ریکارڈ جوڑتا ہے، پھر COUNT, SUM, AVG, MAX, MIN سے خلاصہ نکلتا ہے۔

4.11.6 Data Visualization in MS Access (Charts)

Charts turn numbers into a picture so a pattern jumps out.

- Pick the chart that matches the question:

Chart	Shape	Best for
Bar (horizontal)	Bars sideways	Categories with long names (English, Mathematics, ...)
Column (vertical)	Bars standing	Side-by-side comparison (2024 / 2025 / 2026 admissions)
Pie	Slices of a circle	Parts of one whole (percent of CS vs IT vs Eng). Few slices.
Line	Dots joined	A trend over time (visitors per month)

Exam: "Which chart?" — line for time, pie for composition, bar/column for comparing categories. Mention one reason.

بار لمبے ناموں والی اقسام، کالم سادہ موازنہ، پائی حصوں کا تناسب، لائن وقت کے ساتھ رجحان۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. Best data type for StudentID if you want Access to generate it?
2. Can a form exist without a table or query?
3. Criteria for marks at least 80?
4. Which Total setting goes under Department?
5. Website visitors each month — which chart?

Answers: 1. AutoNumber 2. No — it needs a data source 3. ≥ 80 under Marks 4. Group By 5. Line

TONIGHT'S 5-MARK

Write a 5-mark answer on **Creating Queries in MS Access** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

CHECKPOINT TEST

40 minutes, closed book. Chapter 4: Data and Analysis. 8 MCQ + 3 short. Mark below 60% means tomorrow morning is Golden repair, not a new chapter.

Multiple choice

1. Processed data used for a decision is called:
 - a. a field
 - b. information
 - c. a query
 - d. redundancy
2. A complete row about one student is a:
 - a. field
 - b. report
 - c. record
 - d. schema
3. A primary key cannot contain:
 - a. numbers
 - b. text
 - c. NULL
 - d. an AutoNumber
4. EmpLicense# after EmpID is chosen as PK is an:
 - a. foreign key
 - b. alternate key
 - c. orphan
 - d. report
5. A foreign key:
 - a. must be unique in the child table
 - b. may contain duplicates
 - c. cannot exist in an RDBMS
 - d. is always AutoNumber
6. The ER symbol for a relationship is a:
 - a. rectangle
 - b. oval
 - c. diamond
 - d. circle
7. The most common cardinality is:
 - a. 1:1
 - b. 1:M
 - c. M:N stored as-is
 - d. 0:0
8. M:N is stored in an RDBMS by:
 - a. duplicating both tables
 - b. a junction table
 - c. deleting one entity
 - d. using a pie chart

Short questions

1. Differentiate data and information with one example each.
2. List four advantages of a DBMS.
3. Define record and field with reference to a Student table.

Checkpoint answers — mark after the 40 minutes.

MCQ: 1. b 2. c 3. c 4. b 5. b 6. c 7. b 8. b

1. Data = raw facts (80, 75, 90). Information = processed meaning ("average is 81.7%, the class is ready").
2. Less redundancy, consistency, security (authorised access), integrity (valid, unique keys). (Sharing / backup also accepted.)
3. Record = one student's full row (101, Ali, CS, A). Field = one column such as Name.

Chapter 5: Application and Impacts of Computing

Application and Impacts of Computing (1/3) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Introduction to Computing Impacts · Artificial Intelligence (AI) · Internet of Things (IoT) & Components · Data Analytics & Components · Comparison: IoT vs AI vs Data Analytics

★ **Must lock:** Artificial Intelligence (AI), Internet of Things (IoT) & Components

If behind, skip non-Golden. Phone in another room.

5.1 Introduction to Computing Impacts

Computing means using computers and computer-based devices to store, process and share information.

- Schools and colleges — online admissions, e-books, online results of the Sindh boards.
- Hospitals — computerised patient records, digital X-rays, online appointment systems.
- Banks — ATMs, mobile banking apps, Raast instant payments, JazzCash and Easypaisa wallets.
- Farms — weather forecasts on mobile phones, soil sensors, tube-well control.

کمپیوٹنگ آج ہماری روزمرہ زندگی کا اہم حصہ بن چکی ہے۔

5.1.1 Artificial Intelligence (AI) ★ Golden

Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding language and making decisions.

IoT (Eyes) — collects data → Analytics — sorts data → AI (Brain) — decides → Action

- Artificial — it is created by humans; it is not natural.
- Intelligence — it can perform tasks that require thinking.
- Face recognition on a mobile phone to unlock the screen.
- YouTube suggesting videos based on your interests.

Remember: Eyes – Sort – Brain – Do. IoT is the eyes, Analytics sorts, AI is the brain, and the actuator does the action. Think of a cricket fielder: his eyes see the...

Exam: Common questions: "Define AI and give two examples" (2–3 marks), and "Explain how AI, IoT and Data Analytics work together" (4–5 marks). To score full marks: (1) write...

مصنوعی ذہانت (AI) کو "مصنوعی" اس لیے کہتے ہیں کہ اسے انسانوں نے بنایا ہے، اور "ذہانت" اس لیے کہ یہ ایسے کام کر سکتی ہے جن کے لیے سوچنا پڑتا ہے۔

5.1.2 Internet of Things (IoT) & Components ★ Golden

The Internet of Things (IoT) is a system in which physical devices are connected to the Internet and can collect and share data to perform tasks automatically.

Sensor (Collects) → Wi-Fi (Connectivity) → Cloud (Processes) → Actuator (Acts) → User App (UI)

- The "things" are ordinary objects — watches, cameras, cars, fans, water pumps — that have been given sensors and an internet connection so...
- Source example: A smartwatch collects your heart-rate data and sends it to your mobile phone, where it is displayed.

#	Component	What it does	Examples
1	Sensors	Devices that collect information from the physical world. They are the "eyes and ears" of the system.	Temperature sensor, motion sensor, fingerprint sensor, smoke sensor, soil moisture sensor
2	Actuators	Devices that perform physical actions in response to system commands. They convert digital signals into physical movement.	Electric motor, switch, automatic door opener, water valve, fan controller
3	Connectivity	The connection that allows devices to send and receive data . Without it, sensors can't talk to the system.	Wi-Fi, Bluetooth, ZigBee, 4G/5G
4	Data Processing	Changing raw data into useful information . Steps: collecting, organising, analysing and deciding.	Checking if the room temperature is too hot to trigger the AC
5	Cloud Computing	Storing data and using services over the internet instead of local hard drives. Handles massive IoT data.	Google Drive, iCloud, online LMS
6	User Interface (UI)	The component that enables interaction between the user and the device . Helps users see information and give commands.	Mobile apps, touch screens, voice commands

Remember: Sensor senses, Actuator acts. Both words give you the answer: sensor = see/sense (input), actuator = act (output). Remember the six components with "Some Ants..."

Exam: "Define IoT and explain its components" is one of the most frequent long questions. Write the definition with the three key ideas — physical devices, connected to the...

انٹرنیٹ آف تھنگز (IoT) ایک ایسا نظام ہے جس میں مختلف آلات (جیسے گھڑیاں، کیمرے، گاڑیاں) انٹرنیٹ سے جڑے ہوتے ہیں اور خودکار طریقے سے...

...

5.1.3 Data Analytics & Components

Data Analytics is the process of collecting, organising, cleaning, studying and presenting data to find useful information.

1. Collection → 2. Storage → 3. Cleaning → 4. Analysis → 5. Visualization

- It helps people make better decisions.
- Data means raw facts and figures (for example, "72, 45, 88").
- Information is data that has been processed so it has meaning ("Ali scored 72%; he passed").

Roll No	Name	Marks (raw)
1	Ayesha	42
2	Bilal	18
3	Hina	35
3	Hina	35
4	Kashif	250
5	Sana	(missing)

Remember: "Cows Sit Calmly And Vote" — Collection, Storage, Cleaning, Analysis, Visualization. Or think of cooking daal: buy the lentils (collect), keep them in a jar...

ڈیٹا اینالیٹکس ڈیٹا کو جمع کرنے، ترتیب دینے، صاف کرنے، اس کا مطالعہ کرنے اور پیش کرنے کا عمل ہے تاکہ مفید معلومات حاصل ہوں اور بہتر فیصلے کیے جا سکیں۔

Comparison: IoT vs AI vs Data Analytics

These three technologies are often used together, so students mix them up.

- The table below, from the source, compares them on four aspects.

Aspect	IoT	Artificial Intelligence (AI)	Data Analytics
Definition	Network of physical devices connected to the internet.	Technology enabling machines to think and decide.	Process of examining data to find patterns.
Main purpose	Data collection and automation.	Intelligent decision making and predictions.	Extracting meaningful insights.
Type of work	Works with physical devices (hardware).	Works with smart software systems.	Works with data (numbers, records).
Output	Raw data from the environment.	Smart decisions or actions.	Reports, graphs, insights.
Role in a smart system	Eyes and hands	Brain	Organiser / reporter
Pakistani example	Safe City cameras in Lahore/Karachi capturing traffic	Software recognising a car's number plate for an e-challan	Monthly report showing which roads have the most violations

Remember: Ask "What is the output?" Raw readings → IoT. Charts and reports → Data Analytics. A decision or prediction → AI.

IoT جسمانی آلات (ہارڈویئر) کا نیٹ ورک ہے جو ماحول سے خام ڈیٹا جمع کرتا ہے۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

- Name four places in Pakistan where computing has become vital.
- Why is AI called "artificial"?
- Classify as sensor or actuator: (a) fingerprint scanner, (b) automatic door opener, (c) motion detector, (d) electric motor.
- List the five components of Data Analytics in order.
- Which of the three works mainly with hardware?

Answers: 1. Any four of: schools, hospitals, banks, farms, industries 2. Because it is created by humans 3. (a) sensor, (b) actuator, (c) sensor, (d) actuator 4. Collection, Storage, Cleaning, Analysis, Visualization 5. IoT

TONIGHT'S 5-MARK

Write a 5-mark answer on **Artificial Intelligence (AI)** using Define · Explain · Example · Diagram · Working. Tomorrow's 0-15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 5: Application and Impacts of Computing

Application and Impacts of Computing (2/3) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Uses of IoT in Different Areas · Scope of AI in Education in Pakistan · Career Connection in IoT and AI · Information Sources (Primary, Secondary, Tertiary)

★ **Must lock:** Information Sources (Primary, Secondary, Tertiary)

If behind, skip non-Golden. Phone in another room.

5.2 Uses of IoT in Different Areas

IoT improves efficiency (doing more with less effort), saves time, and enhances the overall quality of life.

Soil Sensor → Cloud AI (Needs water?) → Water Pump

- It plays a significant role in developing modern infrastructure (the basic systems of a country — roads, power, water, hospitals).
- The source lists six areas.

Remember: Remember the six areas with "CHIEF H": Cities, Health, Industry, Education, Farming, Homes.

IoT کارکردگی بہتر بناتا ہے، وقت بچاتا ہے اور زندگی کا معیار بلند کرتا ہے۔

5.2.1 Scope of AI in Education in Pakistan

Artificial Intelligence is gradually becoming an important part of Pakistan's education system, improving teaching, learning and administration (managing the school: admissions, records, timetables).

- Learning Management Systems (LMS) — online platforms (e.g. Google Classroom, Moodle) where teachers post lessons and assignments; AI...
- Automated grading and digital testing — MCQ tests marked instantly by computer.
- AI-based career counselling — software that studies a student's marks and interests and suggests suitable fields.
- Helps students learn at their own speed.

Practical uses	Benefits	Challenges
LMS	Learn at own speed	Limited rural internet
Automated grading & digital testing	Identifies students needing help	Lack of computers/devices
AI career counselling	Supports students with disabilities	Teacher training needed
—	—	Data privacy & security

Trap: Mistake: Writing that AI will replace teachers.

پاکستان کے تعلیمی نظام میں مصنوعی ذہانت آہستہ آہستہ اہم کردار ادا کر رہی ہے، جیسے لرننگ مینجمنٹ سسٹم (LMS)، خودکار جانچ اور کیریئر کونسلنگ۔

5.2.2 Career Connection in IoT and AI

Learning computing skills opens doors to many employers in Pakistan: software houses (e.g.

- IoT Engineer: designs smart systems and connected devices (e.g. a smart irrigation kit).
- Data Analyst: analyses the data generated by IoT systems and produces reports.
- Programming — Python, C++.
- Networking — Wi-Fi, protocols (rules for how devices communicate).

Remember: Skills = "PNEP": Programming, Networking, Electronics, Problem-solving. You start building the first one this year when you learn Python!

کمپیوٹنگ کی مہارتیں سیکھ کر آپ سافٹ ویئر ہاؤسز، ٹیلی کام کمپنیوں (جیسے جاز، ٹیلی نار)، بینکوں اور سرکاری آئی ٹی محکموں میں کام کر سکتے ہیں۔

5.3 Information Sources (Primary, Secondary, Tertiary) ★ Golden

Information sources are ways to find knowledge and data.

- The internet has vast information, but not all of it is accurate.
- Sources are divided into three main categories.
- Primary sources give original, firsthand information straight from people, events or official records.

Primary (firsthand)	Secondary (explained by others)	Tertiary (summarised, quick reference)
Interview with a flood victim in Dadu	Newspaper report on the 2022 floods	Encyclopedia entry "Floods in Pakistan"
Photograph taken at an event	Textbook chapter explaining the event	Wikipedia article
Personal diary or letter	Biography written by a historian	Dictionary
NADRA official records (CNIC data)	Pakistan Bureau of Statistics report analysing census data	Handbook / almanac of facts
Original survey responses	Research paper analysing the survey	Index or list of research papers
Sensor readings from an IoT weather station	TV news analysis of the weather	A school "fact sheet" on climate
Quaid-e-Azam's original speech recording	Article discussing the speech	Encyclopedia summary of his life

Remember: Rule of thumb (Author – Date – References): If an online article does not state the author's name, the date of publication, or list its references, it should...

Exam: Typical questions: "Differentiate between primary, secondary and tertiary sources with two examples each" (long/short), "Classify the following sources..." (MCQ/short),...

معلومات کے ذرائع علم اور ڈیٹا تلاش کرنے کے طریقے ہیں، جنہیں تین اقسام میں تقسیم کیا جاتا ہے۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. How do smart bins help waste management?
2. Give three practical uses of AI in education.
3. What does an IoT Engineer do?
4. A video interview with a cricketer after a match.

Answers: 1. They send alerts when full, so only full bins are emptied 2. LMS, automated grading and digital testing, AI-based career counselling 3. Designs smart systems and connected devices 4. P

TONIGHT'S 5-MARK

Write a 5-mark answer on **Information Sources (Primary, Secondary, Tertiary)** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 5: Application and Impacts of Computing

Application and Impacts of Computing (3/3) · Checkpoint · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–90 min remaining cards · 90–130 min closed-book chapter checkpoint · 130–180 min mark and rewrite missed Golden's.

Cover: Impacts of Computing in Various Fields · Assistive Technologies · Importance of Assistive Technologies · Career Connection in Assistive Tech

★ **Must lock:** Assistive Technologies

If behind, skip non-Golden. Phone in another room.

5.4 Impacts of Computing in Various Fields

Computing has drastically changed society.

- It makes life faster and more connected, but it also brings challenges like privacy risks and technology addiction.
- For every field, think of both sides.

Field	Positive impact (+)	Negative impact (-)
Education	Online classes, e-books and learning apps make studying highly flexible and accessible.	Excessive screen time reduces attention spans and causes physical health issues (eye strain, back pain).
Communication	Emails and video calls allow instant, global communication at almost zero cost (e.g. a WhatsApp video call to a relative in Dubai).	People interact less in person, leading to social isolation and reduced face-to-face skills.
Business	Online shopping, digital banking and automation dramatically improve speed and efficiency.	Traditional manual jobs are lost to machines; risk of severe cyber-attacks on banks.
Healthcare	Digital patient records and telemedicine allow doctors to treat remote patients accurately.	Patient data privacy is at risk if hospital servers are hacked.
Agriculture	Smart irrigation and weather forecasting massively increase crop yields.	Poor internet access in rural areas means many farmers cannot use these tools.

Remember: For the five fields, remember "E-C-B-H-A" — Education, Communication, Business, Healthcare, Agriculture — and always write one + and one – for each.

کمپیوٹنگ نے معاشرے کو بہت بدل دیا ہے۔

5.5 Assistive Technologies ★ Golden

Assistive technologies are devices, software and systems that help people with disabilities or functional limitations (difficulty in seeing, hearing, moving, speaking, remembering, etc.).

Mobility → Sensory → Communication → Cognitive

- They allow more people to participate equally in school, work and society.
- There are four categories:

Category	Purpose	Examples
1. Mobility aids	Help people move from one place to another.	Smart wheelchairs, stair lifts
2. Sensory aids	Support vision and hearing to improve interaction.	Hearing aids, screen readers (VoiceOver), Braille displays
3. Communication aids	Assist individuals with speech difficulties to express themselves.	Speech-generating devices, text-to-speech apps
4. Cognitive / learning aids	Help people with memory, reading and understanding.	Audiobooks, spell-checkers, digital reminder apps

Remember: Four categories = "My Small Cat Cries": Mobility, Sensory, Communication, Cognitive. Link each to a body function: legs (mobility), eyes/ears (sensory), mouth...

Exam: Expect: "What are assistive technologies? Explain their categories with examples" (long), "Name two sensory aids" (short), or "A screen reader is an example of ___"...

معاون ٹیکنالوجیز (Assistive Technologies) وہ آلات، سافٹ ویئر اور نظام ہیں جو معذور افراد کی مدد کرتے ہیں تاکہ وہ اسکول، کام اور مع...

5.5.1 Importance of Assistive Technologies

Assistive technologies are important for three main reasons: Equal access to education: Screen readers allow blind students to read digital textbooks equally with their peers (classmates).

Equal Access → Independence → Inclusion

- Equal access to education: Screen readers allow blind students to read digital textbooks equally with their peers (classmates).
- Independence in daily life: Mobility aids reduce the need for constant human assistance — a person in a smart wheelchair can go to the...
- Social inclusion: They break down barriers, allowing individuals to work, study and socialise confidently.

Remember: Remember "All": Access, Independence, Inclusion.

معاون ٹیکنالوجیز کی اہمیت تین باتوں میں ہے: تعلیم تک برابر رسائی (ناہینا طلبہ اسکرین ریڈر سے کتابیں پڑھ سکتے ہیں)، روزمرہ زندگی میں...

5.5.2 Career Connection in Assistive Tech

Designing assistive technology is a highly rewarding career path — you earn a living while improving people's lives.

- CareerWhat they do Assistive Software DeveloperCreates tools such as speech-to-text software (e.g. Urdu voice typing for people who cannot use a keyboard).
- Urdu voice typing for people who cannot use a keyboard).
- Biomedical EngineerDesigns artificial limbs and hearing aids.

Career	What they do
Assistive Software Developer	Creates tools such as speech-to-text software (e.g. Urdu voice typing for people who cannot use a keyboard).
Biomedical Engineer	Designs artificial limbs and hearing aids.
UX/UI Accessibility Expert	Ensures websites are easy to navigate for visually impaired users (e.g. adding text descriptions to images so screen readers can read them).

معاون ٹیکنالوجی بنانا ایک قابلِ قدر پیشہ ہے۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. Give one positive and one negative impact of computing in healthcare.
2. Which category does a hearing aid belong to?
3. State three reasons why assistive technologies are important.
4. Who designs artificial limbs and hearing aids?

Answers: 1. + Digital records and telemedicine help treat remote patients; – patient data privacy is at risk if servers are hacked 2. Sensory aids 3. Equal access to education, independence in daily life, social inclusion 4. A biomedical engineer

TONIGHT'S 5-MARK

Write a 5-mark answer on **Assistive Technologies** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

CHECKPOINT TEST

40 minutes, closed book. Chapter 5: Application and Impacts of Computing. 8 MCQ + 3 short. Mark below 60% means tomorrow morning is Golden repair, not a new chapter.

Multiple choice

1. AI is called "artificial" because it:
 - a. runs only on electricity
 - b. is created by humans
 - c. is always wrong
 - d. works without data
2. Which of the following is an actuator?
 - a. Temperature sensor
 - b. Motion sensor
 - c. Water valve
 - d. Wi-Fi router
3. ZigBee is an example of which IoT component?
 - a. Connectivity
 - b. Sensor
 - c. User interface
 - d. Actuator
4. Google Drive and iCloud are examples of:
 - a. Sensors
 - b. Data cleaning
 - c. Actuators
 - d. Cloud computing
5. Removing duplicate student records belongs to which component of Data Analytics?
 - a. Data collection
 - b. Data storage
 - c. Data cleaning
 - d. Data visualization
6. A pie chart showing pass vs fail percentage is an example of:
 - a. Data collection
 - b. Data storage
 - c. Data cleaning
 - d. Data visualization
7. According to the comparison table, the output of IoT is:
 - a. Raw data from the environment
 - b. Smart decisions
 - c. Reports and graphs
 - d. Printed documents
8. In a smart system, AI acts as the:
 - a. eyes
 - b. brain
 - c. hands
 - d. wires

Short questions

1. Define Artificial Intelligence and give two examples from daily life.
2. Define the Internet of Things (IoT) with one example.
3. Differentiate between a sensor and an actuator with two examples of each.

Checkpoint answers — mark after the 40 minutes.

MCQ: 1. b 2. c 3. a 4. d 5. c 6. d 7. a 8. b

1. AI is human-made ("artificial") technology that can perform tasks requiring thinking ("intelligence"); it uses data to make decisions. Examples: face recognition to unlock a phone; YouTube suggesting videos; chatbots answering questions.
2. IoT is a system in which physical devices are connected to the internet and can collect and share data to perform tasks automatically. Example: a smartwatch collects heart-rate data and sends it to a mobile phone for display.
3. A sensor collects information from the physical world (input; the "eyes and ears"), e.g. temperature sensor, motion sensor. An actuator performs a physical action in response to a command, converting digital signals into movement (output), e.g. electric motor, automatic door opener.

Chapter 6: Digital Literacy

Digital Literacy (1/4) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Introduction to Digital Literacy · Digital Literacy in the Information Age · Types of Data · Data-Collection Strategies · Interviews and Surveys

★ **Must lock:** Types of Data, Data-Collection Strategies, Interviews and Surveys

If behind, skip non-Golden. Phone in another room.

6.1 Introduction to Digital Literacy

Digital literacy is the ability to understand and use digital devices, the internet, and online tools in a safe, effective, and responsible manner.

Locate information → Organise it properly → Turn it into something useful → Share it responsibly

- Safe – you protect yourself and others online (for example, you do not share passwords or private details).
- Effective – you get the job done well (for example, you find the right information quickly).
- Responsible – you use information honestly and fairly (for example, you do not copy someone else's work or spread false news).
- find information,

Remember: Remember the four verbs of a digitally literate student: Locate – Organise – Create – Share ("LOCS": think of a lock that keeps your digital life safe).

ڈیجیٹل خواندگی (Digital Literacy) سے مراد ڈیجیٹل آلات، انٹرنیٹ اور آن لائن ٹولز کو محفوظ، مؤثر اور ذمہ دارانہ طریقے سے سمجھنا اور استعمال کرنا ہے۔

6.2 Digital Literacy in the Information Age

The Information Age is the period we live in now, where vast amounts of data are produced, shared and accessed instantly through the internet, mobile devices and digital platforms (websites, apps, social media).

- Simply having access to technology is not enough.
- Owning a smartphone does not make you good at using information.
- You must develop the skills to understand, manage and use information wisely.

Ability	What it means	Example
Technical	Knowing how to operate devices and software.	Creating a Google Form; using a spreadsheet.
Cognitive (thinking)	Thinking critically about information.	Checking whether a news story is true before believing it.
Social	Communicating and behaving well with others online.	Being polite in a class WhatsApp group; respecting privacy.

Remember: Three abilities = "Tech, Think, Talk": Technical, Cognitive (think), Social (talk with others).

معلوماتی دور (Information Age) میں انٹرنیٹ اور موبائل کے ذریعے بے پناہ معلومات فوراً تیار اور شیئر ہوتی ہیں۔

6.3 Types of Data ★ Golden

Data means facts, numbers, words or observations that we collect.

- The data gathered through a survey, an investigation or a classroom project can be described in two ways: qualitatively or quantitatively.
- Knowing the difference is critical, because each type is analysed in a different way.

Type of data	Nature	Examples	Best used for
Qualitative	Descriptive, non-numerical	Opinions, reasons, behaviour	Understanding why and how something happens
Quantitative	Numerical, countable or measurable	Hours, scores, percentages, totals	Comparing results and making graphs/charts

Remember: QUALitative = QUALity = words and feelings. QUANTitative = QUANTity = numbers and counts. Qualitative answers why/how; quantitative answers how many/how much.

Exam: Common questions: "Differentiate between qualitative and quantitative data" (4–5 marks) or "Classify the following data". Write the answer as a table with four rows:...

کیفیتی ڈیٹا (Qualitative Data) الفاظ پر مشتمل ہوتا ہے اور بتاتا ہے کہ لوگ کیا سوچتے، محسوس کرتے یا تجربہ کرتے ہیں، جیسے آن لائن کل...

6.4 Data-Collection Strategies ★ Golden

Digital literacy is not limited to finding information that already exists on the internet.

Interviews → Surveys → Prototypes → Observation → Simulations

- You should also learn how to gather original (new) information yourself.
- A data-collection strategy is a planned method for gathering this information.
- The five common strategies are: Interviews-Surveys-Prototypes-Observation-Simulations

Strategy	How it works (one line)	Usual data type	Best when...
Interview	Planned conversation with a person or small group	Mostly qualitative	You need deep, detailed answers from a few people
Survey	Same questions asked to many people	Quantitative and qualitative	You need answers from many people quickly
Prototype	Users test an early version and give feedback	Both	You want to improve a product or solution
Observation	Watch and record; no questions asked	Notes (qualitative) or counts (quantitative)	You want to see what people actually do
Simulation	Computer model; change variables and see what happens	Mostly quantitative	Real experiments are costly, risky or difficult

Remember: Mnemonic for the five strategies: "I S P O S" – "I Saw People On Screens": Interviews, Surveys, Prototypes, Observation, Simulations.

Exam: "List and explain the data-collection strategies" is a frequent long question. Name all five, give a one-sentence definition and one example for each, and say what type...

ڈیجیٹل خواندگی صرف موجود معلومات تلاش کرنے تک محدود نہیں، بلکہ نئی معلومات خود جمع کرنا بھی اس کا حصہ ہے۔

6.4.1 – 6.4.2 Interviews and Surveys ★ Golden

An interview is a planned conversation used to gather information directly from a person.

- It may be one-to-one (you and one person) or in a small group.
- Questions are planned in advance, so you do not forget anything important.
- Answers are usually qualitative (opinions, reasons, experiences).

Question type	What it looks like	Data it gives
Closed: Yes/No	Do you study with your phone nearby? (Yes / No)	Countable (quantitative)
Closed: Multiple choice	Which activity do you use your phone for most? (Lectures / Notes / Social media / Games)	Categories you can count
Number / scale	How many hours do you use your phone daily? ____	Quantitative
Rating scale	"My phone helps me study." 1 = Strongly disagree ... 5 = Strongly agree	Quantitative (you can average the ratings)
Open-ended	How does your phone help you in studies?	Qualitative (free text)

Remember: Interview = In-depth with few. Survey = Same questions to many. Think of a doctor talking to one patient (interview) vs a census form sent to every house...

Exam: You may be asked to "differentiate between an interview and a survey" or to "identify what is wrong with this survey question and rewrite it". For rewriting, name the...

انٹرویو ایک منصوبہ بند گفتگو ہے جس میں کسی ایک فرد یا چھوٹے گروہ سے براہ راست معلومات حاصل کی جاتی ہیں؛ اس کے جوابات عموماً کیفیتی ہوتے ہیں۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

- Write the definition of digital literacy in one sentence.
- What is the Information Age?
- Is "time spent online in hours" qualitative or quantitative?
- Name the five data-collection strategies.
- Name two digital tools for creating surveys.

Answers: 1. The ability to understand and use digital devices, the internet and online tools in a safe, effective and responsible... 2. A period in which vast amounts of data are produced, shared and accessed instantly through the internet, mobile devices... 3. Quantitative (measurable number) 4. Interviews, surveys, prototypes, observation, simulations 5. Google Forms and Microsoft Forms

TONIGHT'S 5-MARK

Write a 5-mark answer on **Types of Data** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

Chapter 6: Digital Literacy

Digital Literacy (2/4) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Prototypes, Observation and Simulations · Primary and Secondary Data · Designing a Data-Collection Approach · Presenting Data Using Digital Tools: 6.7.1 Spreadsheets and 6.7.2 Presentations

★ **Must lock:** Prototypes, Observation and Simulations, Primary and Secondary Data

If behind, skip non-Golden. Phone in another room.

6.4.3 – 6.4.5 Prototypes, Observation and Simulations ★ Golden

A prototype is a simple early version of a product, model or solution.

- It allows an idea to be tested before time and effort are spent on the final version.
- Users try the prototype and give feedback, and this feedback becomes useful data for improvement.
- Example: Before building a full school app, you draw the screens on paper (or make a few clickable screens in a design tool) and ask five...

Remember: Watch it → Observation. Ask it → Survey/Interview. Try it → Prototype. Model it → Simulation.

Exam: Scenario questions are popular: "Which data-collection method is most suitable for ... ? Give a reason." Always name the method and justify it using its key feature (e.g....

پروٹوٹائپ کسی پروڈکٹ یا حل کا ابتدائی سادہ نمونہ ہے جسے صارفین آزما کر رائے دیتے ہیں تاکہ اسے بہتر بنایا جا سکے۔

6.5 Primary and Secondary Data ★ Golden

Whenever you use data, it is important to know where it came from.

- The source affects how much control the researcher had over collecting it, and how confidently the results can be used.

Aspect	Primary Data	Secondary Data
Source	Collected directly by the researcher.	Already collected by others.
Examples	Survey responses, interviews, observations.	Reports, articles, websites, databases.
Purpose	Prepared for a specific study.	Reused for a new purpose.
Control	High control over questions and collection process.	Limited control because the data already exists.

Remember: Primary = "1st hand" (you collect it first). Secondary = "2nd hand" (like a second-hand book – someone else used it first).

Exam: "Differentiate between primary and secondary data" is one of the most-asked questions in this chapter. Draw the four-row table: Source, Examples, Purpose, Control. Learn...

بنیادی ڈیٹا (Primary Data) محقق خود کسی خاص مقصد کے لیے براہ راست جمع کرتا ہے، مثلاً سروے، انٹرویو یا مشاہدہ۔

6.6 Designing a Data-Collection Approach

Before collecting data, a researcher must have a plan.

Question → Data type → Method → Sample → Tools → Collect → Store + backup → Ethics

- Define the research question clearly. What exactly do you want to find out?
- Decide the type of data needed: quantitative, qualitative, or both.
- Select suitable methods (survey, interview, observation).
- Choose the sample or target group – the people who will give you data.

Remember: Remember the steps with: "Quiet Teachers Make Students Think, Check, Save Ethically" → Question, Type, Method, Sample, Tools, Collect, Store, Ethics.

ڈیٹا جمع کرنے سے پہلے منصوبہ بندی ضروری ہے: تحقیقی سوال واضح کریں، ڈیٹا کی قسم طے کریں، مناسب طریقہ چنیں، نمونہ (Sample) منتخب کریں...

6.7 Presenting Data Using Digital Tools: 6.7.1 Spreadsheets and 6.7.2 Presentations

Collecting data is only one part of the work.

Spreadsheet data → Chart / Graph → Insert → Slides

- You must also present your findings so that others can understand them quickly and easily.
- Digital tools make this much easier.
- The four main tools are spreadsheets, presentations, infographics and reports.

Formula	What it does
=SUM(B2:B4)	Adds all values in B2 to B4.
=AVERAGE(B2:B4)	Mean: sum ÷ number of values.
=MAX(B2:B4) / =MIN(B2:B4)	Largest / smallest value.
=MEDIAN(B2:B4)	Middle value when sorted.
=COUNT(B2:B4)	Counts cells containing numbers.
=COUNTA(C2:C4)	Counts cells that are not empty (text or numbers).
=COUNTIF(C2:C4, "Yes")	Counts cells that match a condition.

Remember: Presentation rule of thumb: one idea per slide, and no more than about six short bullet points. If you are reading long sentences off the slide, the slide has...

ڈیٹا جمع کرنے کے بعد اسے پیش کرنا بھی ضروری ہے۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. Why would an engineer use a simulation instead of a real test?
2. Which type of data gives the researcher high control?
3. What is the first step in designing a data-collection approach?
4. Write the formula to find the average of cells D2 to D20.

Answers: 1. Because the real experiment may be costly, risky or difficult 2. Primary data 3. Define the research question clearly 4. =AVERAGE(D2:D20)

TONIGHT'S 5-MARK

Write a 5-mark answer on **Prototypes, Observation and Simulations** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

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Chapter 6: Digital Literacy

Digital Literacy (3/4) · Teach · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–105 min today's cards · 105–135 min closed-book drill · 135–180 min one 5-mark homework.

Cover: Infographics and Reports · Case Study: Digital Inquiry Project · Steps 1 – 3: Advanced Search and Methodology · Steps 4 – 6: Survey Preparation and Primary Data Collection

★ **Must lock:** Case Study: Digital Inquiry Project, Steps 1 – 3: Advanced Search and Methodology, Steps 4 – 6: Survey Preparation and Primary Data Collection

If behind, skip non-Golden. Phone in another room.

6.7.3 – 6.7.4 Infographics and Reports

An infographic (information + graphic) is a visual way of presenting information, facts or data so that people can understand the main idea quickly and easily.

- Simplify complex concepts.
- Highlight the most important findings and results.
- Use clear visuals, limited text and accurate information.

Situation	Best tool	Why
You must calculate the average marks of 40 students.	Spreadsheet	Formulas calculate instantly.
You will explain results to the class in assembly.	Presentation	Designed for a live audience.
You want a one-page poster for the notice board.	Infographic	Main result understood at a glance.
The principal wants the full study with method and recommendations.	Report	Detailed, structured document.

Remember: Match tool to job: Spreadsheet = Calculate, Presentation = Tell, Infographic = Show quickly, Report = Explain fully.

انفوگرافک معلومات کو تصویری انداز میں پیش کرنے کا طریقہ ہے تاکہ اصل بات جلدی سمجھ آ جائے، اس کے لیے کینوا (Canva) اور پاور بی آئی...

6.8 Case Study: Digital Inquiry Project ★ Golden

This case study shows how to apply all the ideas of this chapter to solve a real-world problem from start to finish.

1 Question → 2 Search → 3 Collect → 4 Analyse → 5 Create

- An inquiry is an investigation to find the answer to a question.
- Research question: "How does daily mobile phone use affect the study habits of students?" Scenario: Students notice classmates using phones...
- Some say phones help them learn; others say phones waste time.

Workflow stage	Steps
Question & Search	1 Advanced searches · 2 Identify data types · 3 Select methods
Collect	4 Prepare the survey · 5 Choose the sample · 6 Collect primary data ethically · 7 Use secondary data
Analyse	8 Organise in a spreadsheet · 9 Create charts · 10 Analyse results
Create	11 Write the conclusion · 12 Recommendations → final digital artefact

Remember: Workflow in five words: Ask → Search → Collect → Analyse → Create. Every research project you ever do will follow this pattern.

Exam: Expect a long question: "Describe the steps of a digital inquiry project using the example of mobile phone use." Write the research question first, then the 12 steps...

اس کیس اسٹڈی میں تحقیقی سوال یہ ہے: "روزانہ موبائل فون کا استعمال طلبہ کی پڑھائی کی عادات پر کیسے اثر ڈالتا ہے؟"

6.8 Steps 1 – 3: Advanced Search and Methodology ★ Golden

Students first search for background information using search engines (such as Google or Bing).

- They use advanced search operators – special symbols or words typed into the search box – to narrow down results and find reliable...
- Suggested queryOperator usedWhat it does "mobile phone use" "study habits" students " " (quotes)Finds the exact phrases "mobile phone use"...
- "academic performance" filetype:pdfFinds only PDF documents (often official reports and research papers).

Suggested query	Operator used	What it does
"mobile phone use" "study habits" students	" " (quotes)	Finds the exact phrases "mobile phone use" and "study habits", not the words scattered randomly.
"academic performance" filetype:pdf	filetype:	Finds only PDF documents (often official reports and research papers).
digital distraction site:.edu	site:	Searches only educational websites (addresses ending in .edu).

Remember: Three operators, three jobs: " " = exact, filetype: = format, site: = where. Note: there is no space after the colon (site:.edu, not site: .edu).

Exam: "Explain any three advanced search operators with examples" is a likely short question. For each, write the operator, an example query, and what it does. Also learn the...

پہلے قدم میں طلبہ سرچ انجن پر جدید سرچ آپریٹرز (Advanced Search Operators) استعمال کرتے ہیں: کوٹیشن مارکس (") بالکل وہی جملہ تلا...

6.8 Steps 4 – 6: Survey Preparation and Primary Data Collection ★ Golden

Design clear, unbiased questions. Each question is linked to the type of data it produces: QuestionType of data How many hours do you use your phone daily?Quantitative Which activity do you use your phone for the most?Quantitative...

- Design clear, unbiased questions.
- Each question is linked to the type of data it produces: QuestionType of data How many hours do you use your phone daily?Quantitative Which...

Question	Type of data
How many hours do you use your phone daily?	Quantitative
Which activity do you use your phone for the most?	Quantitative (categories)
How does your phone help you in studies?	Qualitative
Suggest one way to use phones responsibly.	Qualitative

Remember: Ethics in five words: Consent, No-Sensitive, Anonymous, Private, Educational – "C-NAPE". Think: "Cover the person's nape (neck) – protect them."

Exam: Questions often ask: "What ethical guidelines should be followed when collecting data?" List all five points. Use the words consent and anonymity – examiners look for...

چوتھے قدم میں واضح اور غیر جانبدار سوالات تیار کیے جاتے ہیں۔

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. Name two tools for making infographics.
2. What is the research question in the case study?
3. Write a query that finds the exact phrase "online learning" only on educational websites.
4. What is a sample?

Answers: 1. Any two: Canva, Power BI, Tableau 2. "How does daily mobile phone use affect the study habits of students?" 3. "online learning" site:.edu 4. The group of people who provide the data

TONIGHT'S 5-MARK

Write a 5-mark answer on **Case Study: Digital Inquiry Project** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

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Chapter 6: Digital Literacy

Digital Literacy (4/4) · Checkpoint · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap yesterday ★ · 15–90 min remaining cards · 90–130 min closed-book chapter checkpoint · 130–180 min mark and rewrite missed Golden.

Cover: Steps 7 – 9: Secondary Data and Spreadsheet Organisation · Steps 10 – 12: Data Analysis, Conclusion and Recommendations · The Final Digital Artefact · Careers in Digital Research

★ **Must lock:** Steps 7 – 9: Secondary Data and Spreadsheet Organisation, Steps 10 – 12: Data Analysis, Conclusion and Recommendations, The Final Digital Artefact

If behind, skip non-Golden. Phone in another room.

6.8 Steps 7 – 9: Secondary Data and Spreadsheet Organisation ★ Golden

Students include a reliable online article stating that "excessive screen time lowers concentration." They will compare this secondary data with their own primary survey results.

- If both agree, the conclusion is stronger.
- If they disagree, the students must think about why (for example, a different age group or country).

Student	Hours	Study Use	Main Activity	Distracted?
1	3	Yes	Lectures	Sometimes
2	5	No	Social Media	Yes
3	2	Yes	Notes	No

Remember: Chart chooser: Bar = Compare, Pie = Parts of a whole, Line = Change over time.

Exam: You may get the spreadsheet table and be asked to "write formulas to find the total and average hours" or "count students who use the phone for study". Write the full...

ساتویں قدم میں ایک قابل اعتماد آن لائن مضمون کا ثانوی ڈیٹا ("زیادہ اسکرین ٹائم توجہ کم کرتا ہے") اپنے سروے کے بنیادی ڈیٹا سے موازنہ...

6.8 Steps 10 – 12: Data Analysis, Conclusion and Recommendations ★ Golden

To analyse means to study the data carefully to find patterns and trends.

- Most students (the majority) use phones for 3–5 hours daily.
- Half use it for educational purposes (lectures/notes).
- A clear trend: students who use phones mostly for entertainment report much higher distraction levels than those who use them for study.

Remember: Conclusion formula: "The data shows ... However ... Therefore ..." – state the main finding, mention the other side, then give the overall answer.

Exam: You may be asked to "write a conclusion and two recommendations" for given data. Use the "shows ... however ... therefore" pattern and make sure each recommendation follows...

دسویں قدم میں چارٹس اور اسپریڈشیٹ سے رجحانات تلاش کیے جاتے ہیں: زیادہ تر طلبہ روزانہ تین سے پانچ گھنٹے فون استعمال کرتے ہیں، نصف ا...

6.8 The Final Digital Artefact ★ Golden

A digital artefact is a final product created using digital technologies.

- Students must create one final artefact that answers the original research question.
- It is not just a decorated file; it must present evidence and conclusions.
- Artefact option: Suitable digital tools: Main purpose: Digital Report: MS Word, Google Docs: Explain the full study in a structured text document.

Artefact option	Suitable digital tools	Main purpose
Digital Report	MS Word, Google Docs	Explain the full study in a structured text document.
Presentation	PowerPoint, Canva, Slides	Share key findings with a live audience.
Infographic	Canva, Adobe Express	Show the main results visually and quickly on one page.
Dashboard	Power BI, Tableau	Explore data through interactive charts and filters.

Remember: The structure follows the inquiry itself: What (title, question, intro) → How (method) → Found (primary, secondary, charts) → So what (conclusion,...

Exam: Learn the four artefact options with tools and purpose (a table question), and the ten-part structure in order. "What is a digital artefact?" – stress that it must...

ڈیجیٹل آرٹیفیکٹ (Digital Artefact) ڈیجیٹل ٹیکنالوجی سے بنائی گئی آخری پروڈکٹ ہے جو تحقیقی سوال کا جواب ثبوت اور نتائج کے ساتھ پیش...

Careers in Digital Research

The skills in this chapter lead to real jobs: Career: What they do: Chapter skill used: Market Research Analyst: Designs surveys to collect primary data to guide business decisions. Surveys, primary data UX (User Experience) Researcher: Gathers qualitative and quantitative data to improve app and website designs. Prototypes, interviews, observation Data Journalist: Uses advanced search skills to research stories and presents them via infographics. Advanced search, infographics Digital Marketing Specialist: Collects audience data and presents results through dashboards and reports. Dashboards, reports

Career	What they do	Chapter skill used
Market Research Analyst	Designs surveys to collect primary data to guide business decisions.	Surveys, primary data
UX (User Experience) Researcher	Gathers qualitative and quantitative data to improve app and website designs.	Prototypes, interviews, observation
Data Journalist	Uses advanced search skills to research stories and presents them via infographics.	Advanced search, infographics
Digital Marketing Specialist	Collects audience data and presents results through dashboards and reports.	Dashboards, reports

Remember: Link each career to one tool: Market Researcher → Survey; UX Researcher → Prototype; Data Journalist → Search + Infographic; Digital Marketer → Dashboard.

اس باب کی مہارتوں سے کئی کیریئر بنتے ہیں: مارکیٹ ریسرچ اینالسٹ کاروباری فیصلوں کے لیے سروے بناتا ہے، یو ایکس (UX) ریسرچر ایپس اور...

CLOSED-BOOK DRILL

Closed book, 30 minutes. Then mark with the answers under this box.

1. What is the secondary data used in Step 7?
2. According to Step 10, which group reported higher distraction?
3. Which artefact lets viewers explore data with interactive filters?
4. Which career designs surveys to guide business decisions?

Answers: 1. A reliable online article stating "excessive screen time lowers concentration" 2. Students who use phones mostly for entertainment 3. A dashboard (Power BI, Tableau) 4. Market Research Analyst

TONIGHT'S 5-MARK

Write a 5-mark answer on **Steps 7 – 9: Secondary Data and Spreadsheet Organisation** using Define · Explain · Example · Diagram · Working. Tomorrow's 0–15 min is you reciting it without notes.

پانچ نمبر کا جواب ترکیب کے ساتھ لکھیں؛ کل زبانی سنائیں۔

CHECKPOINT TEST

40 minutes, closed book. Chapter 6: Digital Literacy. 8 MCQ + 3 short. Mark below 60% means tomorrow morning is Golden repair, not a new chapter.

Multiple choice

1. Digital literacy is best described as the ability to:
 - a. type quickly on a keyboard
 - b. understand and use digital devices, the internet and online tools safely, effectively and responsibly
 - c. repair computer hardware
 - d. write computer programs
2. Which of the following is qualitative data?
 - a. Marks obtained in a test
 - b. Number of hours spent online
 - c. Students' opinions about online classes
 - d. Percentage of students using the internet
3. Which method is most suitable when the same questions must be asked to many people?
 - a. Survey
 - b. Interview
 - c. Prototype
 - d. Simulation
4. Simulations are especially useful when:
 - a. you want people's opinions
 - b. the data already exists in a report
 - c. you want to watch people without asking questions
 - d. real experiments are costly, risky or difficult
5. To find out how many cars cross an intersection at 8 AM, the best method is:
 - a. Interview
 - b. Observation
 - c. Survey
 - d. Prototype
6. Which of these is primary data?
 - a. Responses to a Google Form you created for your project
 - b. An official government report
 - c. A published research article
 - d. Web statistics from a website
7. A researcher has limited control over:
 - a. their own interviews
 - b. their own survey questions
 - c. their own observations
 - d. secondary data
8. The first step in designing a data-collection approach is to:
 - a. collect the data
 - b. choose the sample
 - c. define the research question clearly
 - d. store the data with a backup

Short questions

1. Define digital literacy. Name the three kinds of ability it involves.
2. Why is simply having access to technology not enough in the Information Age?
3. Differentiate between qualitative and quantitative data, giving two examples of each.

Checkpoint answers — mark after the 40 minutes.

MCQ: 1. b 2. c 3. a 4. d 5. b 6. a 7. d 8. c

1. Digital literacy is the ability to understand and use digital devices, the internet and online tools in a safe, effective and responsible manner. It involves technical, cognitive and social abilities.
2. In the Information Age vast amounts of data are produced and shared instantly. Owning a device does not mean you can judge or use this information well. Users must develop skills to understand, manage and use information wisely, and to tell reliable information from unreliable.
3. Qualitative data is descriptive and non-numerical; it explains why and how (e.g. students' opinions about online classes; observation notes). Quantitative data is numerical, countable or measurable; it is used for comparing and making charts (e.g. marks in a test; hours spent online).

Golden blitz — Chapters 1–2

★ only · Chapters 1–2 · Golden blitz · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap · 15–150 min ★ Golden cards only (write every table from memory) · 150–180 min one 5-mark Golden answer.

Cover: Analog and Digital Signals · Logic Gates (Basic Gates: AND, OR, NOT) · Logic Diagrams · Karnaugh Maps (K-Maps) · Software Development Life Cycle (SDLC) · The Waterfall Model · The Agile Model · Communication Models and the OSI 7-Layer Model

★ **Must lock:** Analog and Digital Signals, Logic Gates (Basic Gates: AND, OR, NOT), Logic Diagrams, Karnaugh Maps (K-Maps), Software Development Life Cycle (SDLC), The Waterfall Model, The Agile Model, Communication Models and the OSI 7-Layer Model, Algorithm, Algorithm vs Pseudocode, Decomposition, Sorting Algorithms: Bubble Sort, Sorting Algorithms: Selection Sort, Searching Algorithms: Binary Search

If behind, skip non-Golden. Phone in another room.

Write every table and diagram on this day from memory on scrap paper, then check the cards.

1.1.3 Analog and Digital Signals ★ Golden

A signal is a changing voltage (or other quantity) that carries information.

- Analog signals are continuous signals that change smoothly over time, like a sine wave.
- Examples: the human voice, a traditional mercury thermometer.
- Digital signals have only two levels: HIGH (1) and LOW (0).

Feature	Analog data	Digital data
Nature	Continuous flow.	Discrete, separate values (bits).
Wave shape	Sine wave (smooth curve).	Square wave (sudden jumps).
Reliability	Low (affected by noise).	High (resistant to noise).
Examples	Human voice, traditional thermometer.	Computer data.

Remember: Analog = A smooth curve. Digital = Doorsteps. Analog looks like ocean waves; digital looks like a castle wall (square teeth).

Exam: Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least...

اینا لگ سگنل وقت کے ساتھ ہموار انداز میں بدلتا ہے، جیسے سائن ویو (Sine Wave)؛ انسانی آواز اس کی مثال ہے۔

1.1.7 Logic Gates (Basic Gates: AND, OR, NOT) ★ Golden

Logic gates are electronic circuits that perform Boolean operations.

- They take one or more input bits and produce one output bit.
- They are the building blocks of all digital systems — a processor contains billions of them.
- The three basic gates match the three basic operations: the AND gate, the OR gate and the NOT gate (also called an inverter).

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Remember: Shortcut: AND output is 0 if any input is 0. OR output is 1 if any input is 1. Learn just the "special row": AND is 1 only at 11; OR is 0 only at 00.

Exam: You are often asked to "draw the symbol, write the expression and truth table" of a gate (2–3 marks each). Draw neat symbols with labelled inputs A, B and output Y....

لاجک گیٹس الیکٹرانک سرکٹ ہیں جو بولین آپریشن انجام دیتے ہیں اور تمام ڈیجیٹل نظاموں کی بنیادی اینٹیں ہیں۔

1.1.11 Logic Diagrams ★ Golden

A logic diagram shows how logic gates are connected to create a circuit.

- It is a "picture" of a Boolean expression.
- To draw one, use the order of precedence: the operation with the highest priority is drawn first (nearest the inputs), and the last...

Remember: Read the expression and ask "What is done last?" That gate goes on the far right, next to Y. Work backwards to the inputs. Count gates: $Y = A \cdot B + C$ needs 2...

Exam: Typical question: "Draw the logic diagram of $Y = \dots$ " and sometimes "also make its truth table" (4–5 marks). Write the precedence steps in one line, draw neat gate symbols...

لاچک ڈایاگرام دکھاتا ہے کہ گیٹس آپس میں کیسے جڑ کر سرکٹ بناتے ہیں۔

1.1.13 Karnaugh Maps (K-Maps) ★ Golden

A Karnaugh map (K-map) is a graphical tool used to simplify Boolean expressions.

- It is a grid of cells — one cell for each row of the truth table.
- The cells are arranged so that neighbouring (adjacent) cells differ by only ONE bit.
- This ordering is called Gray code.

	B = 0	B = 1
A = 0	Cell 0 ($A'B'$)	Cell 1 ($A'B$)
A = 1	Cell 2 (AB')	Cell 3 (AB)

Remember: Gray code song: "zero-zero, zero-one, one-one, one-zero" — only one bit flips at each step. Imagine the K-map is drawn on a tube, so the left and right edges...

Exam: K-maps are a favourite long question (5–6 marks): "Simplify $F(A,B,C) = \sum(\dots)$ using a K-map." Draw the map with labelled Gray-code headings, fill every cell (write the 0s...)

کارنو میپ (K-Map) بولین ایکسپریشن کو آسان بنانے کا تصویری طریقہ ہے۔

1.2.1 – 1.2.3 Software Development Life Cycle (SDLC) ★ Golden

The Software Development Life Cycle (SDLC) is a structured process that guides software development from beginning to end.

1. Requirement Analysis → 2. System Design → 3. Implementation → 4. Testing → 5. Deployment → 6. Maintenance

- It ensures clear goals, better planning, risk reduction and high quality.
- Without it, teams forget features, go over budget and deliver buggy software.
- SDLC has 6 phases: 1.

Phase	Activities and details	Deliverable (what it produces)
1. Requirement Analysis	Meet clients, understand their needs, list the features. This is the most important phase — mistakes here spread to every later phase.	SRS Document (Software Requirements Specification)
2. System Design	Create the High-Level Design (overall architecture), the Low-Level Design (database, UI screens) and flowcharts.	Design Document
3. Implementation	Write the actual code in programming languages (Python, Java, etc.) based on the design documents.	Working Source Code

Remember: Remember the order with "Real Developers Implement, Test, Deploy, Maintain" (R-D-I-T-D-M). Black box = a closed box (you can't see inside). White box = a glass...

Exam: "Explain the phases of SDLC" is a very common long question (5–8 marks). Name all 6 phases in order, write 1–2 lines of activities for each, and state the deliverable of...

ایس ڈی ایل سی (SDLC) سافٹ ویئر بنانے کا منظم طریقہ ہے جس کے چھ مراحل ہیں: ضروریات کا تجزیہ، سسٹم ڈیزائن، کوڈنگ (Implementation)، ٹ...۔

1.2.4 The Waterfall Model ★ Golden

The Waterfall model is a linear, sequential approach to the SDLC.

- Each phase must be completed before the next one begins, like water falling down steps.
- There is no easy way to go back to a previous phase.

Advantages (Pros)	Disadvantages (Cons)
Simple to understand and manage.	Not flexible to changes.
Well documented at every step.	Testing happens very late in the process.

Remember: Waterfall = water only flows downwards. Once water has fallen it cannot climb back up — once a phase is finished you do not go back.

Exam: Expect "Explain the Waterfall model with its advantages and disadvantages" or "When should the Waterfall model be used?" Draw the step diagram, list the three "when to..."

وائرفال (آبشار) ماڈل ایک سیدھا اور ترتیب وار طریقہ ہے؛ ہر مرحلہ مکمل ہونے کے بعد ہی اگلا شروع ہوتا ہے اور پیچھے جانا مشکل ہوتا ہے۔

1.2.4 The Agile Model ★ Golden

The Agile model is an iterative approach: the software is built in small, repeated cycles called sprints (usually 2–4 weeks long).

Plan → Design → Build → Test → Review & feedback → Next sprint ↻

- Each sprint plans, builds and tests a small part of the software, and then shows it to the customer.
- Customer feedback is gathered continuously and used in the next sprint.
- Plan→Design→Build→Test→Review & feedback→Next sprint ↻

Advantages (Pros)	Disadvantages (Cons)
Highly flexible to changes.	Less documentation.
Early delivery of working software.	Unpredictable end-time and cost.

Remember: Agile = agile athlete: quick and able to change direction. Think of making a pizza where the customer tastes each slice and you adjust the next one.

Exam: Questions ask you to define Agile, explain sprints, list pros/cons, or say when to use it. Always mention the words "iterative", "sprints (2–4 weeks)" and "continuous..."

ایجائل (Agile) ماڈل میں سافٹ ویئر چھوٹے چھوٹے ادوار میں بنتا ہے جنہیں اسپرینٹ (Sprint) کہتے ہیں، جو عموماً دو سے چار ہفتے کے ہوتے ہیں۔

1.3.1– 1.3.4 Communication Models and the OSI 7-Layer Model ★ Golden

A communication model explains how data travels from one device to another through a network.

- Because sending data is complicated, the job is split into layers; each layer does one part of the job and passes the data to the next...
- The OSI (Open Systems Interconnection) model was developed by ISO (International Organization for Standardization) in 1984.
- It divides communication into 7 layers.

Layer	Function	Protocols / examples
7. Application	Provides network services directly to user applications (browser, email program).	HTTP, HTTPS, FTP, SMTP, DNS
6. Presentation	Data translation, encryption (security) and compression.	SSL/TLS, JPEG, MP3, ASCII
5. Session	Opens, maintains and closes communication sessions.	NetBIOS, RPC
4. Transport	Reliable end-to-end delivery, error checking, port numbering.	TCP (reliable), UDP (fast)

Remember: Mnemonic (bottom to top, 1 → 7): "Please Do Not Throw Sausage Pizza Away" = Physical, Data Link, Network, Transport, Session, Presentation, Application. (Top...

Exam: A guaranteed long question: "Explain the seven layers of the OSI model." Draw the 7-layer stack (numbered, layer 7 on top), then write one function and one or two...

کمپیونیکیشن ماڈل بتاتا ہے کہ ڈیٹا نیٹ ورک کے ذریعے ایک ڈیوائس سے دوسری تک کیسے پہنچتا ہے۔

2.2 Algorithm ★ Golden

An algorithm is a step-by-step logical procedure used to solve a problem.

- Input — the data it starts with.
- Clear steps — each step has only one meaning.
- A definite end — it must finish after a limited number of steps.
- Output — the result it produces.

Reason	What it means for you
Structured approach	Algorithms provide a structured approach to problem-solving, making complex tasks manageable.
Optimisation	They optimise time and memory usage, which is crucial for large-scale applications (for example, searching millions of records).
Reusability	They can be reused for similar problems, saving time and effort. A sorting algorithm for marks can also sort prices.
Accuracy and consistency	They ensure accuracy and consistency, reducing human error. The same steps always give the same result.
Blueprint	They act as a blueprint (a plan) before actual code implementation, just like an architect's map before building a house.

Remember: Remember the five reasons as S-O-R-A-B: Structured, Optimised, Reusable, Accurate, Blueprint.

Exam: Very common short question: "Define algorithm and write its importance." Write the definition word for word — a step-by-step logical procedure used to solve a problem;...

الگورتھم (Algorithm) کسی مسئلے کو حل کرنے کا مرحلہ وار منطقی طریقہ ہے۔

2.2.2 Algorithm vs Pseudocode ★ Golden

Both algorithms and pseudocode are used in problem-solving, but they serve different purposes in the design phase.

Problem → Algorithm (what) → Pseudocode (how) → Program (code) → Output

- The algorithm says what must be done.
- The pseudocode shows how it could be implemented, using programming-like keywords such as IF, FOR and WHILE, but it still cannot be run on...

Point	Algorithm	Pseudocode
Definition	A step-by-step logical procedure to solve a problem.	A human-readable representation of an algorithm using plain English & keywords.
Nature	Conceptual and language-independent.	Semi-formal and structured, but not executable code.
Purpose	To define what needs to be done logically.	To present how it can be implemented in a readable format.
Form	Usually described in plain steps, bullets or numbered points.	Written using structured constructs (IF, FOR, WHILE) like programming.
Focus	Focuses on the logic and steps of the problem.	Focuses on representing steps clearly before coding.
Audience	For anyone trying to understand the logic (including non-programmers).	Mainly for programmers or students preparing for coding.
Format rules	No fixed format; can be informal.	Follows standard pseudocode conventions for clarity.

Remember: Teacher's analogy: an algorithm is like a recipe ("Bake a cake at 350 degrees"). Pseudocode is like a structured kitchen manual using standard verbs (SET...

Exam: "Differentiate between algorithm and pseudocode" is a favourite 4–5 mark question. Draw a two-column table and give at least five points (definition, nature, purpose,...

الگورتھم بتاتا ہے کہ کیا کرنا ہے، جبکہ سوڈو کوڈ (Pseudocode) بتاتا ہے کہ اسے کیسے کیا جا سکتا ہے۔

2.3.1 Decomposition ★ Golden

Decomposition is a problem-solving technique in which a complex task or system is broken down into smaller, manageable parts.

Vocabulary → Grammar rules → Tenses → Sentence building

- By dividing a big problem into simpler steps, it becomes easier to understand, develop and solve.
- Each small part is called a sub-task (or sub-problem).
- You solve each sub-task separately, then join the solutions to solve the whole problem.

Sub-task	Description
Learn vocabulary	Learn the words for people, actions and things (e.g., "I", "eat", "apple").
Grammar rules	Learn the sentence structure: Subject → Verb → Object.
Learn tenses	Present, past, future (e.g., "I eat" vs. "I ate").
Sentence building	Combine words using the rules to create valid sentences: "I ate an apple."

Remember: Think of eating a big roti: you cannot eat it in one bite, so you break it into small pieces. Decomposition = "break the roti".

Exam: Expect "Define decomposition with a real-life example" or "Explain the benefits of decomposition". Give the definition, draw the small tree (Complex Problem → Sub-task...

ڈیکمپوزیشن (Decomposition) کا مطلب ہے کسی بڑے اور پیچیدہ مسئلے کو چھوٹے اور آسان حصوں میں تقسیم کرنا۔

2.4.1 Sorting Algorithms: Bubble Sort ★ Golden

Bubble Sort is an elementary sorting algorithm that works by repeatedly comparing adjacent (side-by-side) elements and swapping them if they are in the wrong order.

Compare DATA[j] and DATA[j+1] → Left > Right? → Yes: swap → Move one step right → End of pass: largest is at the end → Repeat N-1 passes

- In each pass, the largest remaining value "bubbles up" to the end of the list, like an air bubble rising to the top of water.
- So after pass 1 the largest value is in the last place, after pass 2 the second largest is in second-last place, and so on.
- A list of N items needs N - 1 passes, and pass K makes N - K comparisons.

Comparison	Pair	Left > Right?	Action	List after
1st vs 2nd	[8, 4], 1, 9, 3	8 > 4 Yes	Swap	4, 8, 1, 9, 3
2nd vs 3rd	4, [8, 1], 9, 3	8 > 1 Yes	Swap	4, 1, 8, 9, 3
3rd vs 4th	4, 1, [8, 9], 3	8 > 9 No	No change	4, 1, 8, 9, 3
4th vs 5th	4, 1, 8, [9, 3]	9 > 3 Yes	Swap	4, 1, 8, 3, 9

Remember: Bubble = neighbours. Bubble Sort only ever compares neighbours (j and j+1). Each pass pushes the biggest bubble to the end. For descending order, just change >...

Exam: The most common question: "Sort the list ... using Bubble Sort. Show each pass." Draw one table per pass with columns Pair, Compare, Action, List after. Mark the element...

ببل سارٹ (Bubble Sort) میں ساتھ ساتھ دو عناصر کا موازنہ کیا جاتا ہے اور اگر وہ غلط ترتیب میں ہوں تو ان کی جگہ بدل دی جاتی ہے۔

2.4.1 Sorting Algorithms: Selection Sort ★ Golden

Selection Sort is a simple sorting algorithm that works by repeatedly selecting the smallest element from the unsorted portion of the list and moving it to its correct position at the front.

Fix position K → Compare DATA[K] with DATA[PTR], PTR = K+1 ... N → DATA[K] > DATA[PTR]? swap → End of pass: smallest is at K → K = K + 1

- Think of it as filling seats from the front: first find who belongs in position 1, then position 2, and so on.
- In the textbook version, the element at position K is compared with every element after it; whenever a smaller one is found, the two are...
- So at the end of pass K, the K-th smallest value sits in position K.

Compare	List	Pos1 > PosX?	Action	List after
Pos1 vs Pos2	[8], [4], 1, 9, 3	8 > 4 Yes	Swap	4, 8, 1, 9, 3
Pos1 vs Pos3	[4], 8, [1], 9, 3	4 > 1 Yes	Swap	1, 8, 4, 9, 3
Pos1 vs Pos4	[1], 8, 4, [9], 3	1 > 9 No	No change	1, 8, 4, 9, 3
Pos1 vs Pos5	[1], 8, 4, 9, [3]	1 > 3 No	No change	1, 8, 4, 9, 3

Remember: Teacher's tip — spot the difference: in Selection Sort the inner loop starts at outer + 1 and compares with the fixed position DATA[outer]. In Bubble Sort the...

Exam: Typical questions: "Sort ... using Selection Sort, showing all steps", "Differentiate between Bubble Sort and Selection Sort" and "Why is Selection Sort not stable?" In...

سلیکشن سارٹ (Selection Sort) میں فہرست کے غیر ترتیب شدہ حصے سے سب سے چھوٹا عدد چن کر اسے سامنے اس کی درست جگہ پر رکھا جاتا ہے۔

2.4.2 Searching Algorithms: Binary Search ★ Golden

Binary Search is a highly efficient search algorithm that works by repeatedly dividing the search interval in half.

BEG = first, END = last → MID = INT((BEG+END)/2) → Item = DATA[MID]? Found → Item < DATA[MID]: END = MID-1 → Else: BEG = MID+1 → Repeat while BEG ≤ END

- BEG (low) — first position of the current part.
- END (high) — last position of the current part.
- MID = INT((BEG + END) / 2) — the middle position. INT means "drop the decimal part".
- Equal → found, stop.

Step	BEG (low)	END (high)	MID	DATA[MID]	Compare	Action
1	1	9	INT(10/2) = 5	5	7 > 5	BEG = 6 (go right)
2	6	9	INT(15/2) = 7	7	7 = 7	Found, LOC = 7

Remember: Binary Search is how you find a word in a dictionary: open the middle, decide "before or after?", and throw away half the book each time. "Sorted first, then..."

Exam: You will usually be given a sorted list and an item and asked to "search using Binary Search, showing all steps". Always draw a table with BEG, END, MID, DATA[MID],...

بائنری سرچ (Binary Search) ایک تیز رفتار طریقہ ہے جو ہر قدم پر تلاش کے حصے کو آدھا کر دیتا ہے۔

Golden blitz — Chapters 3–4

★ only · Chapters 3–4 · Golden blitz · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap · 15–150 min ★ Golden cards only (write every table from memory) · 150–180 min one 5-mark Golden answer.

Cover: Fundamentals of Python Programming · Variables · Data Types in Python · Operators and Operands: Arithmetic and Assignment · Relational, Logical and Membership Operators · Bitwise Operators · Control Structures · Selection Statements: if and if-else

★ **Must lock:** Fundamentals of Python Programming, Variables, Data Types in Python, Operators and Operands: Arithmetic and Assignment, Relational, Logical and Membership Operators, Bitwise Operators, Control Structures, Selection Statements: if and if-else, Selection Statements: if-elif-else and Nested if, Repetition Statements: the for Loop and range(), The while Loop, break and continue, Nested Loops and for vs while, Keys and Integrity Constraints, Entity Relationship Model (ER-Model), Referential Integrity, Creating Queries in MS Access

If behind, skip non-Golden. Phone in another room.

Write every table and diagram on this day from memory on scrap paper, then check the cards.

3.4 Fundamentals of Python Programming ★ Golden

The basic structure of a Python program is simple and clear.

- A program takes data, processes it using statements, and finally shows output.
- Python runs the statements top to bottom, one after another (sequentially).
- A statement is one complete instruction, usually written on one line, such as `print("Hi")` or `x = 5`.

```
# This is a comment. It explains the code below.
print("Hello, World!")
if 5 > 2:
    # The space before 'print' is the indentation
    print("Five is greater than two!")
```

Remember: Think of indentation like the points and sub-points in your notebook. Sub-points are written a little to the right to show they "belong" to the main point. A...

Exam: Common questions: "What is the purpose of comments?", "Why is indentation important in Python?", "How is Python different from C++/Java in defining blocks?" Write:...

پائنتھن پروگرام اوپر سے نیچے ترتیب سے چلتا ہے۔

3.4.1 Variables ★ Golden

A variable is a named space in computer memory used to store data.

- Can contain letters, numbers and underscores (_) only — no spaces or symbols like -, @, \$.
- Must start with a letter or underscore, NOT a number (1name is invalid).
- Are case-sensitive: age and Age are two completely different variables.
- Cannot be a Python keyword (a reserved word such as if, for, while, else, True).

```
name = "Ali"
age = 16
my_height = 5.7
score1 = 100
```

Remember: Choose meaningful names: total_marks is far clearer than tm or x. Use lowercase words joined by underscores (this style is called snake_case).

Exam: A favourite question: "Which of these variable names are valid? Give reasons." Check each name against all four rules and state the rule broken, e.g. "2nd_place —...

ویری ایبل (Variable) میموری میں ایک نام والی جگہ ہے جس میں ڈیٹا رکھا جاتا ہے، بالکل ایک لیبل لگے ڈیے کی طرح۔

3.4.1 Data Types in Python ★ Golden

A data type specifies the kind of data a variable holds — a whole number, a decimal, text, True/False, or a collection of items.

- The type decides what you can do with the data (you can divide numbers, but not names).
- Python has several built-in data types:

Name	Type	Description	Example
Integer	int	Whole numbers without decimals (positive, negative or zero).	age = 16
Floating point	float	Numbers with a decimal point.	height = 5.7
String	str	Text data, enclosed in quotes (" " or ' ').	name = "Sammy" or year = "2026"
Boolean	bool	Logical value: True or False (capital T / F).	is_student = True
List	list	Ordered, changeable collection in square brackets [] .	marks = [85, 90, 75]
Tuple	tuple	Ordered, unchangeable sequence in parentheses () .	coords = (10, 20)
Dictionary	dict	Key : value pairs in curly braces { } .	student = {"name": "Ahmed", "age": 16}
Set	set	Unordered collection of unique items (no duplicates).	vowels = {'a', 'e', 'i'}

Remember: Brackets decide the collection: [] List (changeable), () Tuple (locked), { } with key:value = Dictionary, { } with single items = Set (unique). Think: a list...

Exam: Expect: "Name any four data types with examples" or "Differentiate list and tuple". For a difference question, use a two-column table: list — [], changeable, e.g. marks...

ڈیٹا ٹائپ (Data Type) بتاتی ہے کہ ویری ایبل میں کس قسم کا ڈیٹا ہے۔

3.4.3 Operators and Operands: Arithmetic and Assignment ★ Golden

An operand is the value or variable being worked on.

- An operator is the symbol that performs the action.
- Together they form an expression, which Python evaluates to a single value.
- In 10 + 20: 10 is an operand, + is the operator, 20 is an operand, and the result is 30.

Operator	Purpose	Example	Result
+	Addition	5 + 3	8
-	Subtraction	10 - 4	6
*	Multiplication	4 * 2	8
/	Division (result is always a float)	9 / 2	4.5
//	Floor division (drops the decimal part)	9 // 2	4
%	Modulus (remainder)	9 % 2	1
**	Exponentiation (power)	2 ** 3	8

```
print(5 + 3)
print(10 - 4)
print(4 * 2)
print(9 / 2)
print(9 // 2)
print(9 % 2)
print(2 ** 3)
print(8 / 2)
```

Remember: Three kinds of division: / = exact answer (float), // = quotient only ("floor"), % = remainder only. A number is even when n % 2 == 0.

Exam: You will be asked to evaluate expressions such as 17 // 5, 17 % 5, 2 ** 4 or to trace compound assignment. Show each step and write the final value with its type (e.g....

آپرینڈ (Operand) وہ ویلیو ہے جس پر عمل ہوتا ہے اور آپریٹر (Operator) وہ علامت ہے جو عمل کرتی ہے۔

3.4.3 Relational, Logical and Membership Operators ★ Golden

3. Relational (comparison) operators compare two values and return a Boolean (True or False).

- Relational (comparison) operators compare two values and return a Boolean (True or False).
- They are used in conditions.

Op	Description	Example (a = 10, b = 5)	Result
=	Equal to	a == b	False
≠	Not equal to	a != b	True
>	Greater than	a > b	True
<	Less than	a < b	False
≥	Greater than or equal to	a ≥ 10	True
≤	Less than or equal to	a ≤ b	False

```
a = 10
b = 5
print(a == b)
print(a != b)
print(a > b)
print(a < b)
print(a ≥ 10)
print(a ≤ b)
```

Remember: AND is strict, OR is generous. AND is like a teacher who needs both homework and attendance; OR is like a shop that accepts cash or card. Order of evaluation:...

Exam: Typical: "Evaluate $a > b$ and $b > 3$ for $a = 10, b = 5$ " or "Write the truth table of and/or". Write each sub-condition's value first, then combine. Remember: == compares,...

ریشنل آپریٹرز (Relational Operators) دو ویلیوز کا موازنہ کر کے True یا False دیتے ہیں۔

3.4.3 Bitwise Operators ★ Golden

6. Bitwise operators treat numbers as binary digits (bits: 0s and 1s) and work on them bit by bit, column by column.

- AND (&) — gives 1 only if both bits are 1, otherwise 0.
- OR (|) — gives 1 if at least one bit is 1.
- XOR (^) — gives 1 if the bits are different, 0 if they are the same.
- NOT (~) — inverts all bits (0 → 1, 1 → 0). In Python the result is the number made negative, minus 1: $\sim x = -(x + 1)$. So ~ 10 gives -11.

Bit A	Bit B	A & B	A B	A ^ B	~A
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

```
a = 10
b = 6
print(bin(a), bin(b))
print(a & b)
print(a | b)
print(a ^ b)
print(~a)
```

Remember: Shortcuts: $x > n = x // 2^n$, $\sim x = -x - 1$. For AND/OR/XOR always write both numbers in binary, one under the other, with the same number of bits.

Exam: Exams ask you to "evaluate $12 \& 10$, $12 | 10$, $12 \wedge 10$, ~ 5 , $7 >> 1$ showing binary working". Marks are given for the binary conversion and column working, not just the...

بٹ وائز آپریٹرز (Bitwise Operators) اعداد کو بائری میں بدل کر ہر بٹ پر الگ الگ کام کرتے ہیں۔

3.5 Control Structures ★ Golden

Control structures decide the order in which statements are executed in a program (the "flow of control").

- Sequence — do the steps one after another, top to bottom.
- Selection — choose a path depending on a condition (if, else).
- Repetition — repeat steps (loops: for, while).

Remember: Remember "S-S-R": Sequence, Selection, Repetition — "Step, Select, Repeat".

Exam: "Name and explain the three basic control structures" is a very common short question. Give a one-line definition and a two-line Python example for each.

کنٹرول اسٹرکچرز (Control Structures) طے کرتے ہیں کہ پروگرام کی ہدایات کس ترتیب سے چلیں گی۔

3.5.2 Selection Statements: if and if-else ★ Golden

Selection changes the flow of a program based on a condition — an expression that is either True or False.

- It helps a program respond intelligently to different situations.
- The if statement executes a block of code only if the condition is True.
- If the condition is False, the block is completely skipped.

```
if condition:
    statement(s) # runs only when condition is True
```

Remember: Formatting rule: put a colon (:) after if and else, and indent the code inside the blocks. else never has a condition of its own.

Exam: Asked as: "Write a program that inputs a number and prints whether it is even or odd / positive or negative / pass or fail." Write the full program with `int(input())`,...

سلیکشن (Selection) میں پروگرام شرط کی بنیاد پر فیصلہ کرتا ہے۔

3.5.2 Selection Statements: if-elif-else and Nested if ★ Golden

3. The if-elif-else statement handles many conditions.

- The if-elif-else statement handles many conditions.
- (elif is short for "else if".) Python checks the conditions from top to bottom.
- As soon as it finds a True condition, it runs that block and ignores all the rest.
- If none is True, the else block runs.

marks	>= 80?	>= 70?	>= 60?	Output
92	True	(not checked)	(not checked)	A-1 Grade
75	False	True	(not checked)	A Grade
60	False	False	True	B Grade
41	False	False	False	Fail or below B

Remember: In an if-elif-else ladder, put the strictest condition first (highest marks first). Think of a ladder: you step off at the first rung that fits.

Exam: Grade-calculator programs (if-elif-else) and "largest of three numbers" are very common long questions. Check conditions in the right order, indent nested blocks...

if-elif-else میں کئی شرطیں اوپر سے نیچے ترتیب سے دیکھی جاتی ہیں، اور جو شرط سب سے پہلے درست ہو اسی کا بلاک چلتا ہے، باقی سب چھوڑ دی جاتی ہیں۔

3.5.3 Repetition Statements: the for Loop and range() ★ Golden

A repetition statement (loop) lets a block of code run again and again.

- This greatly reduces the need to write the same code many times.
- Each single run of the loop body is called an iteration.
- The for loop is used when the exact number of iterations is known.

Syntax	How it works	Example	Values produced
<code>range(stop)</code>	From 0 up to, but not including , stop.	<code>range(3)</code>	0 1 2
<code>range(start, stop)</code>	From start up to, but not including, stop.	<code>range(2, 5)</code>	2 3 4
<code>range(start, stop, step)</code>	From start towards stop, jumping by step.	<code>range(1, 6, 2)</code>	1 3 5

```
for i in range(3):
    print(i)
```

Remember: range stops one short: `range(1, 11)` gives 1 to 10. To include a number `n`, write `range(..., n + 1)`. Number of values in `range(a, b)` = `b - a`.

Exam: Expect "predict the output" questions on `range()` with a step, and programs such as "print a table", "sum of first `n` numbers", "even numbers from 1 to 20". Always state...

لوپ (Loop) کوڈ کے ایک بلاک کو بار بار چلاتا ہے۔

3.5.3 The while Loop, break and continue ★ Golden

2. The while loop repeats a block of code as long as a condition remains True.

- break — exits (stops) the loop entirely and immediately.
- continue — skips the rest of the current iteration and jumps to the next one.

count	count <= 3?	Printed	count after update
1	True	1	2
2	True	2	3
3	True	3	4
4	False	—	loop ends

```
count = 1
while count <= 3:
    print(count)
    count += 1 # VERY IMPORTANT: update the condition!
```

Remember: break = "exit the room"; continue = "skip this turn". For every while loop, check the I-T-U rule: Initialise, Test, Update.

Exam: "Differentiate break and continue with examples" and "What is an infinite loop?" are frequent. Use the two side-by-side programs above with their outputs — examiners...

while لوپ اس وقت تک چلتا رہتا ہے جب تک شرط درست (True) رہے، اور یہ تب استعمال ہوتا ہے جب پہلے سے معلوم نہ ہو کہ کتنی بار دہرانا ہے۔

3.5.3 Nested Loops and for vs while ★ Golden

A nested loop is a loop inside another loop.

- The inner loop runs completely, from start to finish, for every single iteration of the outer loop.
- If the outer loop runs 2 times and the inner loop runs 3 times, the inner body runs $2 \times 3 = 6$ times.

for loop	while loop
Used when the number of iterations is known .	Used when the number of iterations is unknown (condition-based).
Automatically updates the loop variable.	Requires a manual update of the counter (risk of an infinite loop).
Cleaner and shorter for counting tasks.	Better for dynamic situations (like "keep playing until the user says Stop").
for i in range(1, 4): print(i)	i = 1 / while i <= 3: / print(i) / i += 1

```
for outer in range(1, 3): # runs 2 times (1, 2)
    for inner in range(1, 4): # runs 3 times per outer (1, 2, 3)
        print(f"Outer: {outer}, Inner: {inner}")
```

Remember: Ask yourself: "Do I know how many times?" Yes → for. "Do I only know when to stop?" → while. Nested loops are like a clock: the minute hand (inner loop) goes...

Exam: "Differentiate between for and while loops" (give three points and an example of each) and "How many times will the inner statement run?" (multiply the counts) are...

نیسٹڈ لوپ (Nested Loop) میں ایک لوپ دوسرے لوپ کے اندر ہوتا ہے، اور باہر والے لوپ کے ہر چکر میں اندر والا لوپ شروع سے آخر تک مکمل چلتا ہے۔

4.3.1 Keys and Integrity Constraints ★ Golden

A key is a field (or a set of fields) that uniquely identifies records and links tables.

- Integrity constraints are the rules that keep those keys honest.
- Any field (or combination) that could uniquely identify a row.
- A table may have several.

Table	Field	Role
Student	RollNo	Primary key
Student	BFormNo	Alternate key
Result	RollNo	Foreign key → Student.RollNo
Result	ExamID + RollNo	Could be a composite primary key

Trap: A phone number is a bad PK: people change numbers, two siblings may share one, and some students have none (NULL).

Exam: Board questions: "Differentiate primary and foreign key" (table of 4 points) and "Why can a PK not be NULL?" Because a NULL would mean "this row has no identity", so the...

کینڈیڈیٹ کی وہ فیلڈ ہے جو ریکارڈ کی شناخت کر سکے۔

4.4 Entity Relationship Model (ER-Model) ★ Golden

The ER model is the architect's blueprint: you draw what will be stored and how it connects before you create tables in Access.

Component	Meaning	Symbol	Example
Entity	A real-world object we store data about	Rectangle	STUDENT, TEACHER, COURSE
Attribute	A property of an entity. PK is underlined	Oval	StudentID, Name, Class
Relationship	How two entities are connected	Diamond	Enrolls, Teaches, Has

Trap: The M:N rule.

Exam: Always state cardinality on the diagram (1 and M at the ends of the line). For M:N, the long question is "how do you resolve it?" — draw the junction entity with two...

ای آر ماڈل ڈیٹا بیس کا خاکہ ہے۔

4.5 Referential Integrity ★ Golden

Referential integrity means every foreign-key value in a child table must match an existing primary-key value in the parent — or be NULL if the design allows it.

- If Result.StudentID is a FK to Student.StudentID, you cannot type Result for student 999 if 999 is not in Student.
- You also cannot delete student 101 while results for 101 still exist — unless you first handle those child rows.

ID	Name
101	Ali
102	Sara

Trap: Cascade Delete is powerful and dangerous.

Exam: Write: definition, why it matters (no orphans), then one numbered example of Cascade Update and one of Cascade Delete. Mention parent vs child.

ریفرنشل انٹیگریٹی کا مطلب ہے بچہ ٹیبل کی فارن کی ہمیشہ والدین کی موجود پرائمری کی طرف اشارہ کرے۔

4.11 Creating Queries in MS Access ★ Golden

A query retrieves, filters and analyses data without changing the original tables.

- Create tab → Query Design.
- Show Table → add Student → Close.
- Double-click the fields you want; they drop into the grid.
- Red Run (!) button executes it.

Goal	Where to type	Example
Single criterion	Criteria row, one column	Department → "Computer Science"
AND (both must match)	Same Criteria row, two columns	Name "Ali" and Department "Computer Science"
OR (either may match)	Criteria row and the or: row	Department "IT" or "Computer Science"

Trap: Text criteria need quotes: "IT".

Exam: Remember: AND = same row in the grid; OR = different rows. Examiners draw a blank grid and ask you to fill Criteria and or:.

کوئری ٹیبل کو بدلے بغیر سوال پوچھتی ہے۔

Golden blitz — Chapters 5–6

★ only · Chapters 5–6 · Golden blitz · 3 hours

TODAY'S 3 HOURS

3 hours. 0–15 min recap · 15–150 min ★ Golden cards only (write every table from memory) · 150–180 min one 5-mark Golden answer.

Cover: Artificial Intelligence (AI) · Internet of Things (IoT) & Components · Information Sources (Primary, Secondary, Tertiary) · Assistive Technologies · Types of Data · Data-Collection Strategies · Interviews and Surveys · Prototypes, Observation and Simulations

★ **Must lock:** Artificial Intelligence (AI), Internet of Things (IoT) & Components, Information Sources (Primary, Secondary, Tertiary), Assistive Technologies, Types of Data, Data-Collection Strategies, Interviews and Surveys, Prototypes, Observation and Simulations, Primary and Secondary Data, Case Study: Digital Inquiry Project, Steps 1 – 3: Advanced Search and Methodology, Steps 4 – 6: Survey Preparation and Primary Data Collection, Steps 7 – 9: Secondary Data and Spreadsheet Organisation, Steps 10 – 12: Data Analysis, Conclusion and Recommendations, The Final Digital Artefact

If behind, skip non-Golden. Phone in another room.

Write every table and diagram on this day from memory on scrap paper, then check the cards.

5.1.1 Artificial Intelligence (AI) ★ Golden

Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding language and making decisions.

IoT (Eyes) — collects data → Analytics — sorts data → AI (Brain) — decides → Action

- Artificial — it is created by humans; it is not natural.
- Intelligence — it can perform tasks that require thinking.
- Face recognition on a mobile phone to unlock the screen.
- YouTube suggesting videos based on your interests.

Remember: Eyes – Sort – Brain – Do. IoT is the eyes, Analytics sorts, AI is the brain, and the actuator does the action. Think of a cricket fielder: his eyes see the...

Exam: Common questions: "Define AI and give two examples" (2–3 marks), and "Explain how AI, IoT and Data Analytics work together" (4–5 marks). To score full marks: (1) write...

مصنوعی ذہانت (AI) کو "مصنوعی" اس لیے کہتے ہیں کہ اسے انسانوں نے بنایا ہے، اور "ذہانت" اس لیے کہ یہ ایسے کام کر سکتی ہے جن کے لیے سوچنا پڑتا ہے۔

5.1.2 Internet of Things (IoT) & Components ★ Golden

The Internet of Things (IoT) is a system in which physical devices are connected to the Internet and can collect and share data to perform tasks automatically.

Sensor (Collects) → Wi-Fi (Connectivity) → Cloud (Processes) → Actuator (Acts) → User App (UI)

- The "things" are ordinary objects — watches, cameras, cars, fans, water pumps — that have been given sensors and an internet connection so...
- Source example: A smartwatch collects your heart-rate data and sends it to your mobile phone, where it is displayed.

#	Component	What it does	Examples
1	Sensors	Devices that collect information from the physical world. They are the "eyes and ears" of the system.	Temperature sensor, motion sensor, fingerprint sensor, smoke sensor, soil moisture sensor
2	Actuators	Devices that perform physical actions in response to system commands. They convert digital signals into physical movement.	Electric motor, switch, automatic door opener, water valve, fan controller
3	Connectivity	The connection that allows devices to send and receive data . Without it, sensors can't talk to the system.	Wi-Fi, Bluetooth, ZigBee, 4G/5G
4	Data Processing	Changing raw data into useful information . Steps: collecting, organising, analysing and deciding.	Checking if the room temperature is too hot to trigger the AC
5	Cloud Computing	Storing data and using services over the internet instead of local hard drives. Handles massive IoT data.	Google Drive, iCloud, online LMS
6	User Interface (UI)	The component that enables interaction between the user and the device . Helps users see information and give commands.	Mobile apps, touch screens, voice commands

Remember: Sensor senses, Actuator acts. Both words give you the answer: sensor = see/sense (input), actuator = act (output). Remember the six components with "Some Ants..."

Exam: "Define IoT and explain its components" is one of the most frequent long questions. Write the definition with the three key ideas — physical devices, connected to the...

انٹرنیٹ آف تھنگز (IoT) ایک ایسا نظام ہے جس میں مختلف آلات (جیسے گھڑیاں، کیمرے، گاڑیاں) انٹرنیٹ سے جڑے ہوتے ہیں اور خودکار طریقے سے...

5.3 Information Sources (Primary, Secondary, Tertiary) ★ Golden

Information sources are ways to find knowledge and data.

- The internet has vast information, but not all of it is accurate.
- Sources are divided into three main categories.
- Primary sources give original, firsthand information straight from people, events or official records.

Primary (firsthand)	Secondary (explained by others)	Tertiary (summarised, quick reference)
Interview with a flood victim in Dadu	Newspaper report on the 2022 floods	Encyclopedia entry "Floods in Pakistan"
Photograph taken at an event	Textbook chapter explaining the event	Wikipedia article
Personal diary or letter	Biography written by a historian	Dictionary
NADRA official records (CNIC data)	Pakistan Bureau of Statistics report analysing census data	Handbook / almanac of facts
Original survey responses	Research paper analysing the survey	Index or list of research papers
Sensor readings from an IoT weather station	TV news analysis of the weather	A school "fact sheet" on climate
Quaid-e-Azam's original speech recording	Article discussing the speech	Encyclopedia summary of his life

Remember: Rule of thumb (Author – Date – References): If an online article does not state the author's name, the date of publication, or list its references, it should...

Exam: Typical questions: "Differentiate between primary, secondary and tertiary sources with two examples each" (long/short), "Classify the following sources..." (MCQ/short)...

معلومات کے ذرائع علم اور ڈیٹا تلاش کرنے کے طریقے ہیں، جنہیں تین اقسام میں تقسیم کیا جاتا ہے۔

5.5 Assistive Technologies ★ Golden

Assistive technologies are devices, software and systems that help people with disabilities or functional limitations (difficulty in seeing, hearing, moving, speaking, remembering, etc.).

Mobility → Sensory → Communication → Cognitive

- They allow more people to participate equally in school, work and society.
- There are four categories:

Category	Purpose	Examples
1. Mobility aids	Help people move from one place to another.	Smart wheelchairs, stair lifts
2. Sensory aids	Support vision and hearing to improve interaction.	Hearing aids, screen readers (VoiceOver), Braille displays
3. Communication aids	Assist individuals with speech difficulties to express themselves.	Speech-generating devices, text-to-speech apps
4. Cognitive / learning aids	Help people with memory, reading and understanding.	Audiobooks, spell-checkers, digital reminder apps

Remember: Four categories = "My Small Cat Cries": Mobility, Sensory, Communication, Cognitive. Link each to a body function: legs (mobility), eyes/ears (sensory), mouth...

Exam: Expect: "What are assistive technologies? Explain their categories with examples" (long), "Name two sensory aids" (short), or "A screen reader is an example of ___"...

معاون ٹیکنالوجیز (Assistive Technologies) وہ آلات، سافٹ ویئر اور نظام ہیں جو معذور افراد کی مدد کرتے ہیں تاکہ وہ اسکول، کام اور مع... کے ساتھ

6.3 Types of Data ★ Golden

Data means facts, numbers, words or observations that we collect.

- The data gathered through a survey, an investigation or a classroom project can be described in two ways: qualitatively or quantitatively.
- Knowing the difference is critical, because each type is analysed in a different way.

Type of data	Nature	Examples	Best used for
Qualitative	Descriptive, non-numerical	Opinions, reasons, behaviour	Understanding why and how something happens
Quantitative	Numerical, countable or measurable	Hours, scores, percentages, totals	Comparing results and making graphs/charts

Remember: QUALitative = QUALity = words and feelings. QUANTitative = QUANTity = numbers and counts. Qualitative answers why/how; quantitative answers how many/how much.

Exam: Common questions: "Differentiate between qualitative and quantitative data" (4–5 marks) or "Classify the following data". Write the answer as a table with four rows:...

کیفیتی ڈیٹا (Qualitative Data) الفاظ پر مشتمل ہوتا ہے اور بتاتا ہے کہ لوگ کیا سوچتے، محسوس کرتے یا تجربہ کرتے ہیں، جیسے آن لائن کل... کے ساتھ

6.4 Data-Collection Strategies ★ Golden

Digital literacy is not limited to finding information that already exists on the internet.

Interviews → Surveys → Prototypes → Observation → Simulations

- You should also learn how to gather original (new) information yourself.
- A data-collection strategy is a planned method for gathering this information.
- The five common strategies are: Interviews-Surveys-Prototypes-Observation-Simulations

Strategy	How it works (one line)	Usual data type	Best when...
Interview	Planned conversation with a person or small group	Mostly qualitative	You need deep, detailed answers from a few people
Survey	Same questions asked to many people	Quantitative and qualitative	You need answers from many people quickly
Prototype	Users test an early version and give feedback	Both	You want to improve a product or solution
Observation	Watch and record; no questions asked	Notes (qualitative) or counts (quantitative)	You want to see what people actually do
Simulation	Computer model; change variables and see what happens	Mostly quantitative	Real experiments are costly, risky or difficult

Remember: Mnemonic for the five strategies: "I S P O S" – "I Saw People On Screens": Interviews, Surveys, Prototypes, Observation, Simulations.

Exam: "List and explain the data-collection strategies" is a frequent long question. Name all five, give a one-sentence definition and one example for each, and say what type...

ڈیجیٹل خواندگی صرف موجود معلومات تلاش کرنے تک محدود نہیں، بلکہ نئی معلومات خود جمع کرنا بھی اس کا حصہ ہے۔

6.4.1 – 6.4.2 Interviews and Surveys ★ Golden

An interview is a planned conversation used to gather information directly from a person.

- It may be one-to-one (you and one person) or in a small group.
- Questions are planned in advance, so you do not forget anything important.
- Answers are usually qualitative (opinions, reasons, experiences).

Question type	What it looks like	Data it gives
Closed: Yes/No	Do you study with your phone nearby? (Yes / No)	Countable (quantitative)
Closed: Multiple choice	Which activity do you use your phone for most? (Lectures / Notes / Social media / Games)	Categories you can count
Number / scale	How many hours do you use your phone daily? ____	Quantitative
Rating scale	"My phone helps me study." 1 = Strongly disagree ... 5 = Strongly agree	Quantitative (you can average the ratings)
Open-ended	How does your phone help you in studies?	Qualitative (free text)

Remember: Interview = In-depth with few. Survey = Same questions to many. Think of a doctor talking to one patient (interview) vs a census form sent to every house...

Exam: You may be asked to "differentiate between an interview and a survey" or to "identify what is wrong with this survey question and rewrite it". For rewriting, name the...

انٹرویو ایک منصوبہ بند گفتگو ہے جس میں کسی ایک فرد یا چھوٹے گروہ سے براہ راست معلومات حاصل کی جاتی ہیں؛ اس کے جوابات عموماً کیفیتی ہوتے ہیں۔

6.4.3 – 6.4.5 Prototypes, Observation and Simulations ★ Golden

A prototype is a simple early version of a product, model or solution.

- It allows an idea to be tested before time and effort are spent on the final version.
- Users try the prototype and give feedback, and this feedback becomes useful data for improvement.
- Example: Before building a full school app, you draw the screens on paper (or make a few clickable screens in a design tool) and ask five...

Remember: Watch it → Observation. Ask it → Survey/Interview. Try it → Prototype. Model it → Simulation.

Exam: Scenario questions are popular: "Which data-collection method is most suitable for ... ? Give a reason." Always name the method and justify it using its key feature (e.g....

پروٹوٹائپ کسی پروڈکٹ یا حل کا ابتدائی سادہ نمونہ ہے جسے صارفین آزما کر رائے دیتے ہیں تاکہ اسے بہتر بنایا جا سکے۔

6.5 Primary and Secondary Data ★ Golden

Whenever you use data, it is important to know where it came from.

- The source affects how much control the researcher had over collecting it, and how confidently the results can be used.

Aspect	Primary Data	Secondary Data
Source	Collected directly by the researcher.	Already collected by others.
Examples	Survey responses, interviews, observations.	Reports, articles, websites, databases.
Purpose	Prepared for a specific study.	Reused for a new purpose.
Control	High control over questions and collection process.	Limited control because the data already exists.

Remember: Primary = "1st hand" (you collect it first). Secondary = "2nd hand" (like a second-hand book – someone else used it first).

Exam: "Differentiate between primary and secondary data" is one of the most-asked questions in this chapter. Draw the four-row table: Source, Examples, Purpose, Control. Learn...

بنیادی ڈیٹا (Primary Data) محقق خود کسی خاص مقصد کے لیے براہ راست جمع کرتا ہے، مثلاً سروے، انٹرویو یا مشاہدہ۔

6.8 Case Study: Digital Inquiry Project ★ Golden

This case study shows how to apply all the ideas of this chapter to solve a real-world problem from start to finish.

1 Question → 2 Search → 3 Collect → 4 Analyse → 5 Create

- An inquiry is an investigation to find the answer to a question.
- Research question: "How does daily mobile phone use affect the study habits of students?" Scenario: Students notice classmates using phones...
- Some say phones help them learn; others say phones waste time.

Workflow stage	Steps
Question & Search	1 Advanced searches · 2 Identify data types · 3 Select methods
Collect	4 Prepare the survey · 5 Choose the sample · 6 Collect primary data ethically · 7 Use secondary data
Analyse	8 Organise in a spreadsheet · 9 Create charts · 10 Analyse results
Create	11 Write the conclusion · 12 Recommendations → final digital artefact

Remember: Workflow in five words: Ask → Search → Collect → Analyse → Create. Every research project you ever do will follow this pattern.

Exam: Expect a long question: "Describe the steps of a digital inquiry project using the example of mobile phone use." Write the research question first, then the 12 steps...

اس کیس اسٹڈی میں تحقیقی سوال یہ ہے: "روزانہ موبائل فون کا استعمال طلبہ کی پڑھائی کی عادات پر کیسے اثر ڈالتا ہے؟"

6.8 Steps 1–3: Advanced Search and Methodology ★ Golden

Students first search for background information using search engines (such as Google or Bing).

- They use advanced search operators – special symbols or words typed into the search box – to narrow down results and find reliable...
- Suggested queryOperator usedWhat it does "mobile phone use" "study habits" students" " (quotes)Finds the exact phrases "mobile phone use"...
- "academic performance" filetype:pdfFinds only PDF documents (often official reports and research papers).

Suggested query	Operator used	What it does
"mobile phone use" "study habits" students	" " (quotes)	Finds the exact phrases "mobile phone use" and "study habits", not the words scattered randomly.
"academic performance" filetype:pdf	filetype:	Finds only PDF documents (often official reports and research papers).
digital distraction site:.edu	site:	Searches only educational websites (addresses ending in .edu).

Remember: Three operators, three jobs: " " = exact, filetype: = format, site: = where. Note: there is no space after the colon (site:.edu, not site: .edu).

Exam: "Explain any three advanced search operators with examples" is a likely short question. For each, write the operator, an example query, and what it does. Also learn the...

پہلے قدم میں طلبہ سرچ انجن پر جدید سرچ آپریٹرز (Advanced Search Operators) استعمال کرتے ہیں: کوٹیشن مارکس (" ") بالکل وہی جملہ تلا...

6.8 Steps 4–6: Survey Preparation and Primary Data Collection ★ Golden

Design clear, unbiased questions. Each question is linked to the type of data it produces: QuestionType of data How many hours do you use your phone daily?Quantitative Which activity do you use your phone for the most?Quantitative...

- Design clear, unbiased questions.
- Each question is linked to the type of data it produces: QuestionType of data How many hours do you use your phone daily?Quantitative Which...

Question	Type of data
How many hours do you use your phone daily?	Quantitative
Which activity do you use your phone for the most?	Quantitative (categories)
How does your phone help you in studies?	Qualitative
Suggest one way to use phones responsibly.	Qualitative

Remember: Ethics in five words: Consent, No-Sensitive, Anonymous, Private, Educational – "C-NAPE". Think: "Cover the person's nape (neck) – protect them."

Exam: Questions often ask: "What ethical guidelines should be followed when collecting data?" List all five points. Use the words consent and anonymity – examiners look for...

چوتھے قدم میں واضح اور غیر جانبدار سوالات تیار کیے جاتے ہیں۔

6.8 Steps 7 – 9: Secondary Data and Spreadsheet Organisation ★ Golden

Students include a reliable online article stating that "excessive screen time lowers concentration." They will compare this secondary data with their own primary survey results.

- If both agree, the conclusion is stronger.
- If they disagree, the students must think about why (for example, a different age group or country).

Student	Hours	Study Use	Main Activity	Distracted?
1	3	Yes	Lectures	Sometimes
2	5	No	Social Media	Yes
3	2	Yes	Notes	No

Remember: Chart chooser: Bar = Compare, Pie = Parts of a whole, Line = Change over time.

Exam: You may get the spreadsheet table and be asked to "write formulas to find the total and average hours" or "count students who use the phone for study". Write the full...

ساتویں قدم میں ایک قابلِ اعتماد آن لائن مضمون کا ثانوی ڈیٹا ("زیادہ اسکرین ٹائم توجہ کم کرتا ہے") اپنے سروے کے بنیادی ڈیٹا سے موازنہ...

6.8 Steps 10 – 12: Data Analysis, Conclusion and Recommendations ★ Golden

To analyse means to study the data carefully to find patterns and trends.

- Most students (the majority) use phones for 3–5 hours daily.
- Half use it for educational purposes (lectures/notes).
- A clear trend: students who use phones mostly for entertainment report much higher distraction levels than those who use them for study.

Remember: Conclusion formula: "The data shows ... However ... Therefore ..." – state the main finding, mention the other side, then give the overall answer.

Exam: You may be asked to "write a conclusion and two recommendations" for given data. Use the "shows ... however ... therefore" pattern and make sure each recommendation follows...

دسویں قدم میں چارٹس اور اسپریڈشیٹ سے رجحانات تلاش کیے جاتے ہیں: زیادہ تر طلبہ روزانہ تین سے پانچ گھنٹے فون استعمال کرتے ہیں، نصف ا...

6.8 The Final Digital Artefact ★ Golden

A digital artefact is a final product created using digital technologies.

- Students must create one final artefact that answers the original research question.
- It is not just a decorated file; it must present evidence and conclusions.
- Artefact option Suitable digital tools Main purpose Digital Report MS Word, Google Docs Explain the full study in a structured text document.

Artefact option	Suitable digital tools	Main purpose
Digital Report	MS Word, Google Docs	Explain the full study in a structured text document.
Presentation	PowerPoint, Canva, Slides	Share key findings with a live audience.
Infographic	Canva, Adobe Express	Show the main results visually and quickly on one page.
Dashboard	Power BI, Tableau	Explore data through interactive charts and filters.

Remember: The structure follows the inquiry itself: What (title, question, intro) → How (method) → Found (primary, secondary, charts) → So what (conclusion,...

Exam: Learn the four artefact options with tools and purpose (a table question), and the ten-part structure in order. "What is a digital artefact?" – stress that it must...

ڈیجیٹل آرٹیفیکٹ (Digital Artefact) ڈیجیٹل ٹیکنالوجی سے بنائی گئی آخری پروڈکٹ ہے جو تحقیقی سوال کا جواب ثبوت اور نتائج کے ساتھ پیش...

Night-before recaps

Say every chapter recap aloud, book closed · Recap · 3 hours

TODAY'S 3 HOURS

3 hours. 0–20 min skim the recap pages · 20–150 min say each chapter aloud, book closed · 150–180 min rewrite any line you could not say.

Cover: Say every chapter recap aloud, book closed

★ **Must lock:** All chapters

Phone in another room.

No new cards. Recite, close the booklet, recite again.

Chapter 1 recap — Computer Systems

- Discrete = counted (stairs). Continuous = measured (ramp). Digital systems use bits 0 and 1; they are reliable, accurate, easy to design and programmable.
- Analog signal = smooth sine wave. Digital signal = square wave HIGH/LOW. Digital data resists noise.
- AND · output 1 only if all inputs 1. OR + output 1 if any input 1. NOT ' inverts. NAND = NOT-AND, NOR = NOT-OR (universal). XOR 1 when inputs differ. XNOR 1 when inputs same.
- Truth table rows = 2^n . SOP = sum of minterms where output is 1. POS = product of maxterms where output is 0.
- K-map: group 1, 2, 4, 8 adjacent 1s (power of 2, wrap-around allowed). Each group drops the variables that change.
- SDLC: requirement → design → implementation → testing → deployment → maintenance. Waterfall = sequential; Agile = short sprints and feedback.
- OSI 7: Physical, Data Link, Network, Transport, Session, Presentation, Application. Mnemonic: Please Do Not Throw Sausage Pizza Away. TCP/IP 4: Network Access, Internet, Transport, Application.

Chapter 2 recap — Computational Thinking & Algorithm

- CT pillars: decomposition, pattern recognition, abstraction, algorithms.
- Algorithm = finite, ordered, unambiguous steps. Pseudocode = structured English, not a programming language.
- Bubble Sort: compare neighbours, swap if out of order; largest "bubbles" to the end each pass. $O(n^2)$.
- Selection Sort: find the minimum in the unsorted part and swap it to the front. $O(n^2)$.
- Linear Search: check one by one. Works on unsorted data. $O(n)$.
- Binary Search: list MUST be sorted. Repeatedly test the middle. $O(\log n)$. Trace with low / mid / high.
- Judge an algorithm on correctness, efficiency (time and space) and clarity. Prefer binary search on large sorted lists; linear search on small or unsorted lists.

Chapter 3 recap — Programming Fundamentals

- Python: high-level, interpreted. Variables are names; types int, float, str, bool, list, dict. `input()` returns str.
- Arithmetic + - * / // % **. Relational. Logical and or not . Bitwise & | ^ ~ << >> .
- Selection: if / if-else / if-elif-else / nested. Test from the top; first True wins.
- for = known count + range. while = unknown count; always update the condition. break leaves; continue skips one.
- Libraries: import math / random / datetime. Third-party via pip: NumPy, Pandas, Matplotlib.
- Syntax = will not run. Runtime = crash + traceback. Logical = runs, wrong answer.

Chapter 4 recap — Data and Analysis

- Data = raw facts. Information = processed data used for decisions. Database = organised related data. DBMS = software that manages it (reduces redundancy, improves security, sharing, integrity).
- Table / relation; row / record / tuple; column / field / attribute. Primary key uniquely identifies a row. Foreign key refers to another table's primary key.
- ER model: entity (rectangle), attribute (oval), relationship (diamond), cardinality 1:1, 1:M, M:N.
- Referential integrity: you cannot orphan a child row. Cascade Update / Cascade Delete copy or remove matching child rows.
- Access objects: tables, forms, queries, reports. Queries filter with criteria, wildcards (*, ?), AND/OR. Totals: Count, Sum, Avg, Group By. Charts visualise the result.

Chapter 5 recap — Application and Impacts of Computing

- IoT = connected devices. Six parts: sensor, actuator, connectivity, processing, cloud, UI. AI = machines that learn and decide. Analytics = collect → store → clean → analyse → visualise.
- Together: IoT (eyes) → Analytics (sort) → AI (brain) → actuator (action).
- Sources: primary (firsthand), secondary (explained by others), tertiary (summaries). No author / date / references → do not trust.
- Assistive tech: mobility, sensory, communication, cognitive. Goal: equal access, independence, inclusion.

Chapter 6 recap — Digital Literacy

- Digital literacy = find, evaluate, create and share information with digital tools, safely.
- Qualitative = words/descriptions. Quantitative = numbers. Strategies: interview, survey, prototype, observation, simulation.
- Primary data you collect; secondary data already exists. Good survey questions are clear, single-idea, unbiased.
- Present with spreadsheets (formulas such as `=AVERAGE()`), slides, infographics and reports. Cite sources. Follow the 12-step inquiry project from question to artefact.

DAY 28 OF 30 · WEEK 6

Mock paper — hall conditions

2 hours 30 minutes · 75 marks · no notes · Mock · 3 hours

TODAY'S 3 HOURS

3 hours. Sit the mock in 2 hours 30 minutes. Hall conditions: no notes, no phone, no Day 29 peeking. Use the leftover 30 minutes to pack the paper — do not mark it today.

Cover: 2 hours 30 minutes · 75 marks · no notes

★ **Must lock:** Whole syllabus

Phone in another room.

Invigilate yourself. No phone, no notes, no Day 29 peeking.

Mock paper — Computer Science XI

Time: 2 hours 30 minutes | Total: 75 marks | Answer all questions. Calculators allowed. Write Python with correct indentation.

Section A — Multiple choice (15 × 1 = 15)

1. Which quantity is discrete?
 - a. Room temperature
 - b. Number of desks
 - c. Car speed
 - d. Time of day
2. NAND is called universal because:
 - a. it has two inputs
 - b. any gate can be built from it
 - c. its output is always 1
 - d. it is faster than AND
3. The Boolean expression for XOR is:
 - a. $A \cdot B$
 - b. $A + B$
 - c. $A \oplus B$
 - d. $(A \cdot B)'$
4. OSI layer that chooses the best path (IP addresses) is:
 - a. Transport
 - b. Network
 - c. Data Link
 - d. Session
5. Which SDLC model delivers working software in short sprints?
 - a. Waterfall
 - b. Agile
 - c. Spiral only
 - d. Big Bang
6. Binary Search requires the list to be:
 - a. unsorted
 - b. sorted
 - c. all unique and unsorted
 - d. of even length
7. Worst-case time of Bubble Sort is:
 - a. $O(n)$
 - b. $O(\log n)$
 - c. $O(n^2)$
 - d. $O(1)$
8. `input()` in Python returns a:
 - a. int
 - b. str
 - c. float
 - d. bool
9. `17 // 5` equals:
 - a. 3.4
 - b. 3
 - c. 2
 - d. 3.0
10. A program that runs but prints the wrong total has a:
 - a. syntax error
 - b. runtime error
 - c. logical error
 - d. hardware fault
11. A field that uniquely identifies each record is a:
 - a. foreign key
 - b. primary key
 - c. composite description
 - d. query
12. An actuator in IoT:
 - a. collects temperature
 - b. opens a water valve
 - c. stores cloud files
 - d. draws a chart
13. Wikipedia is a:
 - a. primary source
 - b. secondary source
 - c. tertiary source
 - d. sensor
14. Marks out of 100 are:
 - a. qualitative data
 - b. quantitative data
 - c. tertiary data
 - d. metadata only
15. Which Access object is best for printing a formatted class list?
 - a. Table
 - b. Form
 - c. Query
 - d. Report

Section B — Short questions (10 × 3 = 30)

1. Differentiate analog and digital signals with one example of each.
2. Draw the symbol and truth table of a 2-input NOR gate.
3. List the four pillars of computational thinking in one sentence each.
4. Why can you not use Binary Search on an unsorted attendance list?
5. Write a Python `if-elif-else` that prints Pass if marks ≥ 40 , else Fail.
6. Define primary key and foreign key.
7. Name the six components of IoT.
8. Classify: (a) a photograph you took, (b) a textbook, (c) an encyclopedia.
9. Give two good and two bad survey questions about mobile-phone use in college.
10. State three advantages of a DBMS over paper files.

Section C — Long questions (3 × 10 = 30). Attempt any three.

1. Simplify $F(A,B,C) = \Sigma(1,3,5,7)$ using a 3-variable K-map. Show the grouping and the final expression. Draw the logic diagram.
2. Dry-run Bubble Sort on the list [5, 1, 4, 2]. Show every pass. Then search for 4 using Binary Search on the sorted result, with a low/mid/high table.
3. Write a complete Python program that reads n subject marks, prints the average, and prints "A" if average ≥ 80 , "B" if ≥ 70 , "C" if ≥ 60 , else "D". Include a sample run of four marks: 80, 70, 90, 60.
4. Design an ER diagram and relational schema for a school library (Student, Book, Issue). Mark keys. Write one Access/SQL query that lists all books issued to roll number 101.
5. Explain how IoT, Data Analytics and AI work together on a Sindh farm. Then discuss two positive and two negative impacts of computing on education in Pakistan.

Mark the mock · repair list

Mark to the key. List every ★ Golden you missed. · Mark · 3 hours

TODAY'S 3 HOURS

3 hours. 0–90 min mark the mock to the key · 90–150 min fill the repair list · 150–180 min rewrite one missed Golden from memory.

Cover: Mark to the key. List every ★ Golden you missed.

★ **Must lock:** Missed Golden

Phone in another room.

An answer with no working, no diagram or no example cannot score full marks.

Mock paper — answers

Section A

1. b
2. b
3. c
4. b
5. b
6. b
7. c
8. b
9. b
10. c
11. b
12. b
13. c
14. b
15. d

Section B

1. Analog: continuous, sine wave, e.g. human voice. Digital: two levels 0/1, square wave, e.g. computer data. Digital resists noise.
2. NOR symbol: OR-gate shape with a bubble on the output. $Y = (A+B)'$. Truth: 00→1, 01→0, 10→0, 11→0.
3. Decomposition = split a problem. Pattern recognition = reuse similarities. Abstraction = ignore extra detail. Algorithms = write ordered steps.
4. Binary Search repeatedly halves a sorted range. If the list is unsorted, the middle value does not tell you which half to keep, so the method is invalid.
5. `m = int(input("Marks: "))` then `if m ≥ 40: print("Pass") else print("Fail")`.
6. Primary key: unique identifier of a row (e.g. StudentID). Foreign key: a field that refers to the primary key of another table (e.g. StudentID in Issuance).
7. Sensors, actuators, connectivity, data processing, cloud computing, user interface.
8. (a) primary (b) secondary (c) tertiary.
9. Good: "How many hours per day do you use a phone for study?" / "Which apps do you use for class notes?" Bad: "Don't you agree phones are destroying students?" (leading) / "Do you like phones and also use laptops?" (double-barrelled).
10. Less redundancy, easier search and update, better security and multi-user access, integrity constraints. (Any three.)

Section C — outline

- 1. K-map 1s at m1, m3, m5, m7 form one group of 4 covering B=1 (or C=1 depending on labelling — with standard ABC order the 1s are all cells where C=1). F = C. Diagram: a single wire C to output (or a buffer). Show the map with the group circled.
- 2. Pass 1: [1,4,2,5]; Pass 2: [1,2,4,5]; Pass 3: already sorted. Sorted [1,2,4,5]. Search 4: low=0 high=3 mid=1 (2<4) → low=2; mid=2 (4=4) found at index 2.
- 3. Read n, loop n times adding `float(input())`, avg = total/n, then if-elif for A/B/C/D. Sample: total 300, avg 75.0, grade B.
- 4. Entities Student(StudentID PK, Name, Class), Book(BookID PK, Title, Author), Issue(IssueID PK, StudentID FK, BookID FK, Date). Query: `SELECT Book.Title FROM Issue INNER JOIN Book ON Issue.BookID = Book.BookID WHERE StudentID = 101;`
- 5. Soil sensor (IoT) → cloud spreadsheet of moisture (Analytics) → model decides "too dry" (AI) → pump (actuator). Education +: LMS, remote classes. Education -: screen addiction, urban/rural device gap.

REPAIR LIST

Repair list. Copy every Golden you scored 0 or half on. Tomorrow you redo only this list — not the whole book.

#	Golden topic missed	What I will rewrite from memory
1		
2		
3		
4		
5		
6		

Repair day · then stop

Redo the Day 29 list only. Sleep by 10 pm. · Repair · 3 hours

TODAY'S 3 HOURS

3 hours. Redo only the Day 29 repair list. Stop at 6 pm. Sleep by 10 pm. No new topic.

Cover: Redo the Day 29 list only. Sleep by 10 pm.

★ **Must lock:** Missed Golden only

Phone in another room.

No new chapter. No new non-Golden card. Redo each missed Golden: say the definition, draw the table, write one 5-mark answer.

Stop at 6 pm. Pack your pen, ruler and this booklet's Golden cards in your head — not in the hall.

نیا باب نہ کھولیں۔ صرف چھوٹی گولڈن دہرائیں۔ دس بجے سو جائیں۔